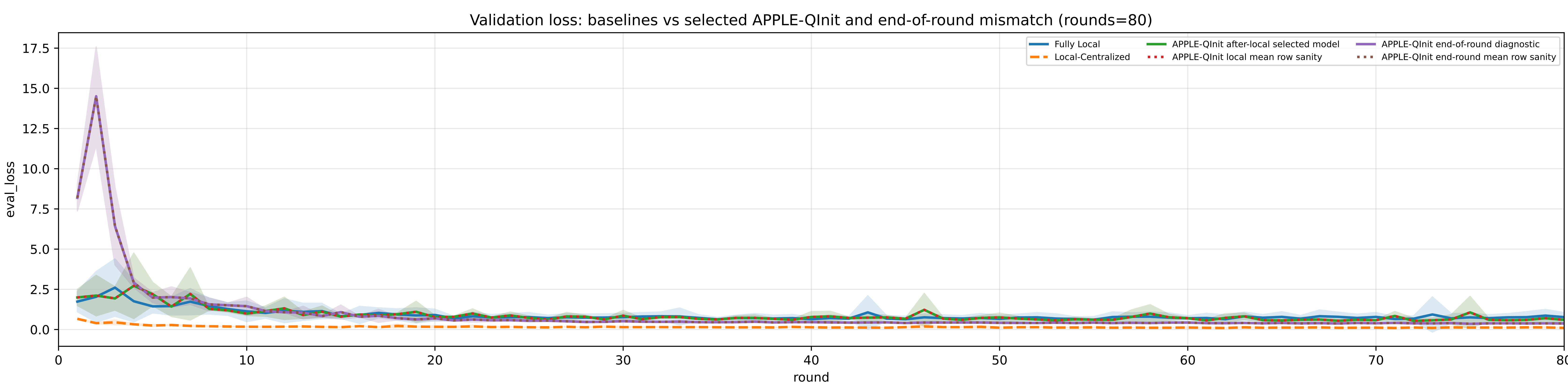
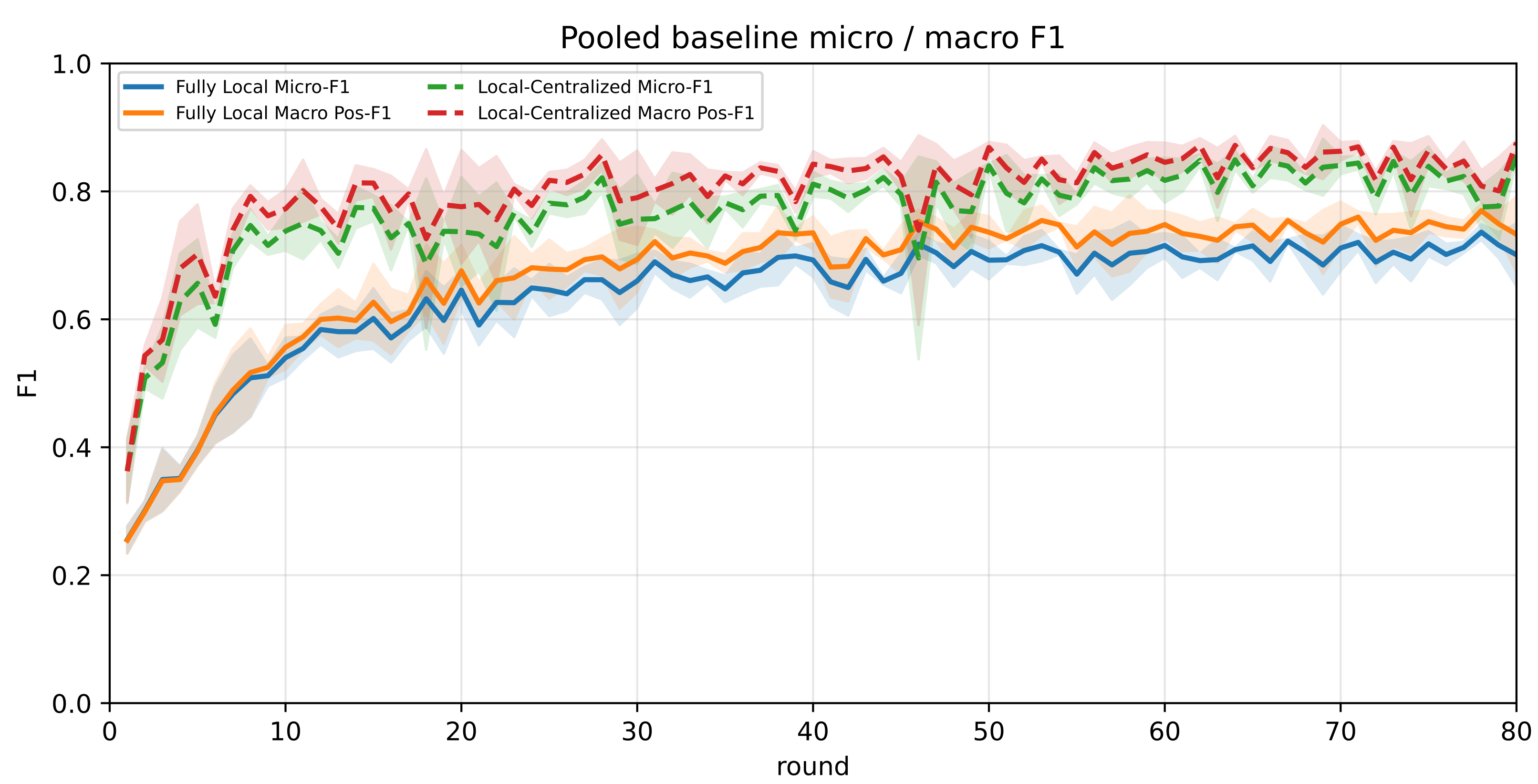
Controlled q-label heterogeneity: iid | APPLE-QInit diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

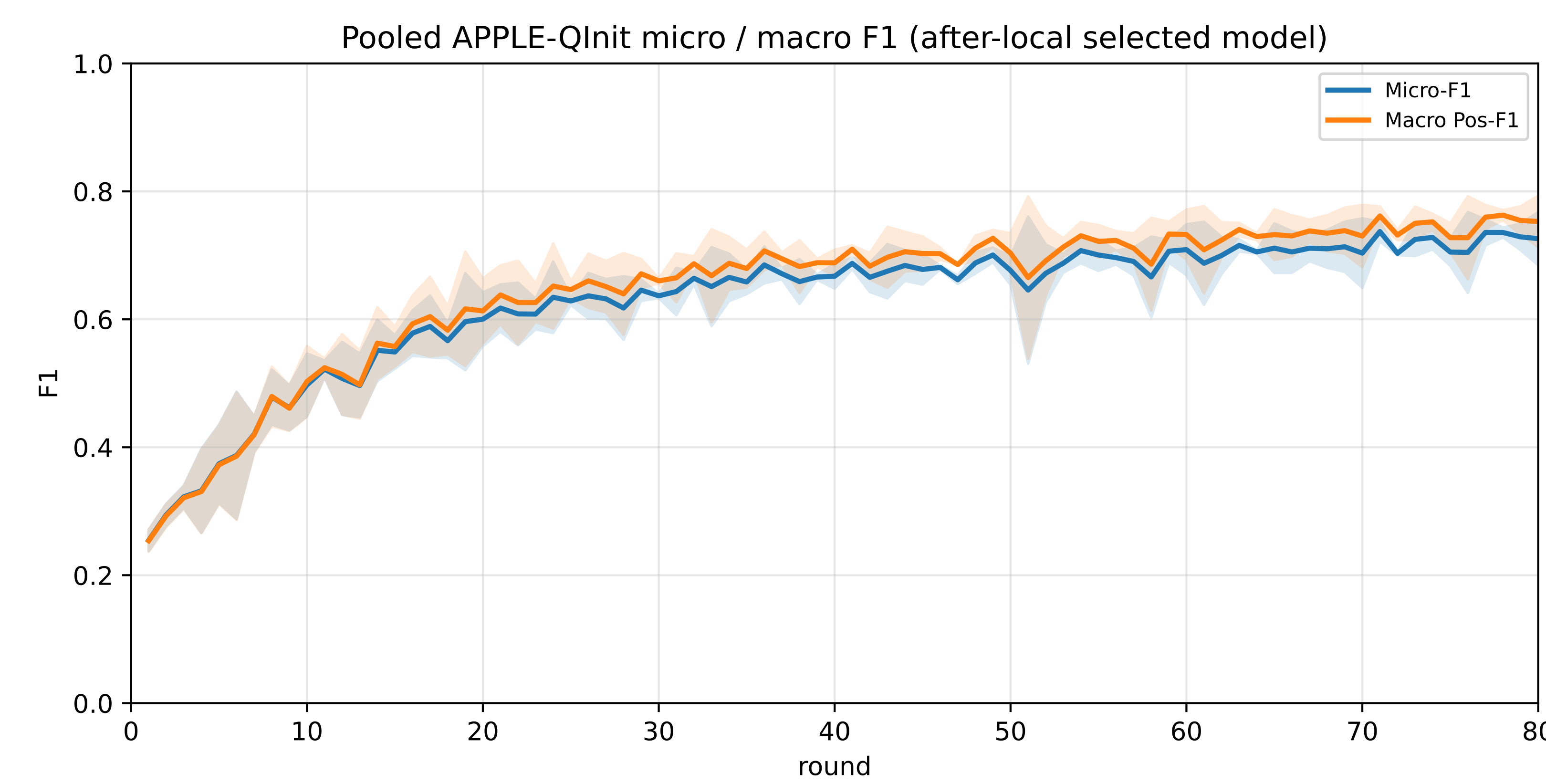
family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

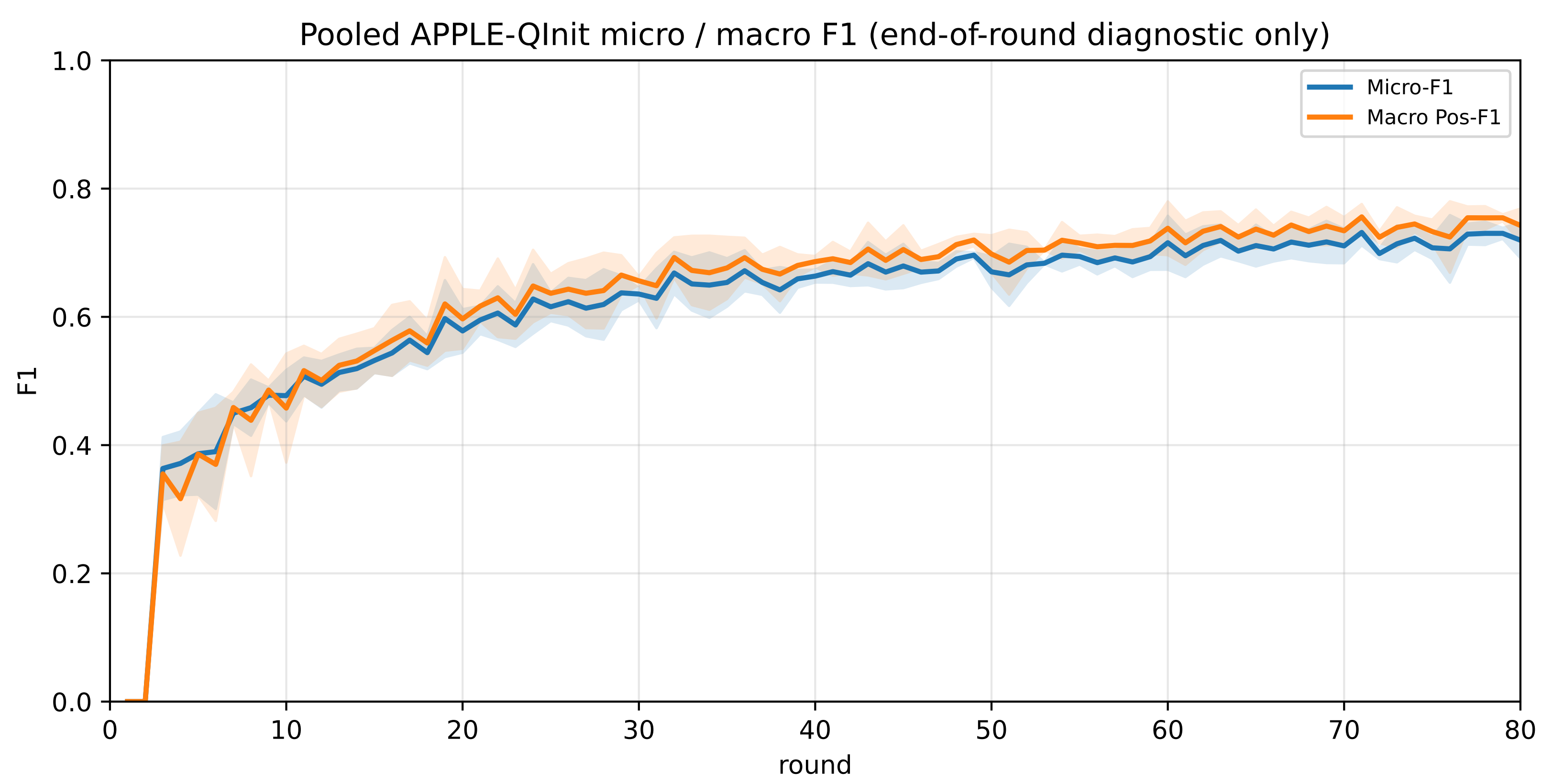
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

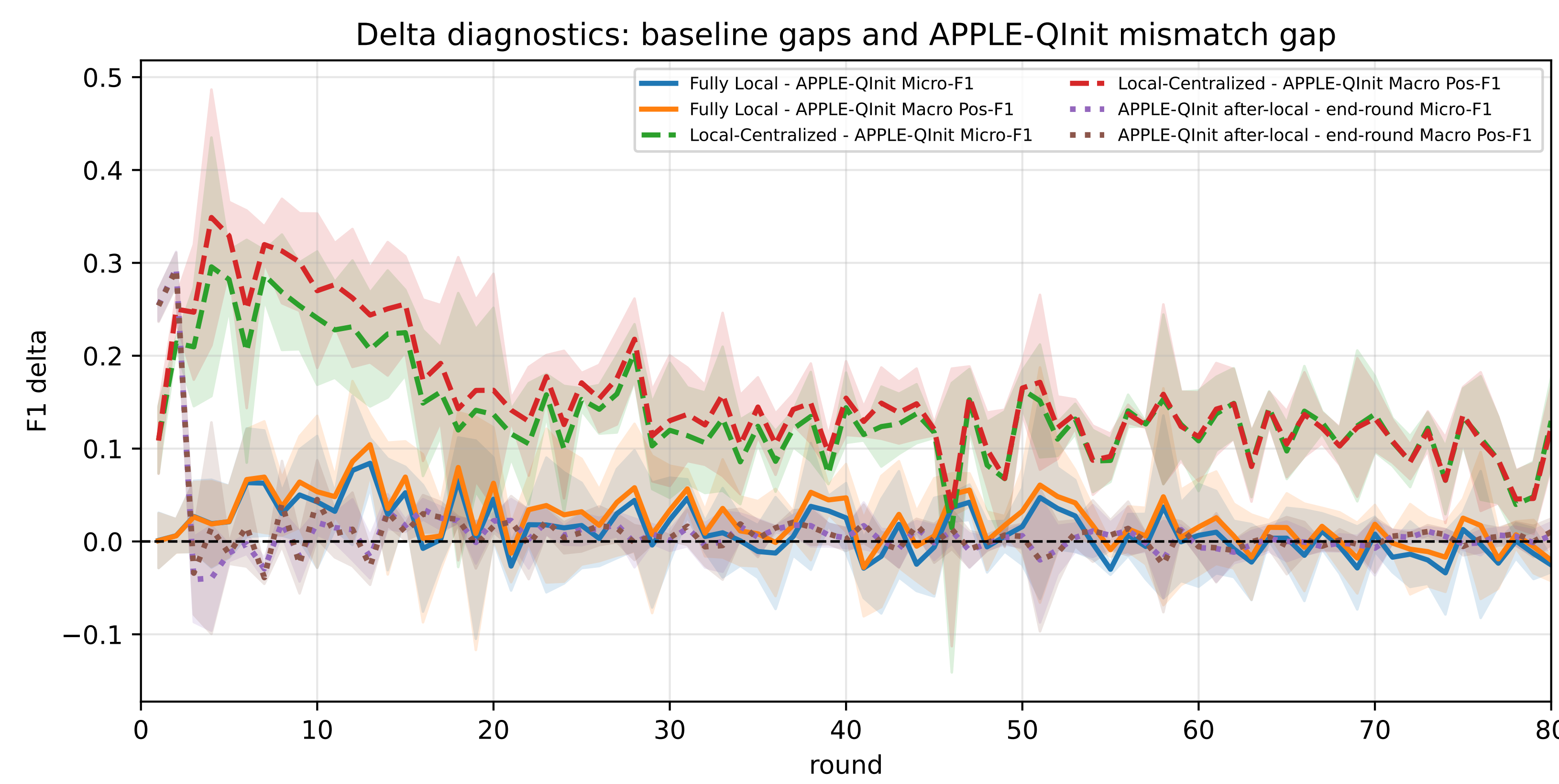
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	67.9% ± 1.6%	72.3% ± 1.5%	90.0% ± 1.5%	85.0% ± 1.7%	67.5% ± 3.5%	66.8% ± 2.2%	50.9% ± 1.2%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
APPLE-QInit	69.5% ± 0.0%	72.6% ± 0.0%	88.6% ± 0.4%	81.9% ± 1.5%	69.4% ± 1.9%	64.0% ± 1.7%	56.8% ± 1.4%











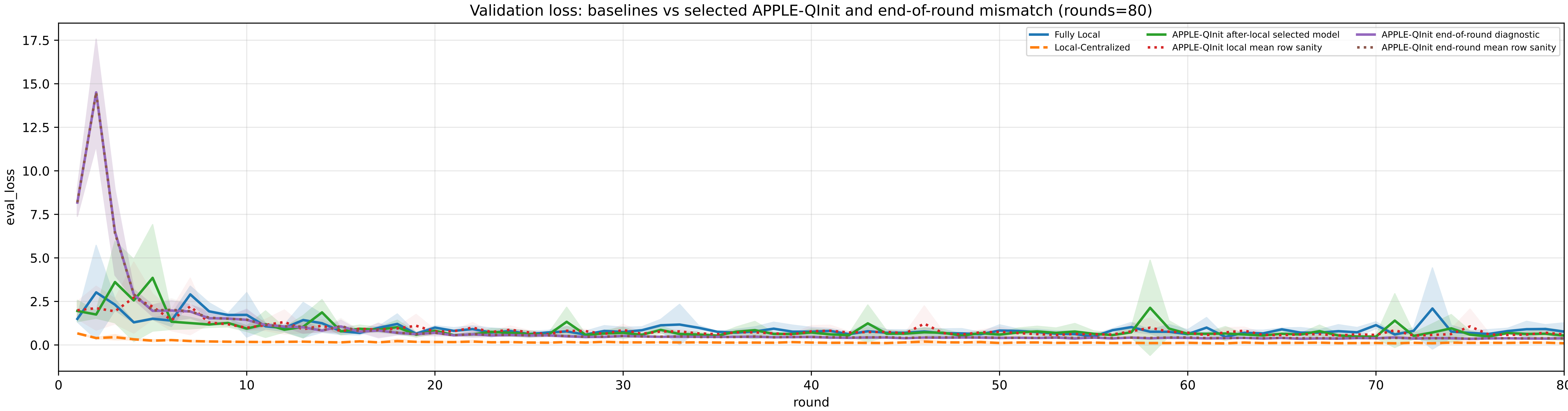
Controlled q-label heterogeneity: iid | APPLE-QInit diagnostics | family=q_task_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

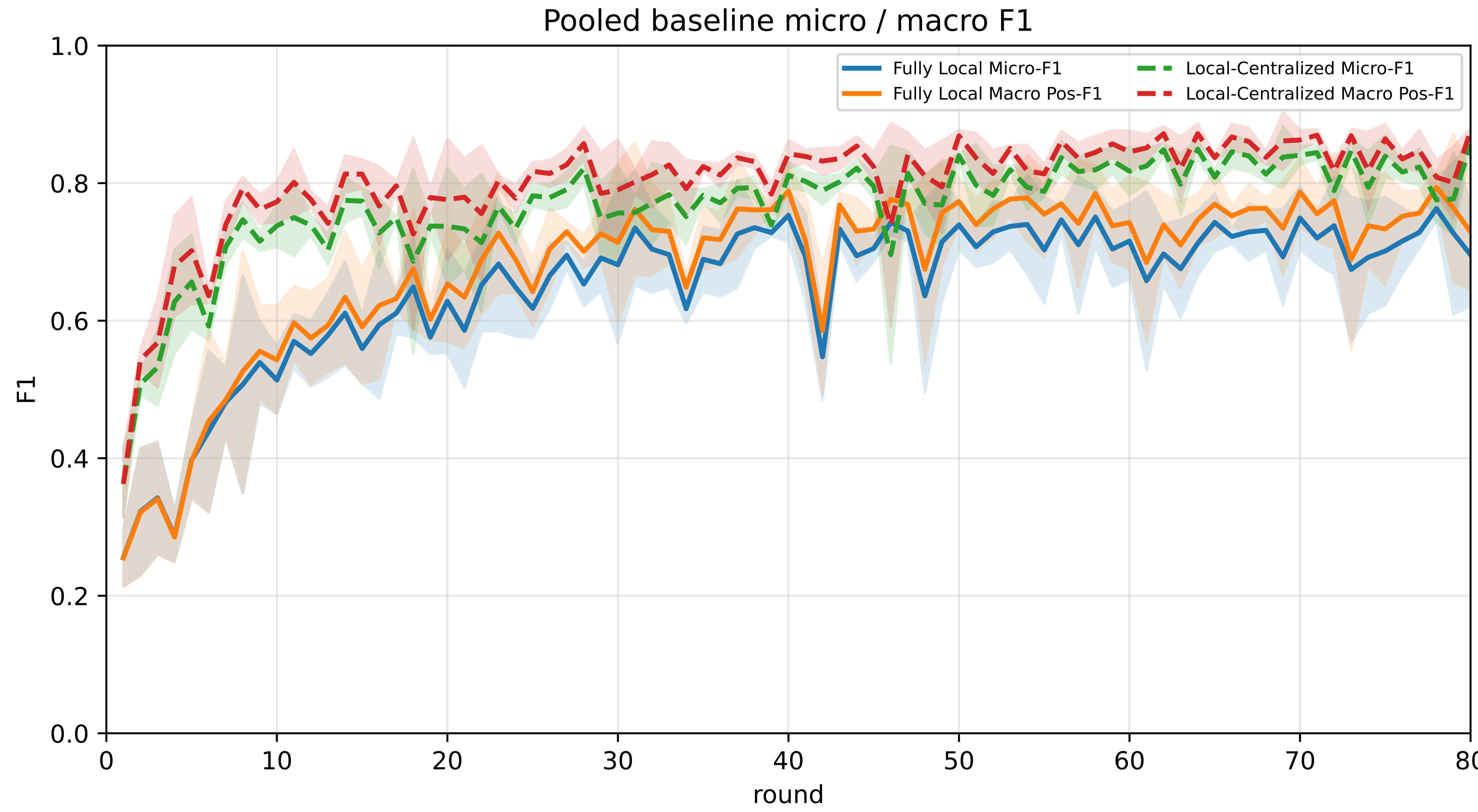
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	70.1% ± 3.2%	74.3% ± 2.5%	93.1% ± 1.1%	86.7% ± 1.6%	70.3% ± 4.6%	67.5% ± 2.2%	53.8% ± 5.0%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
APPLE-QInit	69.2% ± 4.1%	72.4% ± 4.3%	88.9% ± 3.2%	81.5% ± 8.4%	69.6% ± 6.6%	65.9% ± 4.9%	55.9% ± 3.3%

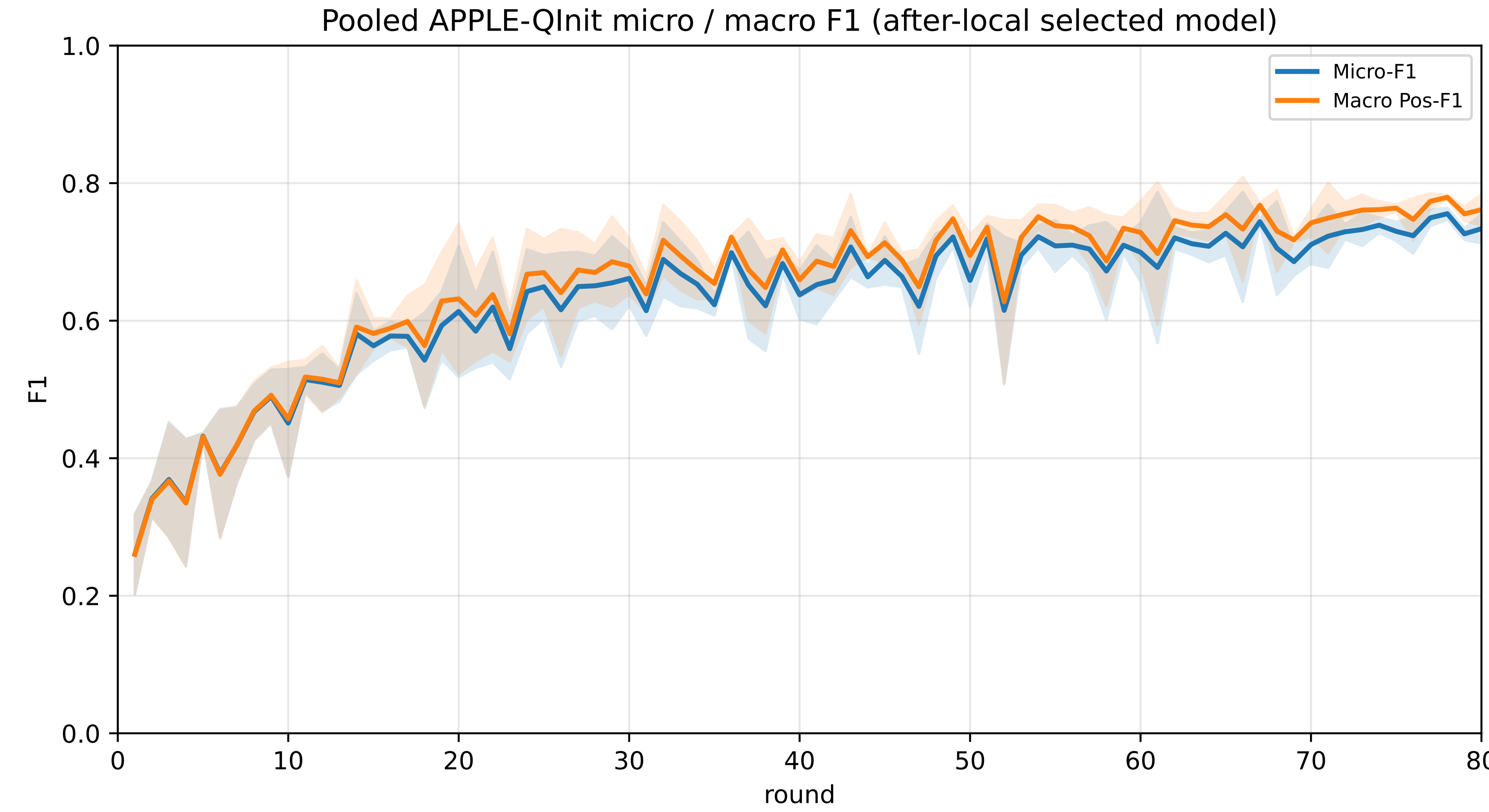
Validation loss: baselines vs selected APPLE-QInit and end-of-round mismatch (rounds=80)



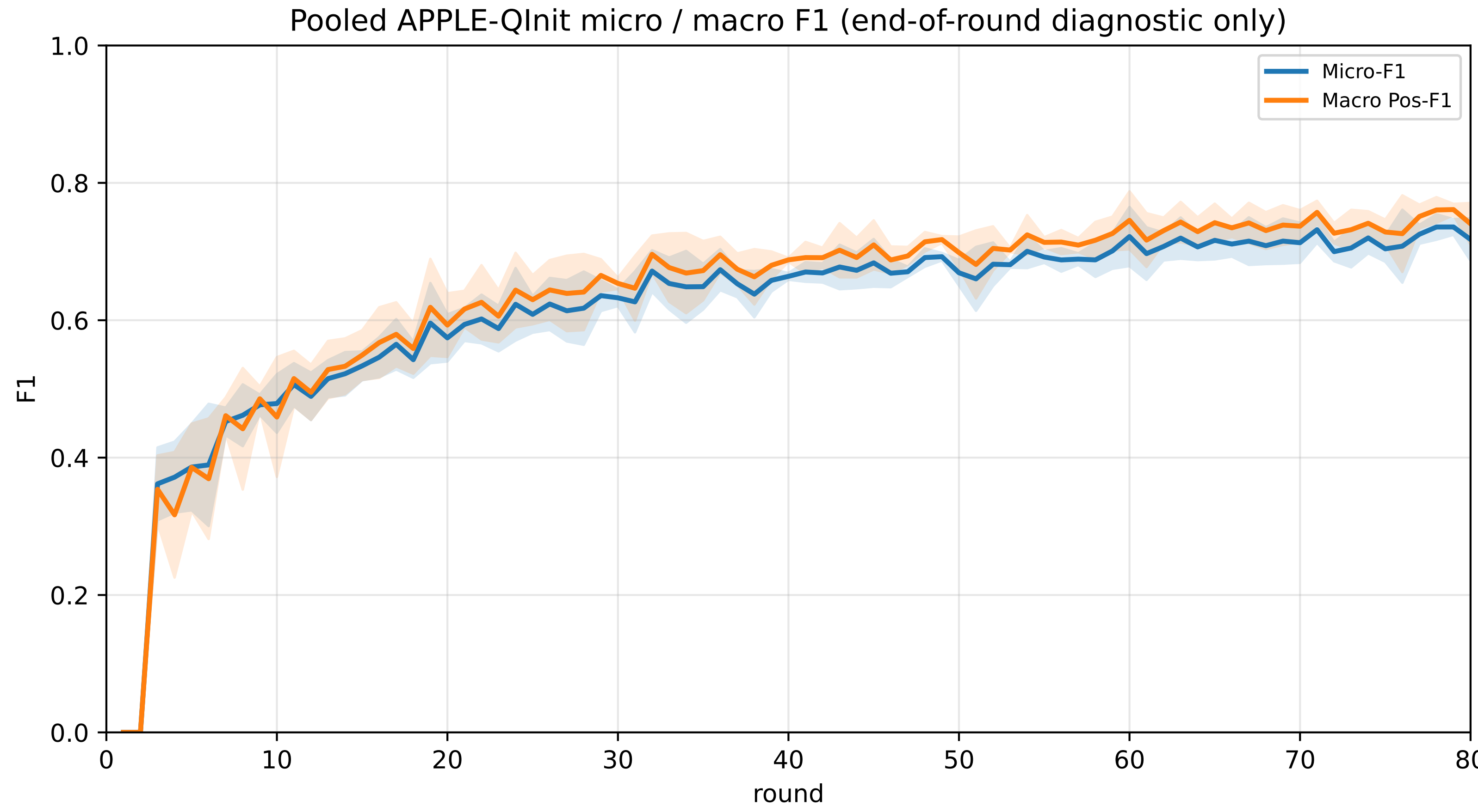
Pooled baseline micro / macro F1



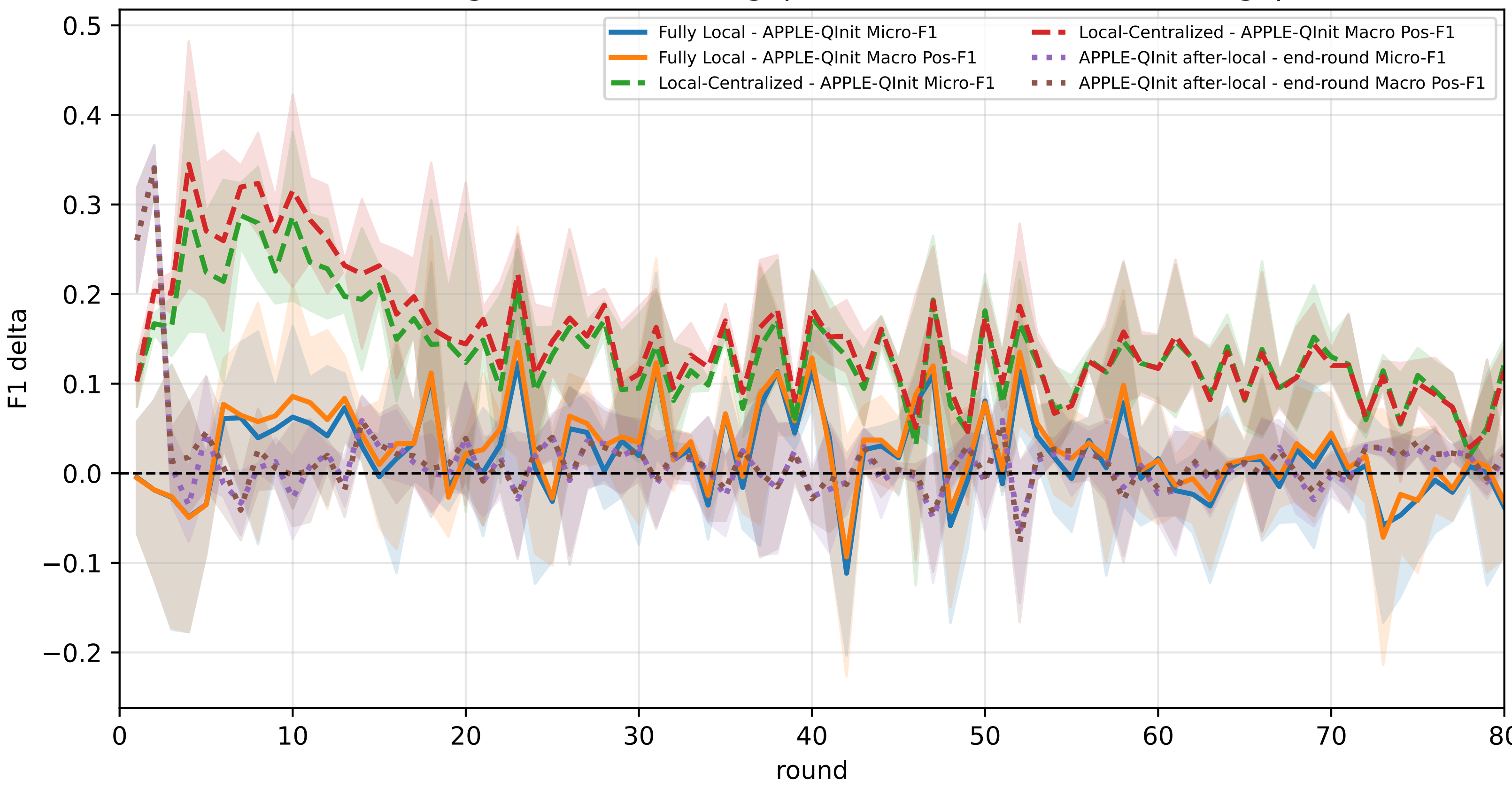
Pooled APPLE-QInit micro / macro F1 (after-local selected model)



Pooled APPLE-QInit micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QInit mismatch gap



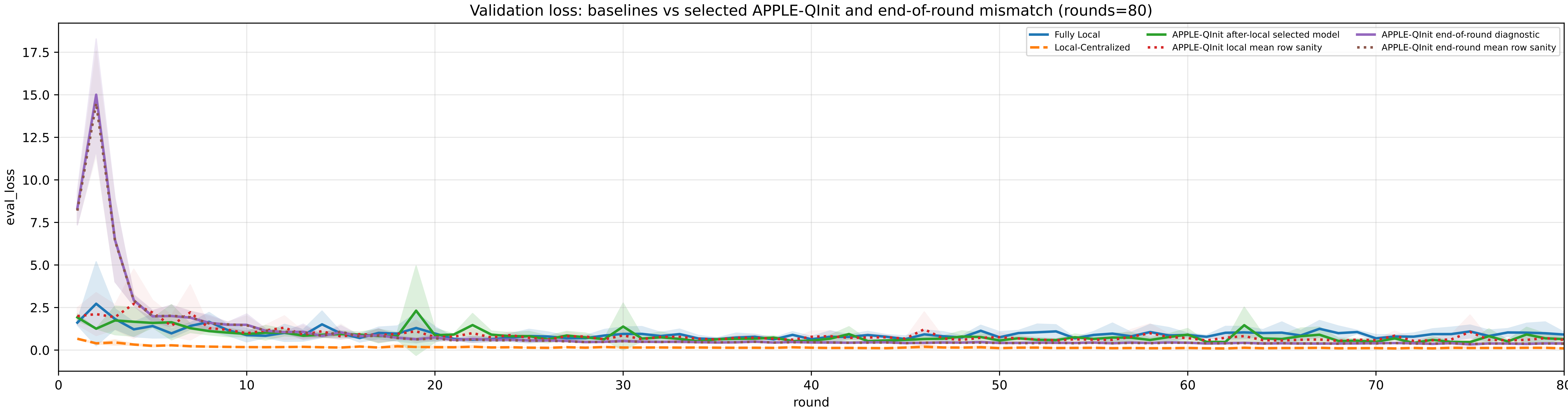
Controlled q-label heterogeneity: iid | APPLE-QInit diagnostics | family=q_task_cycle3 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

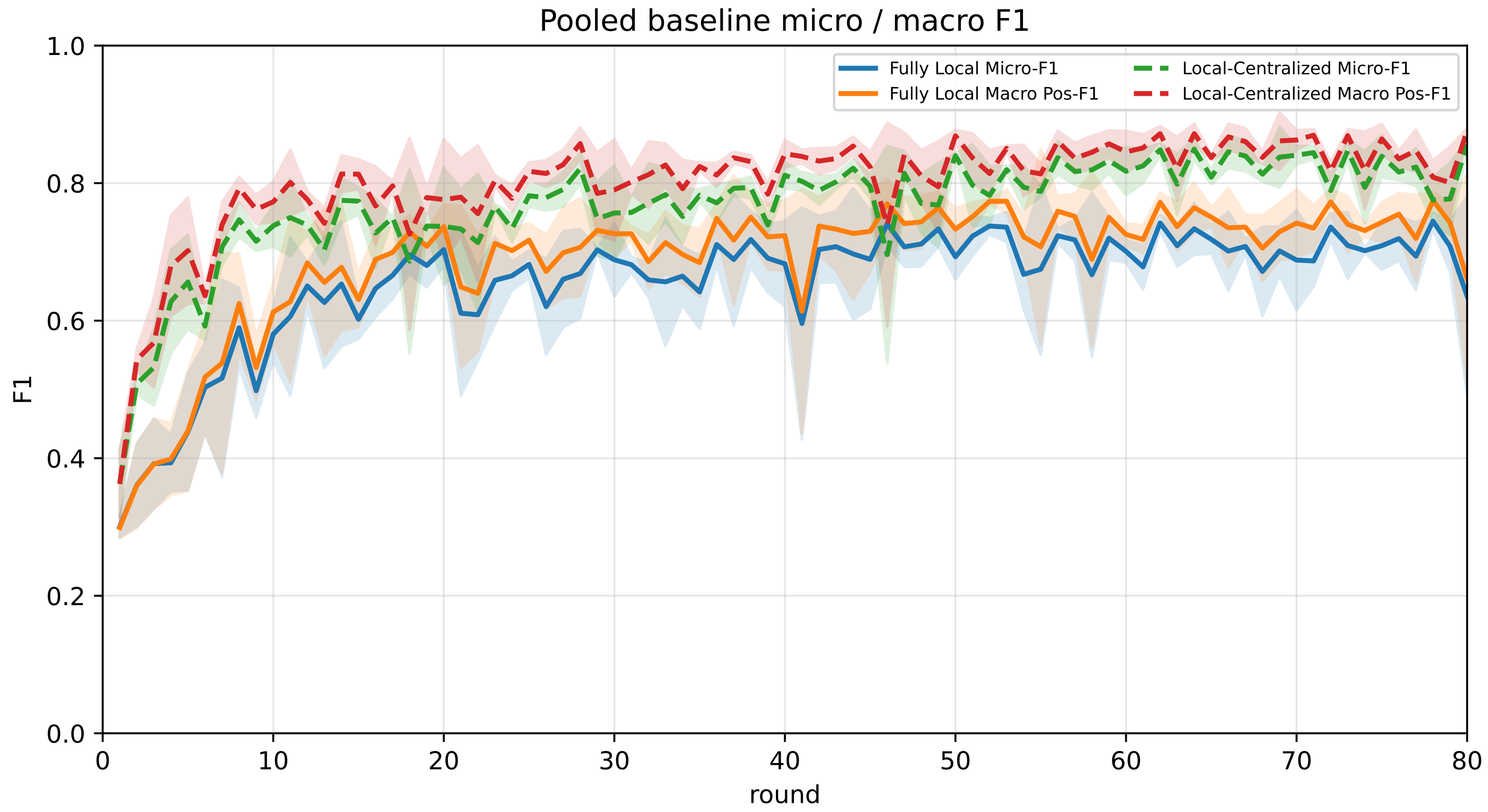
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	69.5% ± 0.8%	73.5% ± 0.6%	90.4% ± 0.9%	85.2% ± 1.6%	67.7% ± 0.6%	71.2% ± 1.1%	52.9% ± 2.1%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
APPLE-QInit	68.3% ± 5.0%	71.6% ± 3.9%	88.9% ± 1.5%	83.3% ± 1.2%	65.9% ± 9.0%	62.6% ± 8.4%	57.5% ± 2.3%

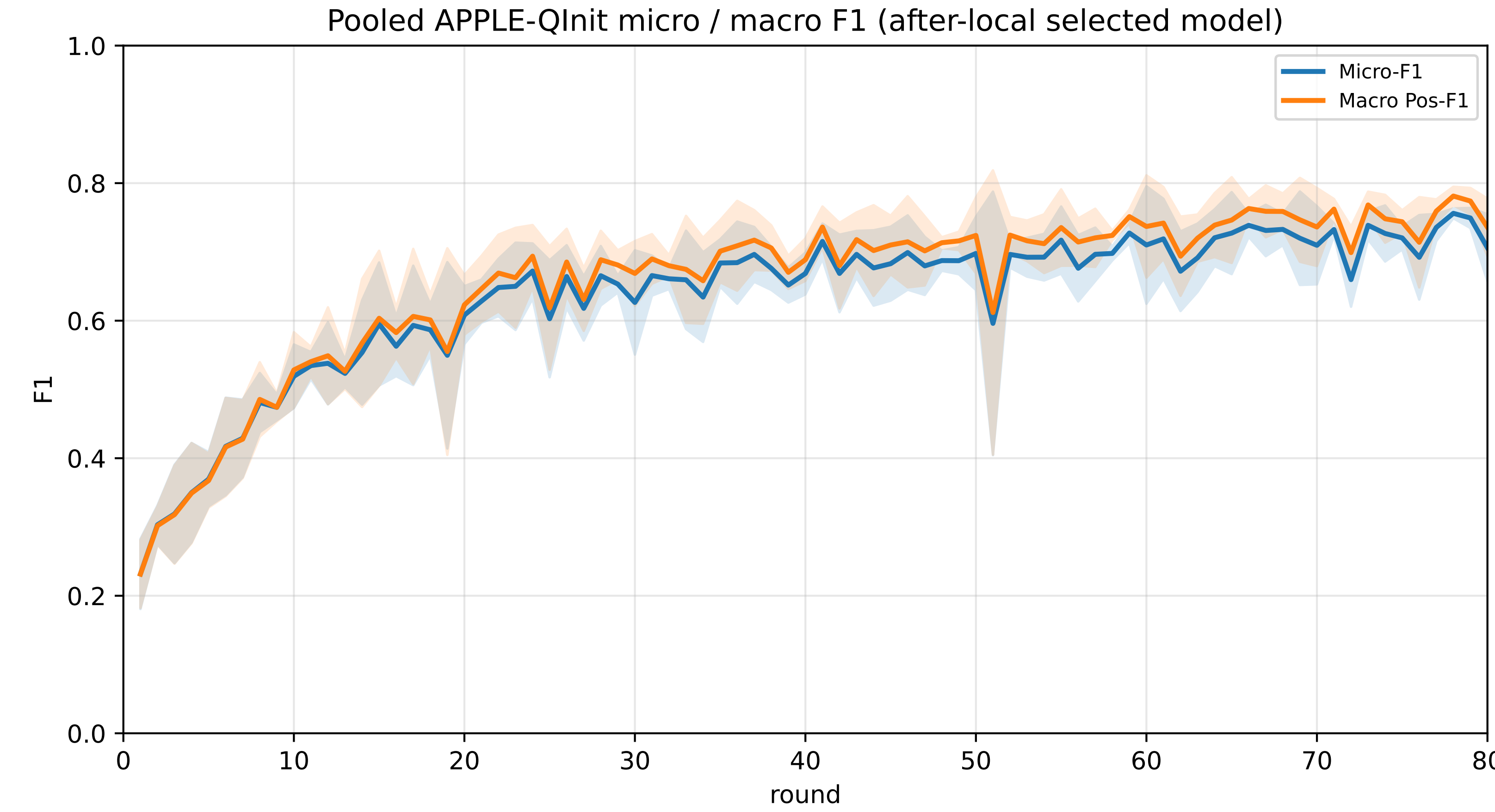
Validation loss: baselines vs selected APPLE-QInit and end-of-round mismatch (rounds=80)



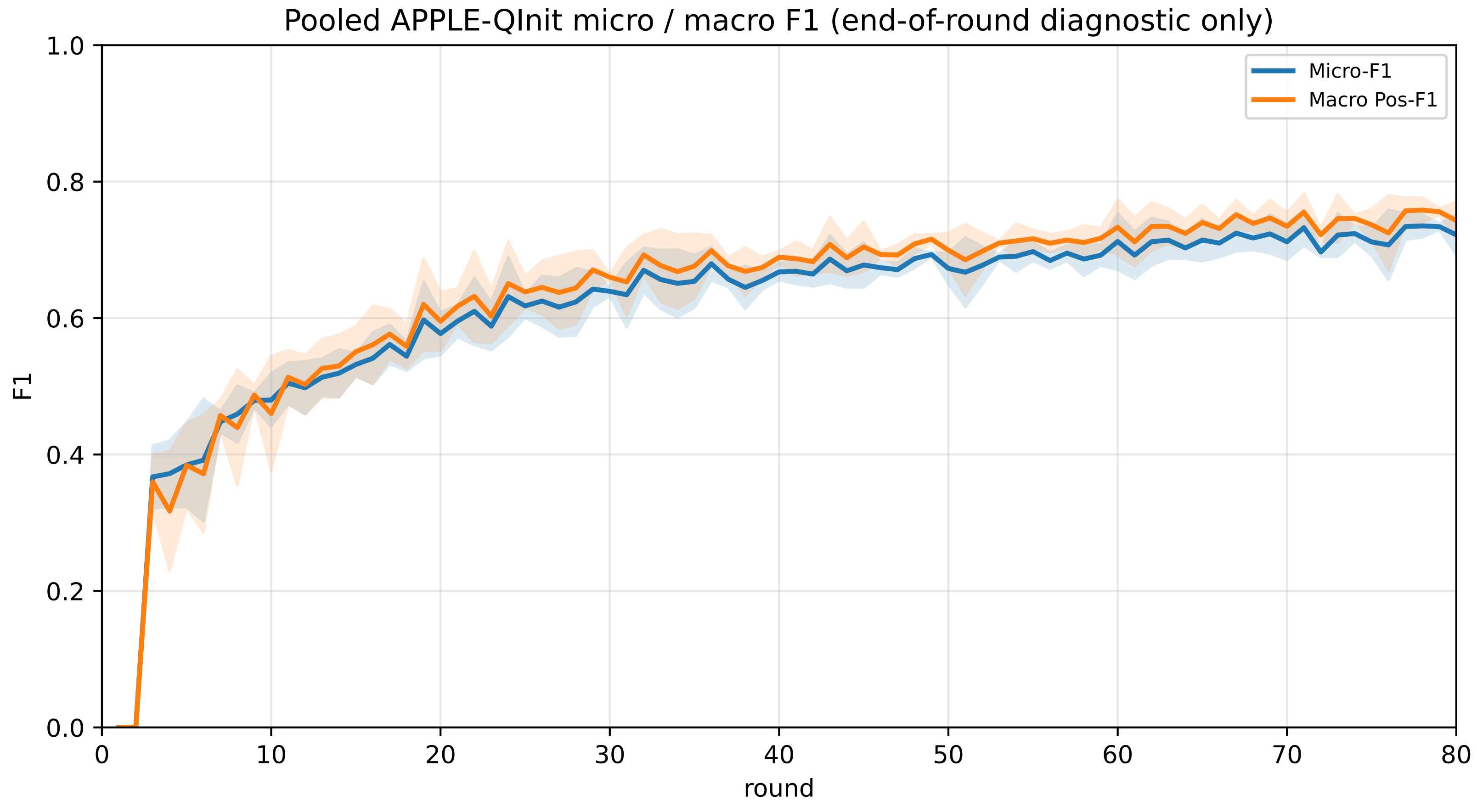
Pooled baseline micro / macro F1



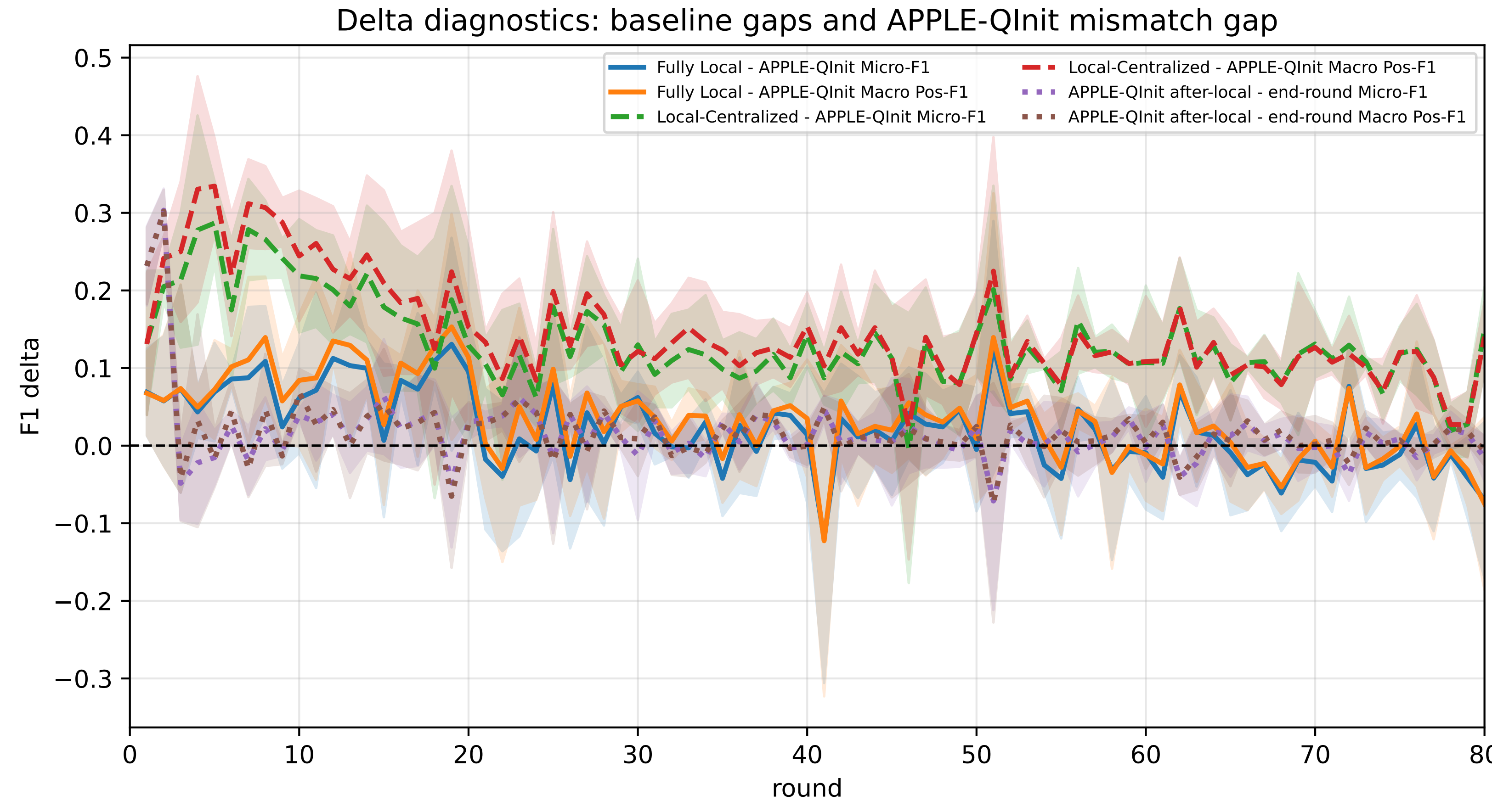
Pooled APPLE-QInit micro / macro F1 (after-local selected model)



Pooled APPLE-QInit micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QInit mismatch gap



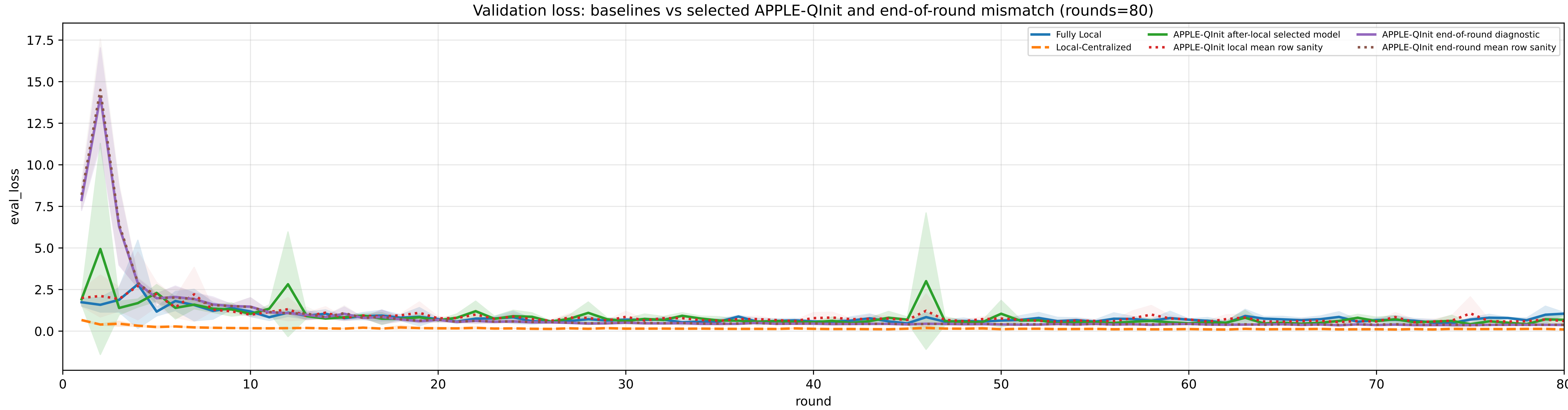
**Controlled q-label heterogeneity: iid | APPLE-QInit diagnostics | family=q_task_cycle4 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti**

family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

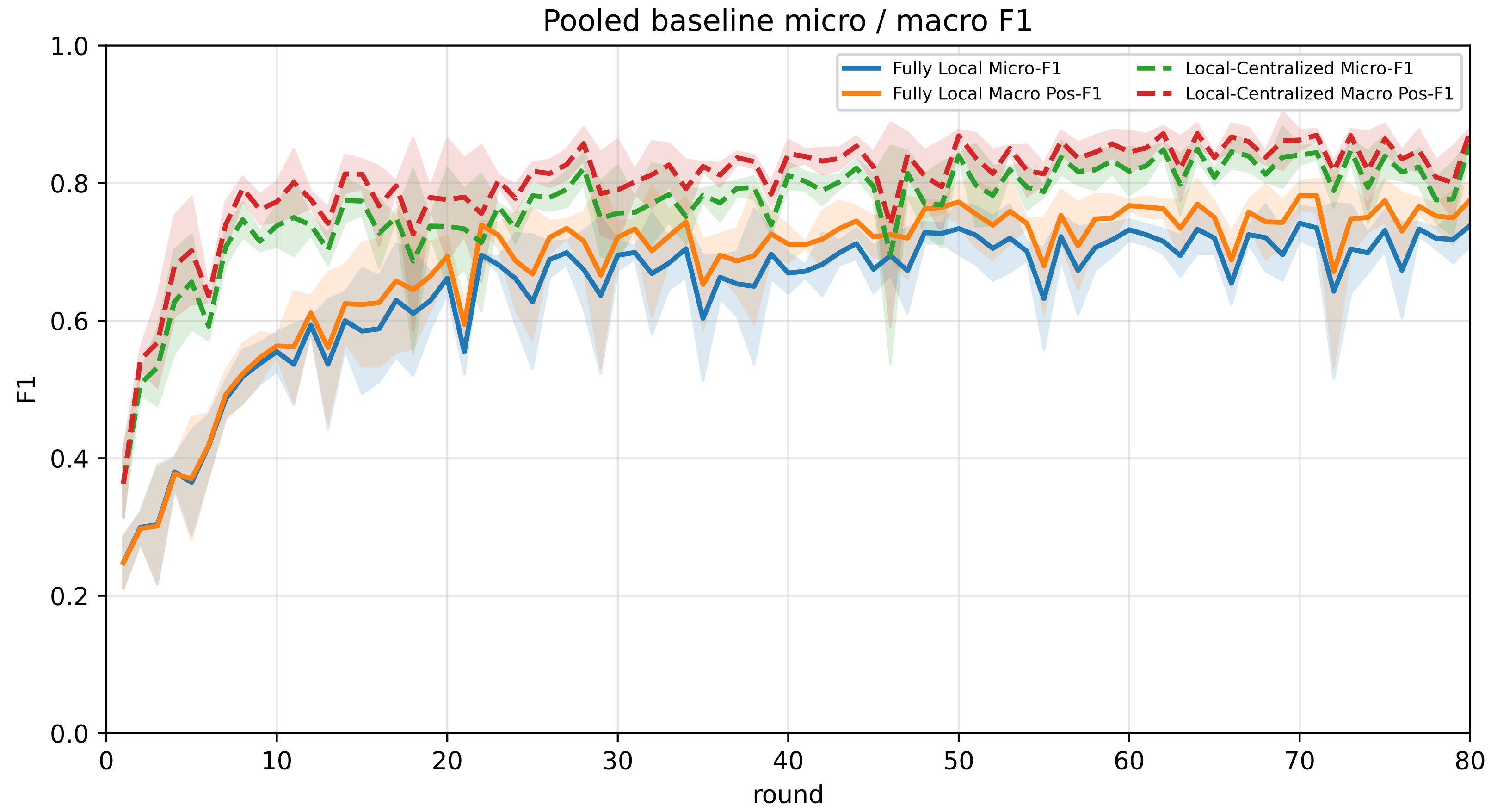
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

Algorithm	**Micro-F1**	**Macro Pos-F1**	**cycle2 pos-F1**	**cycle3 pos-F1**	**cycle4 pos-F1**	**cycle5 pos-F1**	**cycle6 pos-F1**
Fully Local	64.8% ± 3.3%	70.1% ± 2.7%	92.2% ± 3.4%	82.7% ± 3.5%	65.7% ± 8.3%	60.9% ± 7.0%	48.9% ± 2.7%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
APPLE-QInit	69.0% ± 3.0%	72.6% ± 3.0%	87.8% ± 5.3%	86.0% ± 3.2%	72.2% ± 3.0%	61.1% ± 7.0%	55.8% ± 1.1%

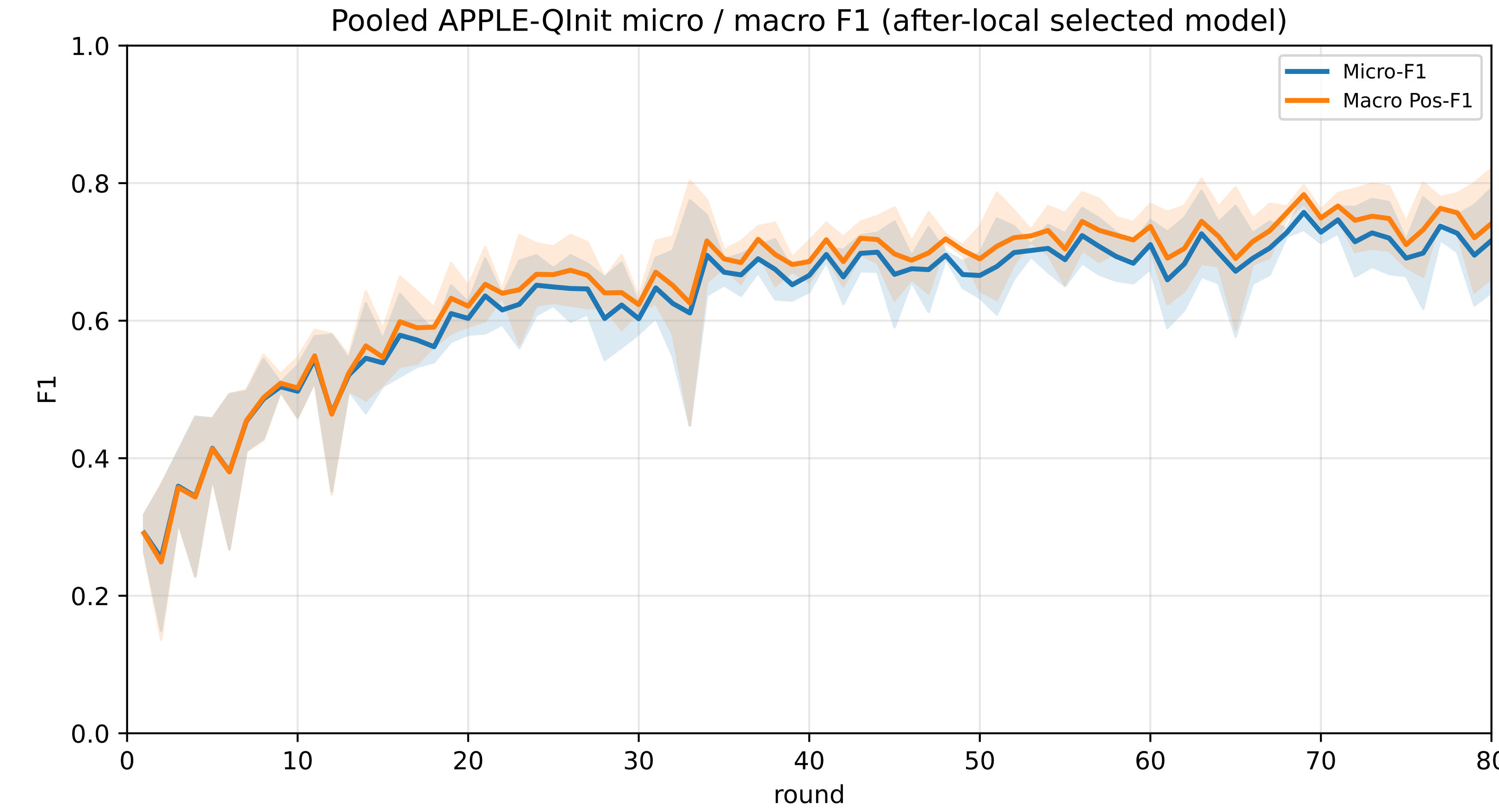
Validation loss: baselines vs selected APPLE-QInit and end-of-round mismatch (rounds=80)



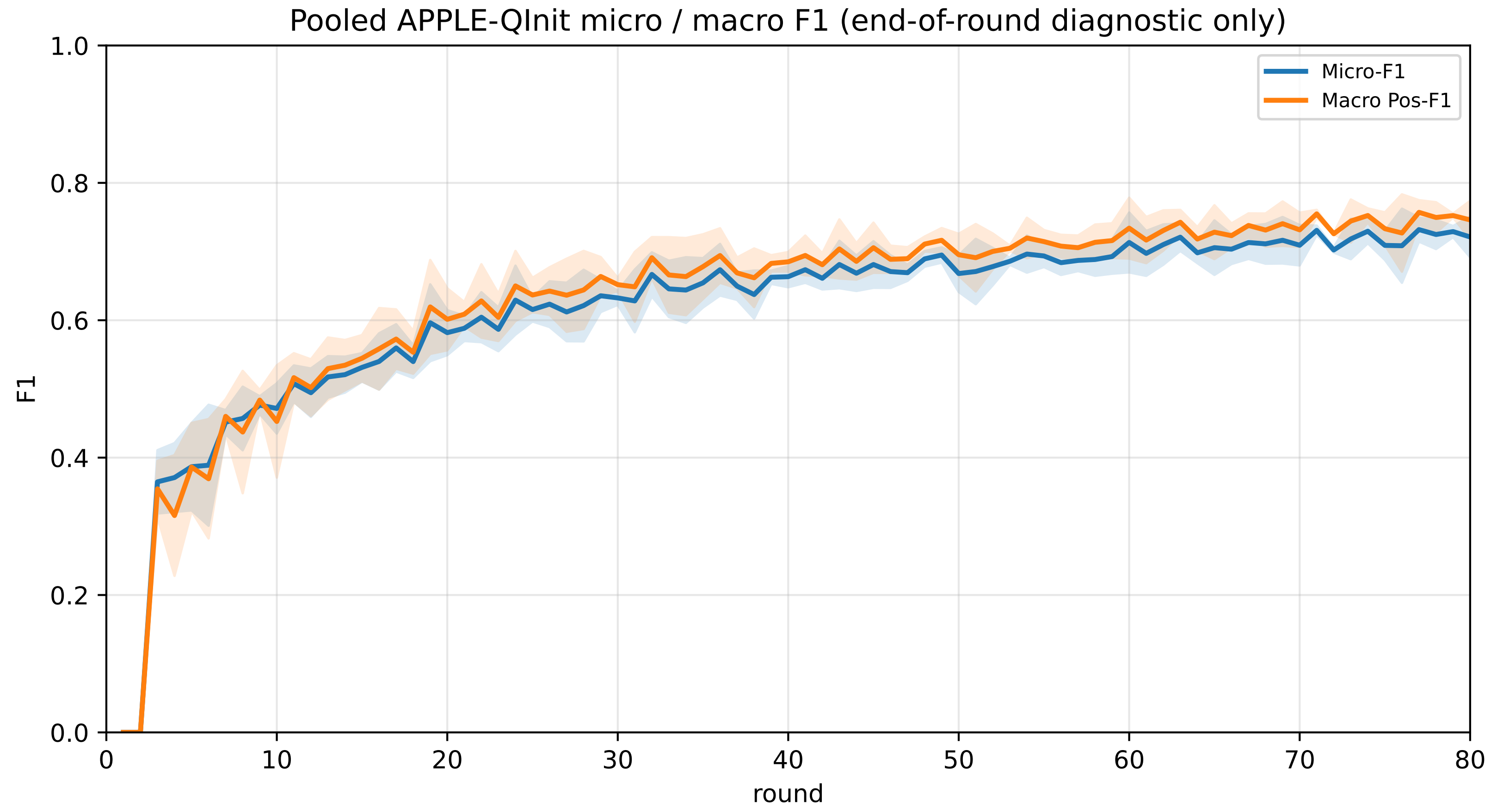
Pooled baseline micro / macro F1



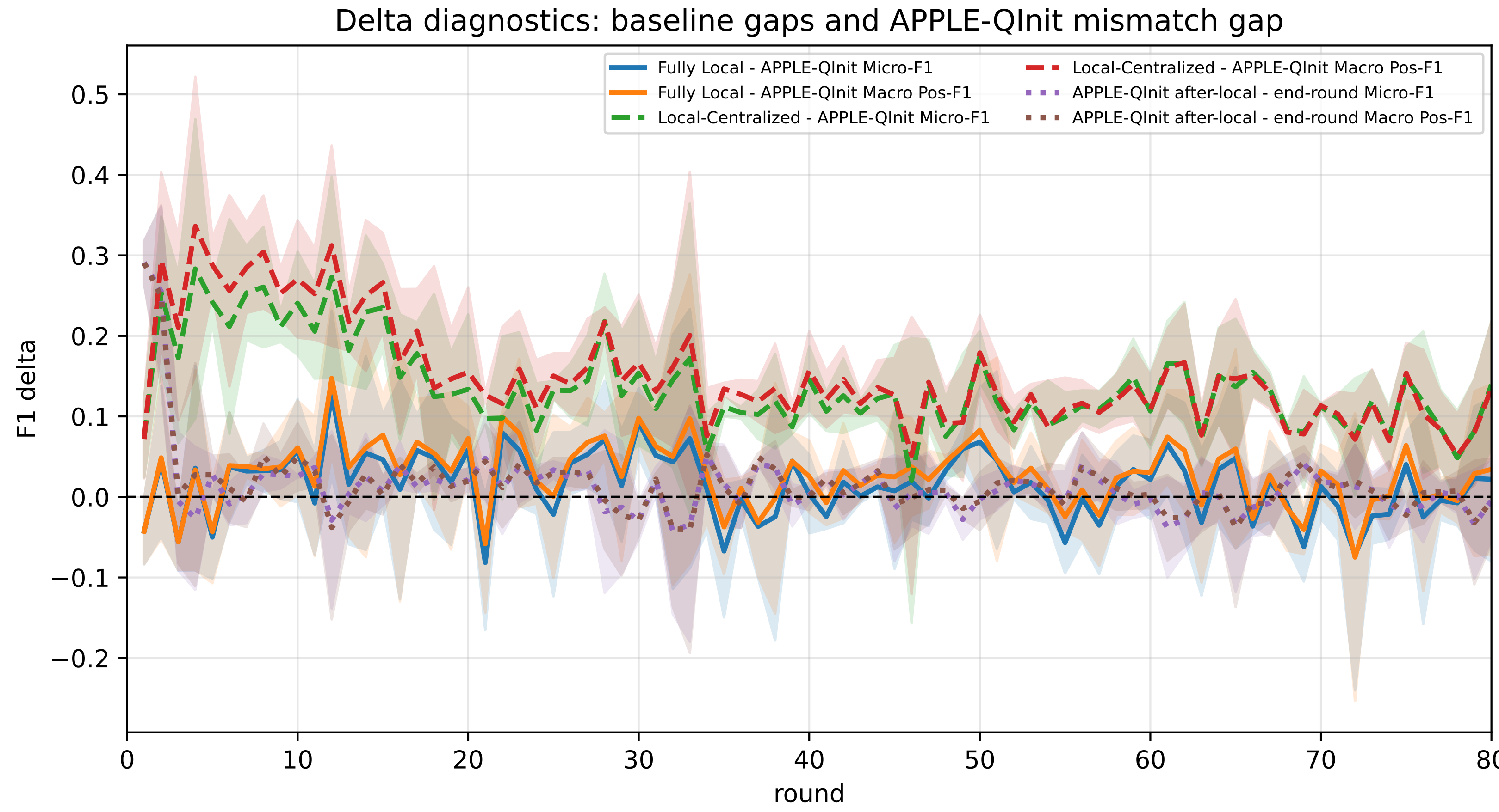
Pooled APPLE-QInit micro / macro F1 (after-local selected model)



Pooled APPLE-QInit micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QInit mismatch gap

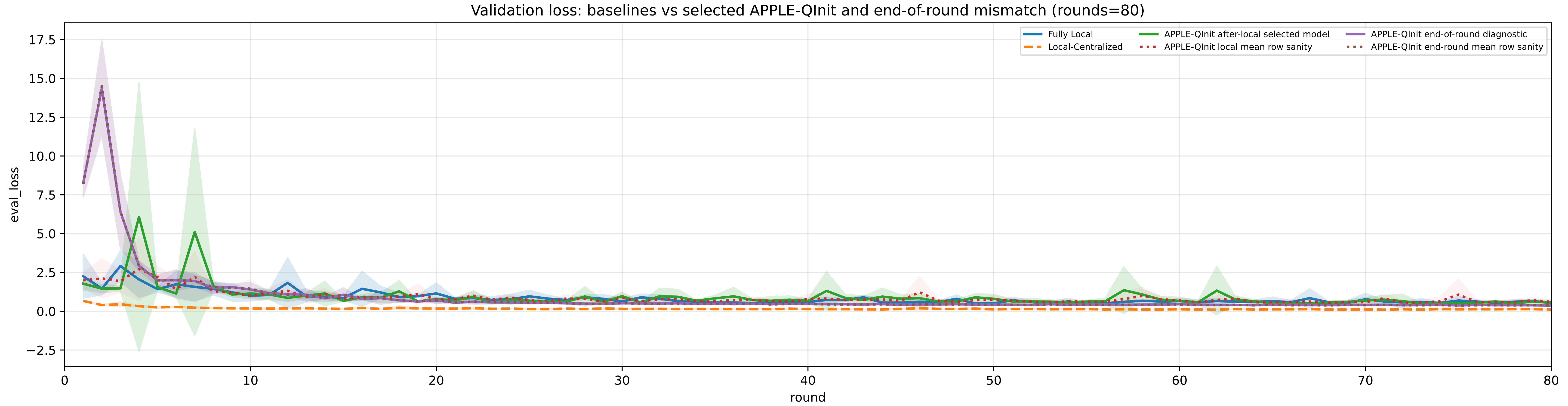


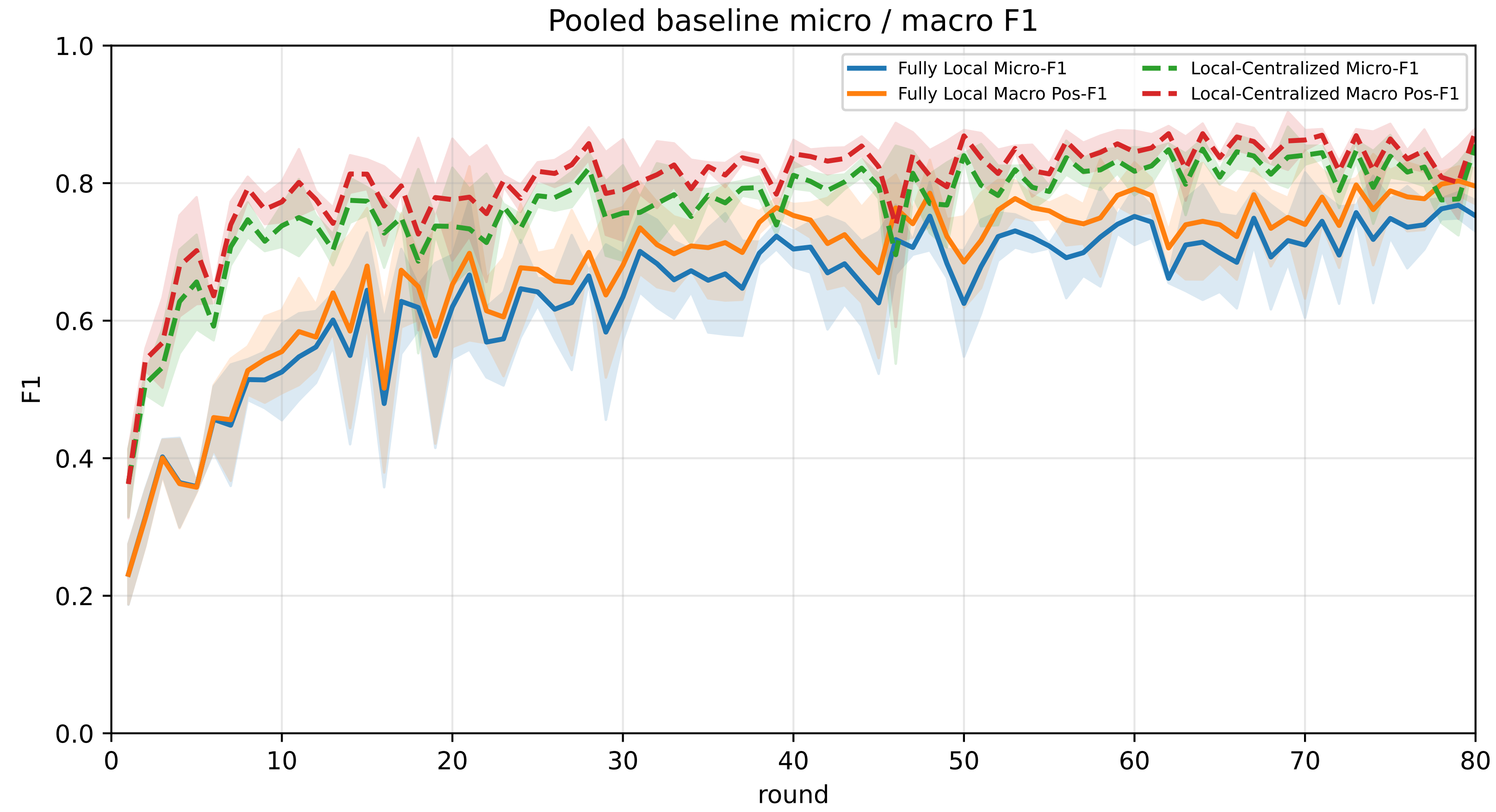
Controlled q-label heterogeneity: iid | APPLE-QInit diagnostics | family=q_task_cycle5 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

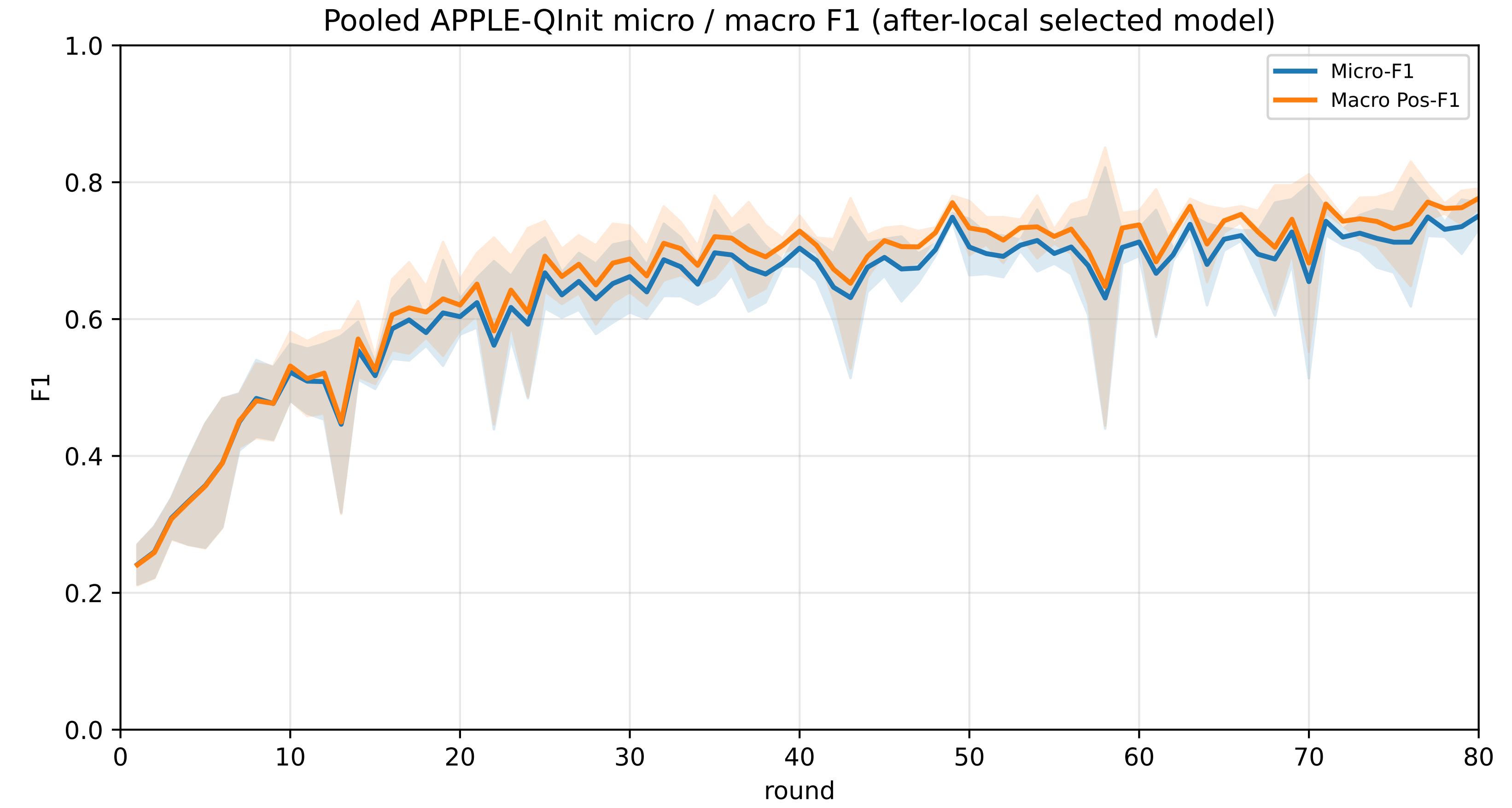
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

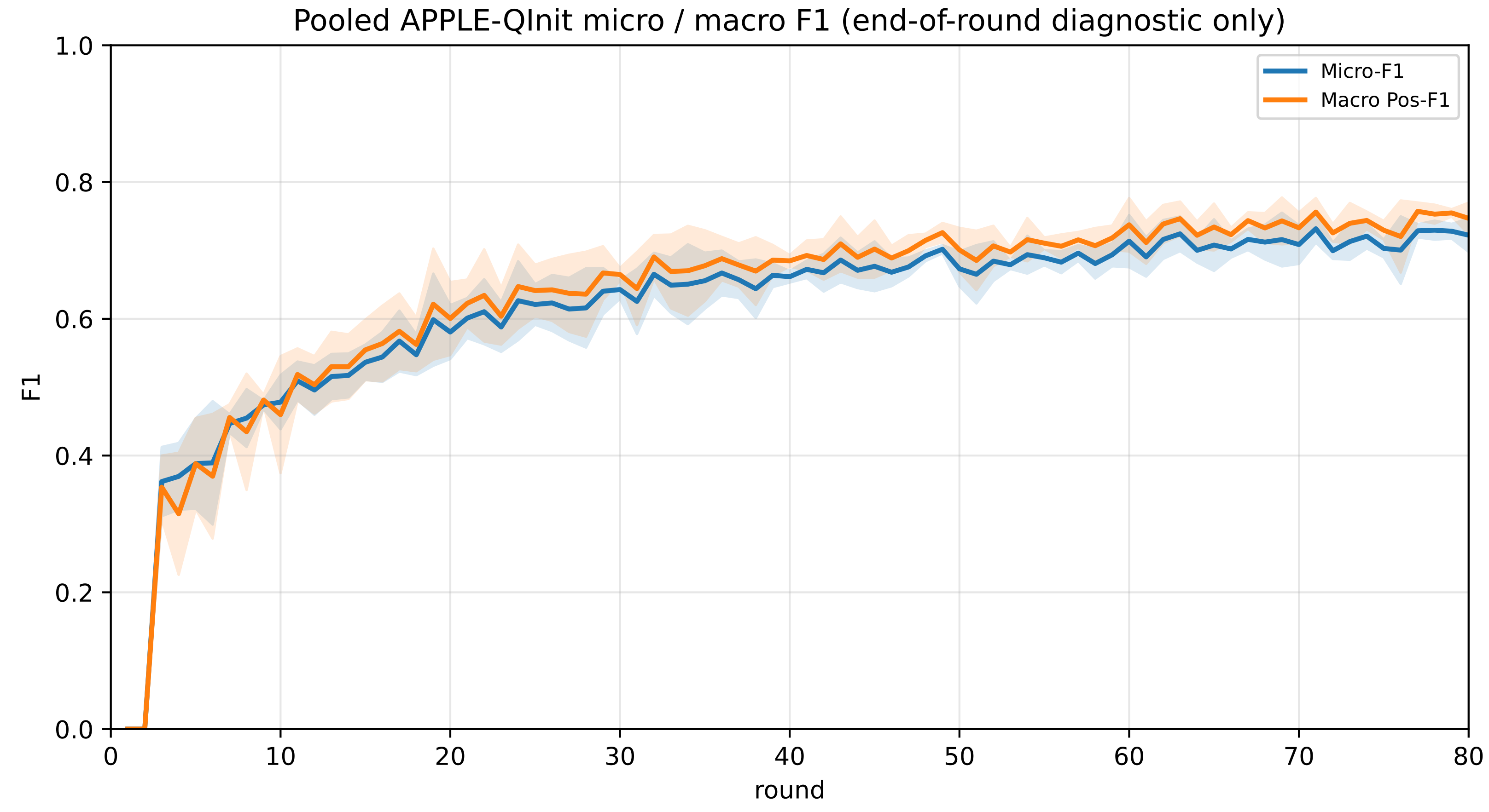
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

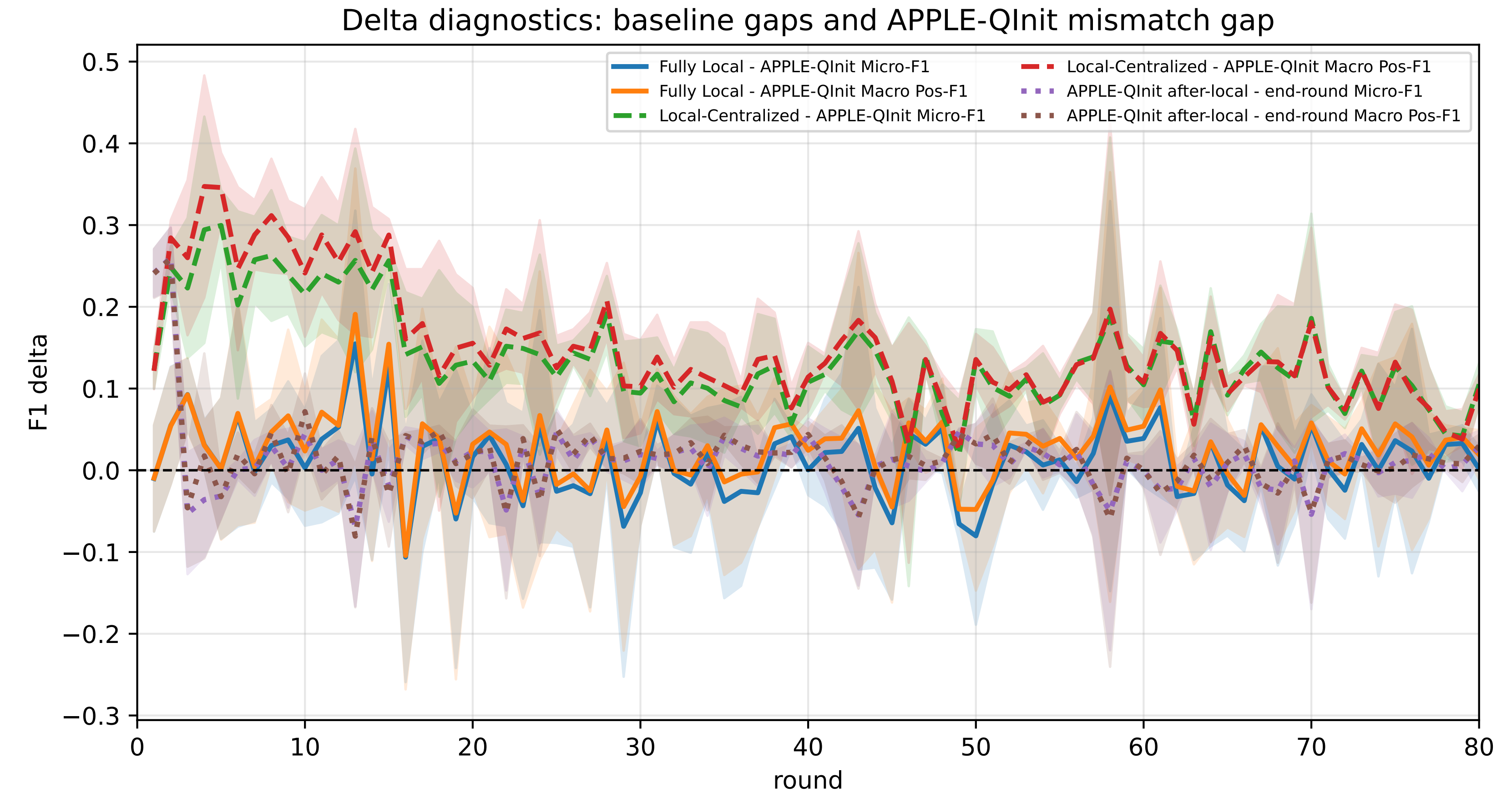
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	70.5% ± 1.9%	74.2% ± 2.2%	87.8% ± 6.8%	86.4% ± 2.4%	74.0% ± 2.0%	69.7% ± 1.8%	53.3% ± 1.7%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
APPLE-QInit	72.6% ± 1.5%	75.1% ± 1.5%	89.3% ± 1.3%	83.5% ± 4.4%	73.4% ± 2.4%	69.0% ± 1.8%	60.2% ± 2.4%











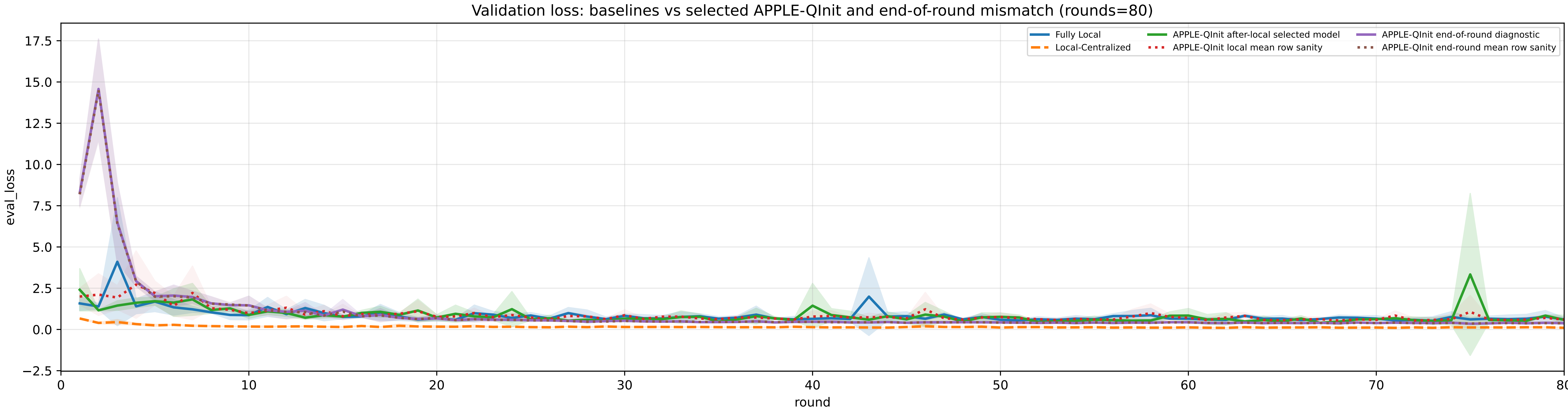
Controlled q-label heterogeneity: iid | APPLE-QInit diagnostics | family=q_task_cycle6 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

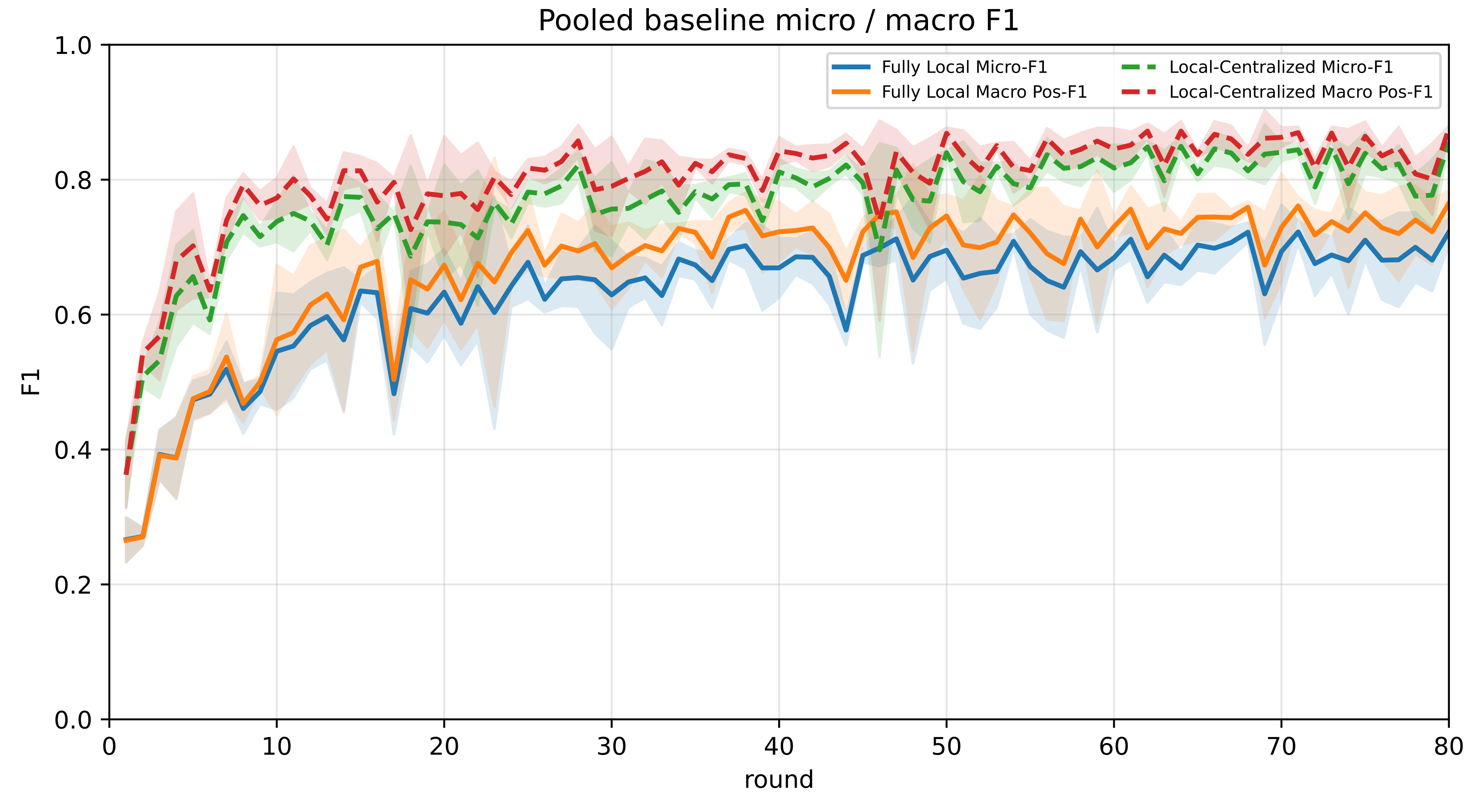
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.5% ± 3.5%	69.6% ± 3.3%	87.3% ± 2.7%	84.4% ± 3.4%	62.5% ± 7.2%	67.0% ± 5.0%	46.8% ± 1.7%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
APPLE-QInit	68.5% ± 5.5%	71.4% ± 4.8%	88.8% ± 1.5%	78.0% ± 8.3%	69.2% ± 4.7%	65.5% ± 6.4%	55.7% ± 5.6%

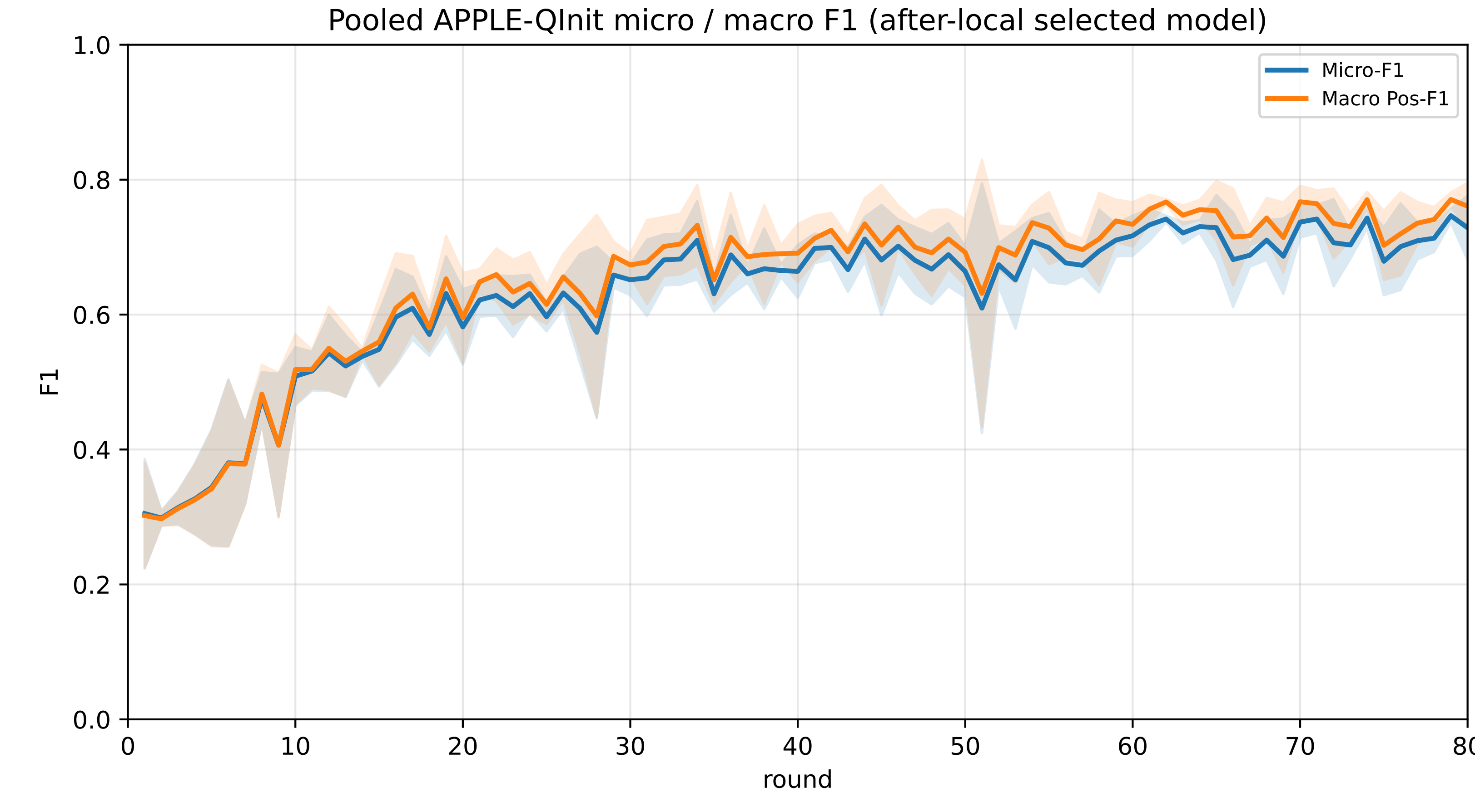
Validation loss: baselines vs selected APPLE-QInit and end-of-round mismatch (rounds=80)



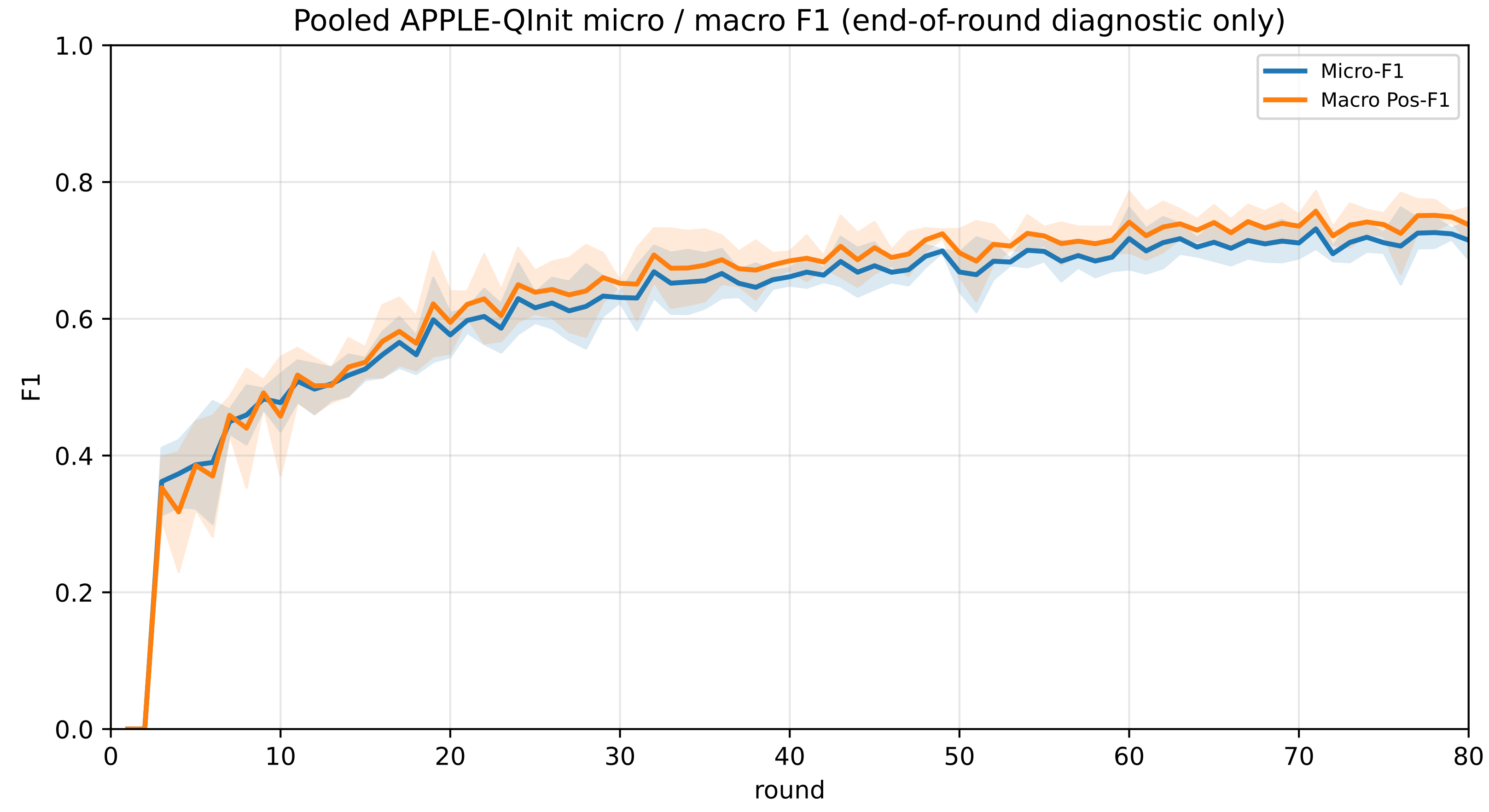
Pooled baseline micro / macro F1



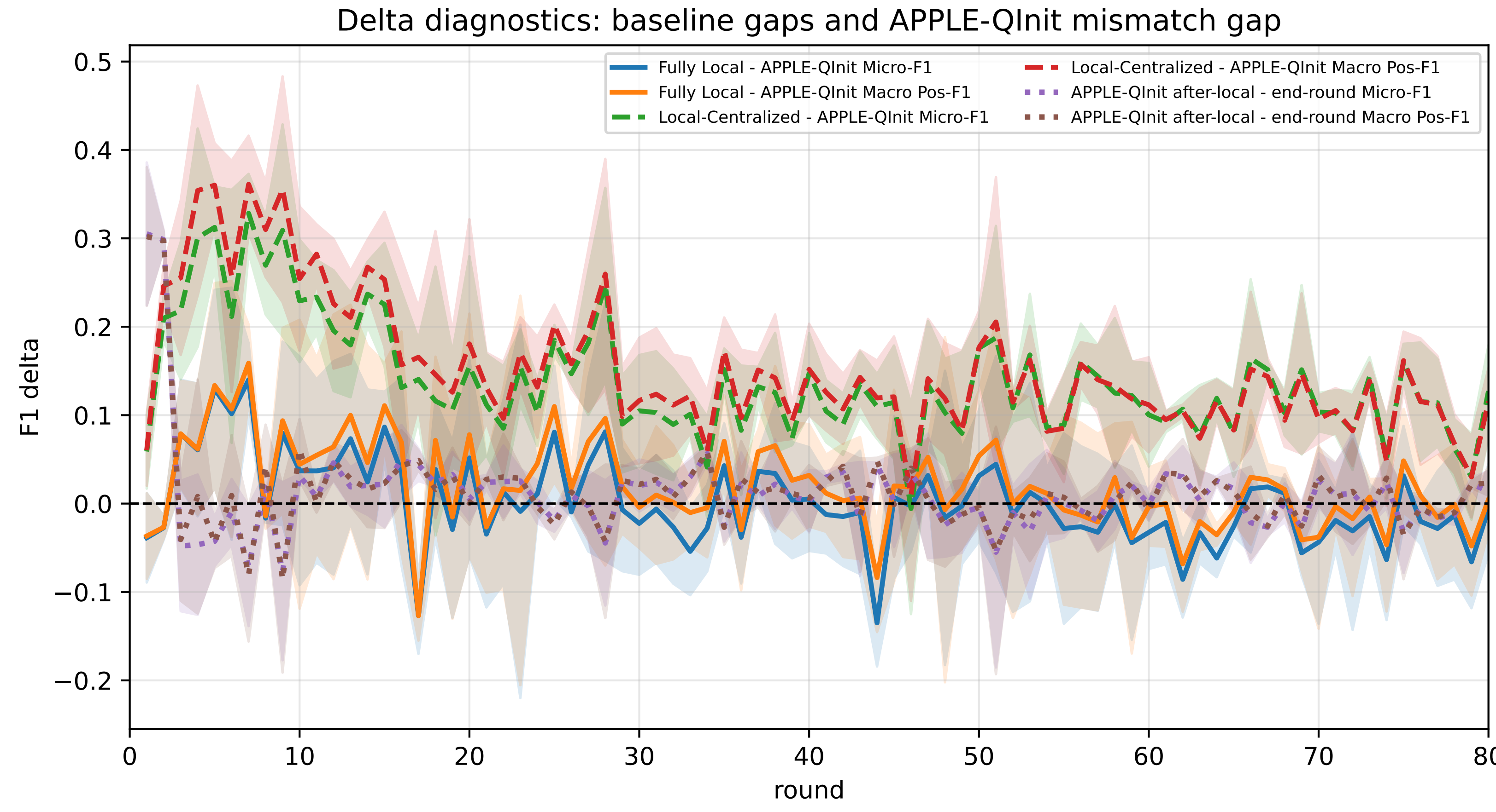
Pooled APPLE-QInit micro / macro F1 (after-local selected model)



Pooled APPLE-QInit micro / macro F1 (end-of-round diagnostic only)

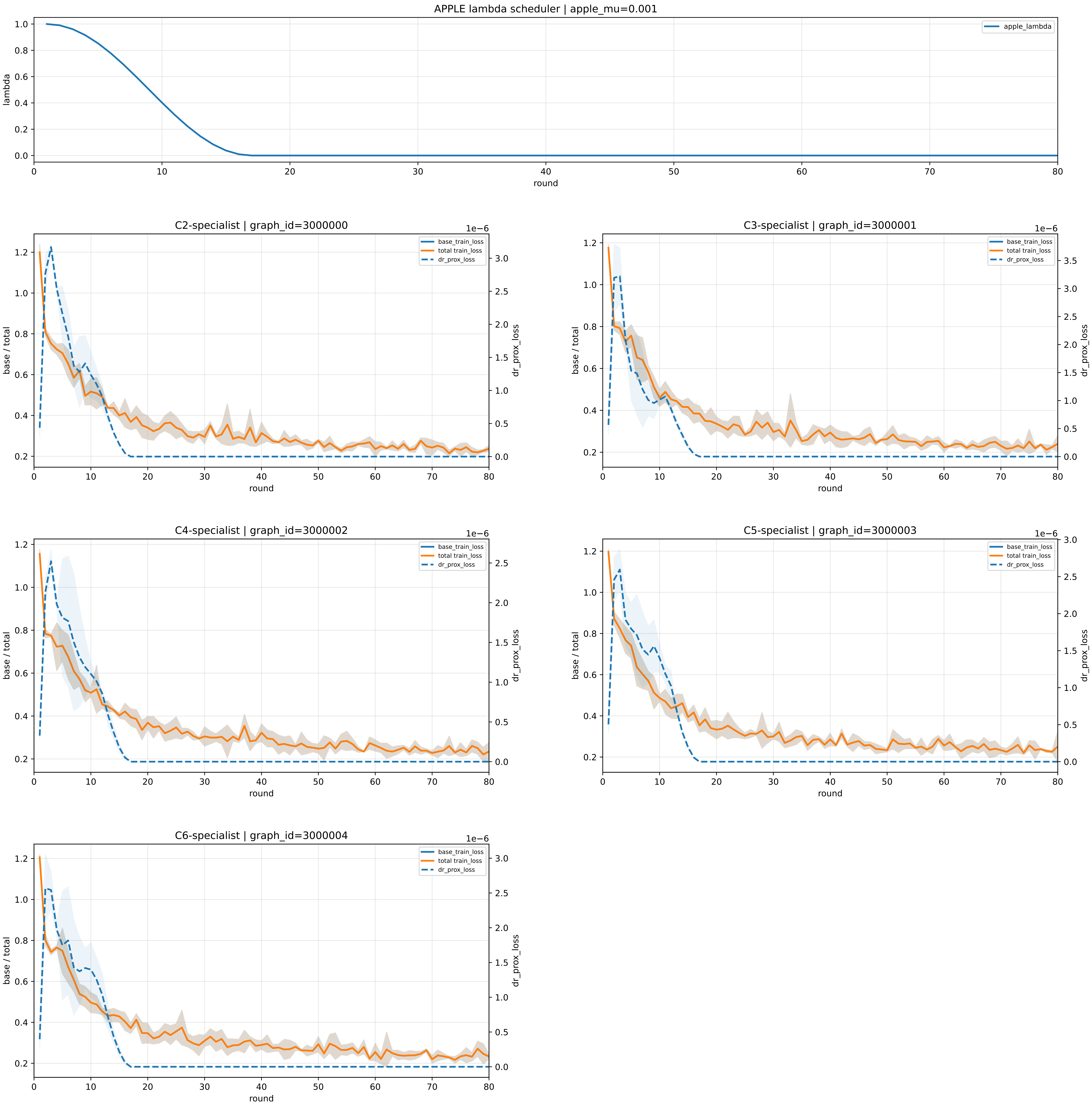


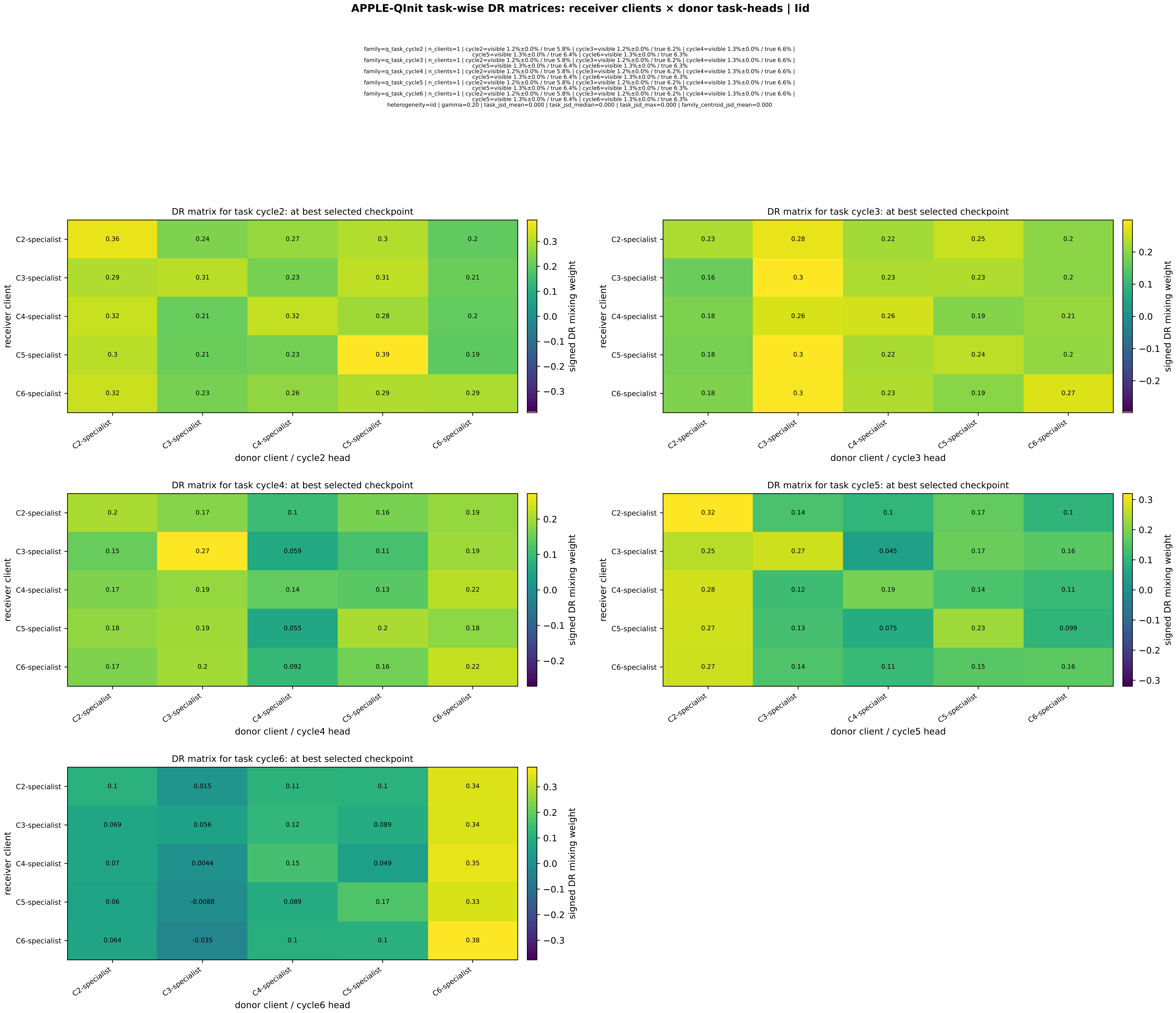
Delta diagnostics: baseline gaps and APPLE-QInit mismatch gap



APPLE-QInit training-loss decomposition | Iid heterogeneity

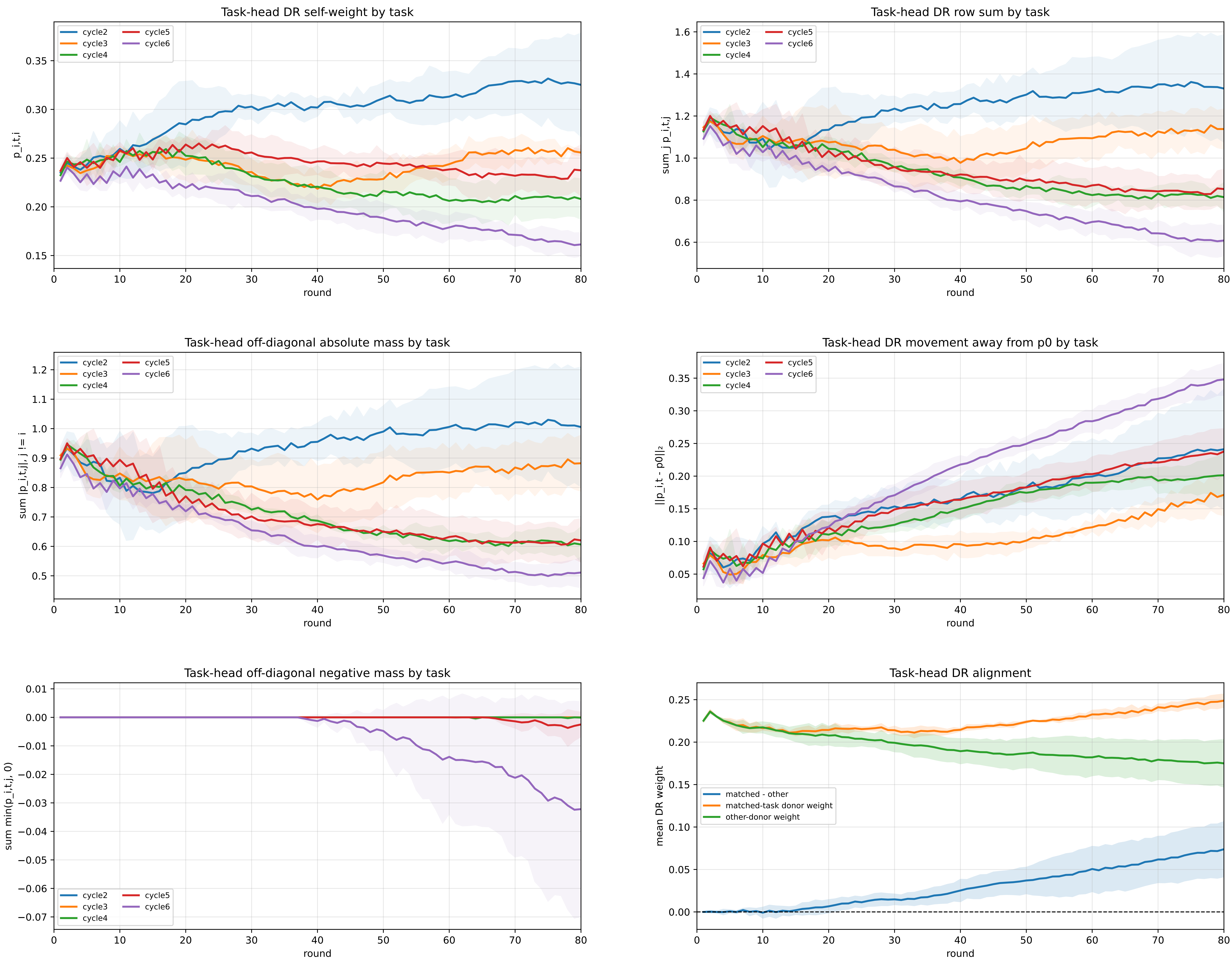
family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000




APPLE-QInit task-wise DR matrices: receiver clients × donor task-heads | Iid
family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000
DR matrix for task cycle2: at best selected checkpoint
receiver client
C2-specialist 0.36 0.24 0.27 0.3 0.2
C3-specialist 0.29 0.31 0.23 0.31 0.21
C4-specialist 0.32 0.21 0.32 0.28 0.2
C5-specialist 0.3 0.21 0.23 0.39 0.19
C6-specialist 0.32 0.23 0.26 0.29 0.29
C2-specialist C3-specialist C4-specialist C5-specialist C6-specialist
donor client / cycle2 head
signed DR mixing weight
DR matrix for task cycle3: at best selected checkpoint
receiver client
C2-specialist 0.23 0.28 0.22 0.25 0.2
C3-specialist 0.16 0.3 0.23 0.23 0.2
C4-specialist 0.18 0.26 0.26 0.19 0.21
C5-specialist 0.18 0.3 0.22 0.24 0.2
C6-specialist 0.18 0.3 0.23 0.19 0.27
C2-specialist C3-specialist C4-specialist C5-specialist C6-specialist
donor client / cycle3 head
signed DR mixing weight
DR matrix for task cycle4: at best selected checkpoint
receiver client
C2-specialist 0.2 0.17 0.1 0.16 0.19
C3-specialist 0.15 0.27 0.059 0.11 0.19
C4-specialist 0.17 0.19 0.14 0.13 0.22
C5-specialist 0.18 0.19 0.055 0.2 0.18
C6-specialist 0.17 0.2 0.092 0.16 0.22
C2-specialist C3-specialist C4-specialist C5-specialist C6-specialist
donor client / cycle4 head
signed DR mixing weight
DR matrix for task cycle5: at best selected checkpoint
receiver client
C2-specialist 0.32 0.14 0.1 0.17 0.1
C3-specialist 0.25 0.27 0.045 0.17 0.16
C4-specialist 0.28 0.12 0.19 0.14 0.11
C5-specialist 0.27 0.13 0.075 0.23 0.099
C6-specialist 0.27 0.14 0.11 0.15 0.16
C2-specialist C3-specialist C4-specialist C5-specialist C6-specialist
donor client / cycle5 head
signed DR mixing weight
DR matrix for task cycle6: at best selected checkpoint
receiver client
C2-specialist 0.1 0.015 0.11 0.1 0.34
C3-specialist 0.069 0.056 0.12 0.089 0.34
C4-specialist 0.07 0.0044 0.15 0.049 0.35
C5-specialist 0.06 -0.0088 0.089 0.17 0.33
C6-specialist 0.064 -0.035 0.1 0.1 0.38
C2-specialist C3-specialist C4-specialist C5-specialist C6-specialist
donor client / cycle6 head
signed DR mixing weight

APPLE-QInit task-wise DR dynamics | Iid

family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000



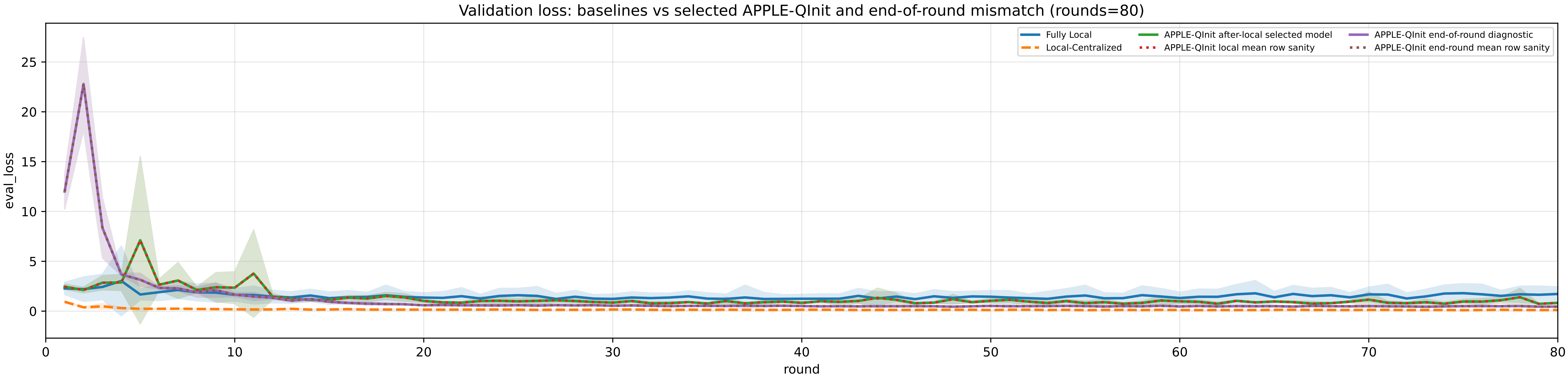
Controlled q-label heterogeneity: Mid | APPLE-QInit diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

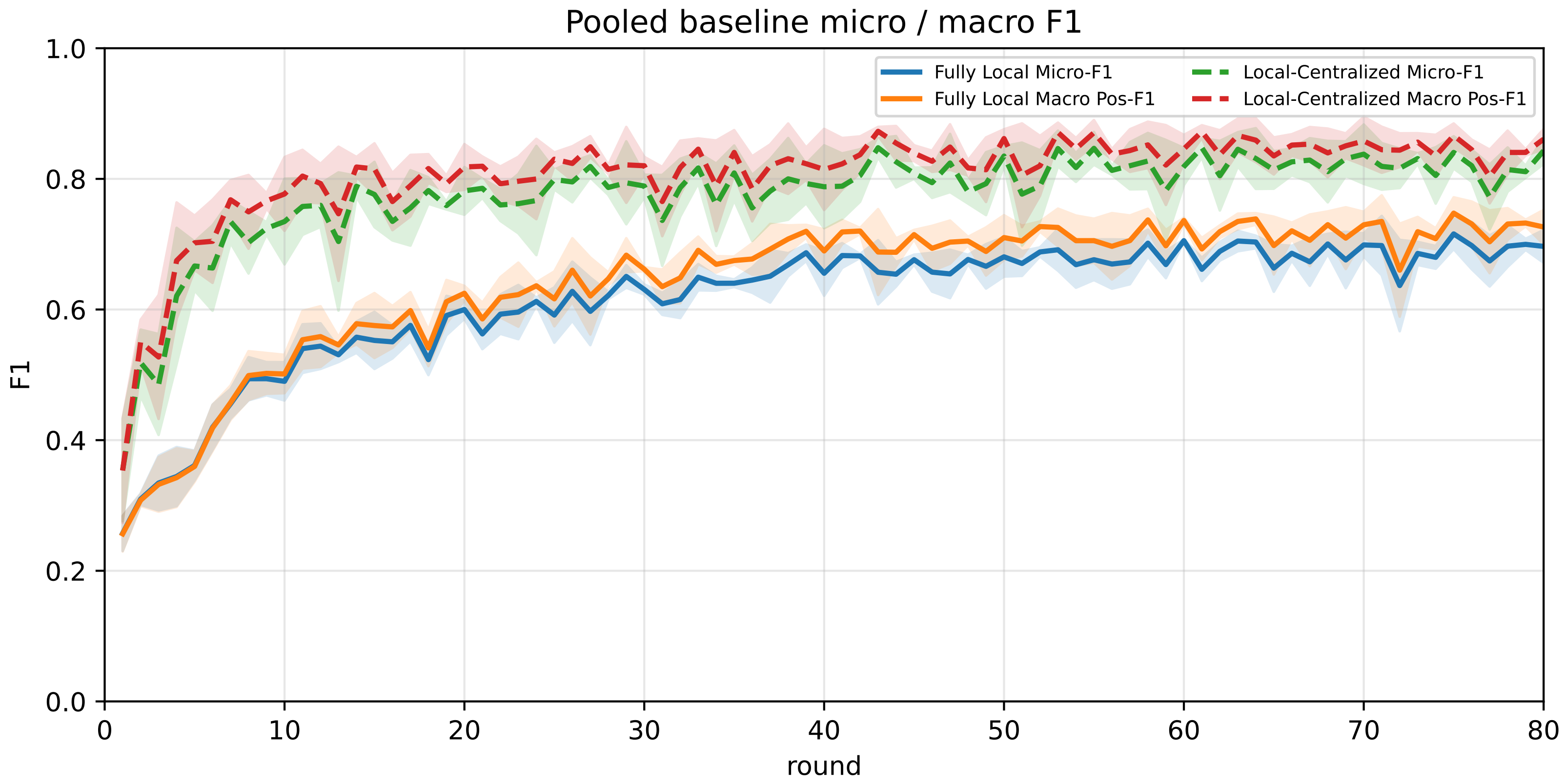
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.6% ± 4.0%	66.0% ± 4.3%	81.6% ± 7.0%	78.2% ± 5.1%	64.0% ± 3.6%	58.6% ± 5.2%	44.4% ± 3.0%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
APPLE-QInit	68.0% ± 4.2%	70.8% ± 4.4%	85.2% ± 5.4%	77.7% ± 7.8%	68.7% ± 3.8%	66.5% ± 1.9%	54.7% ± 3.9%

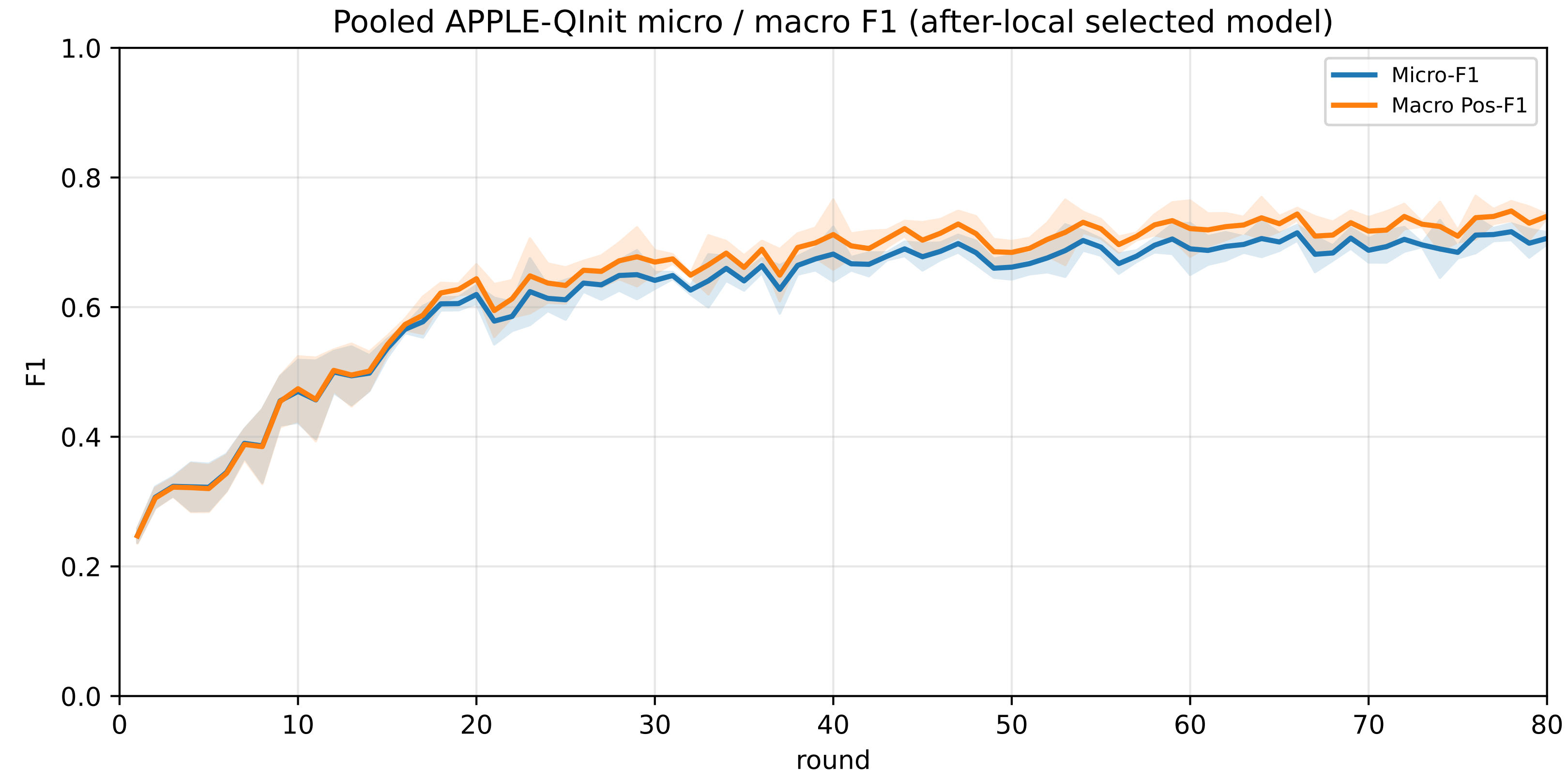
Validation loss: baselines vs selected APPLE-QInit and end-of-round mismatch (rounds=80)



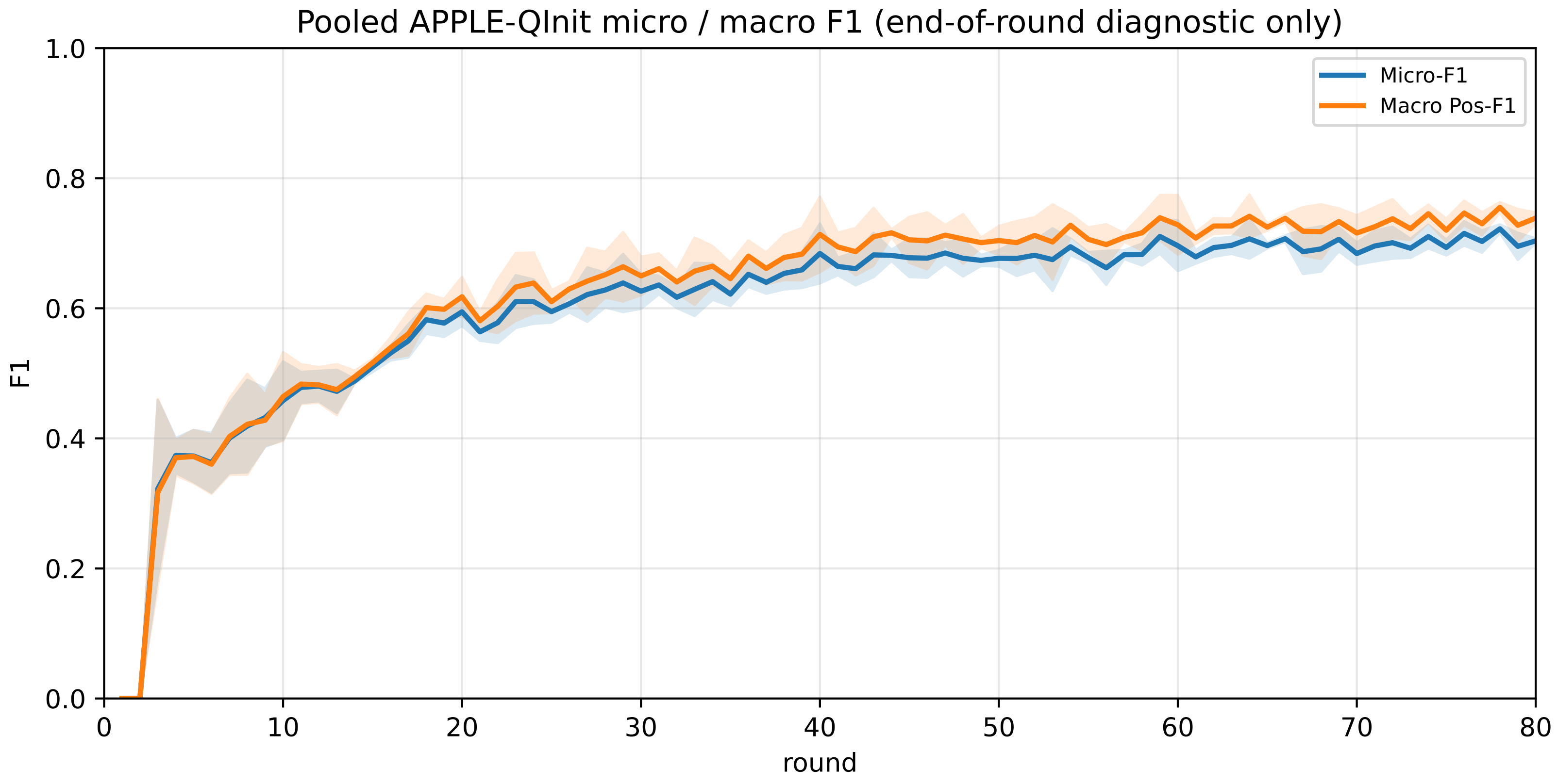
Pooled baseline micro / macro F1



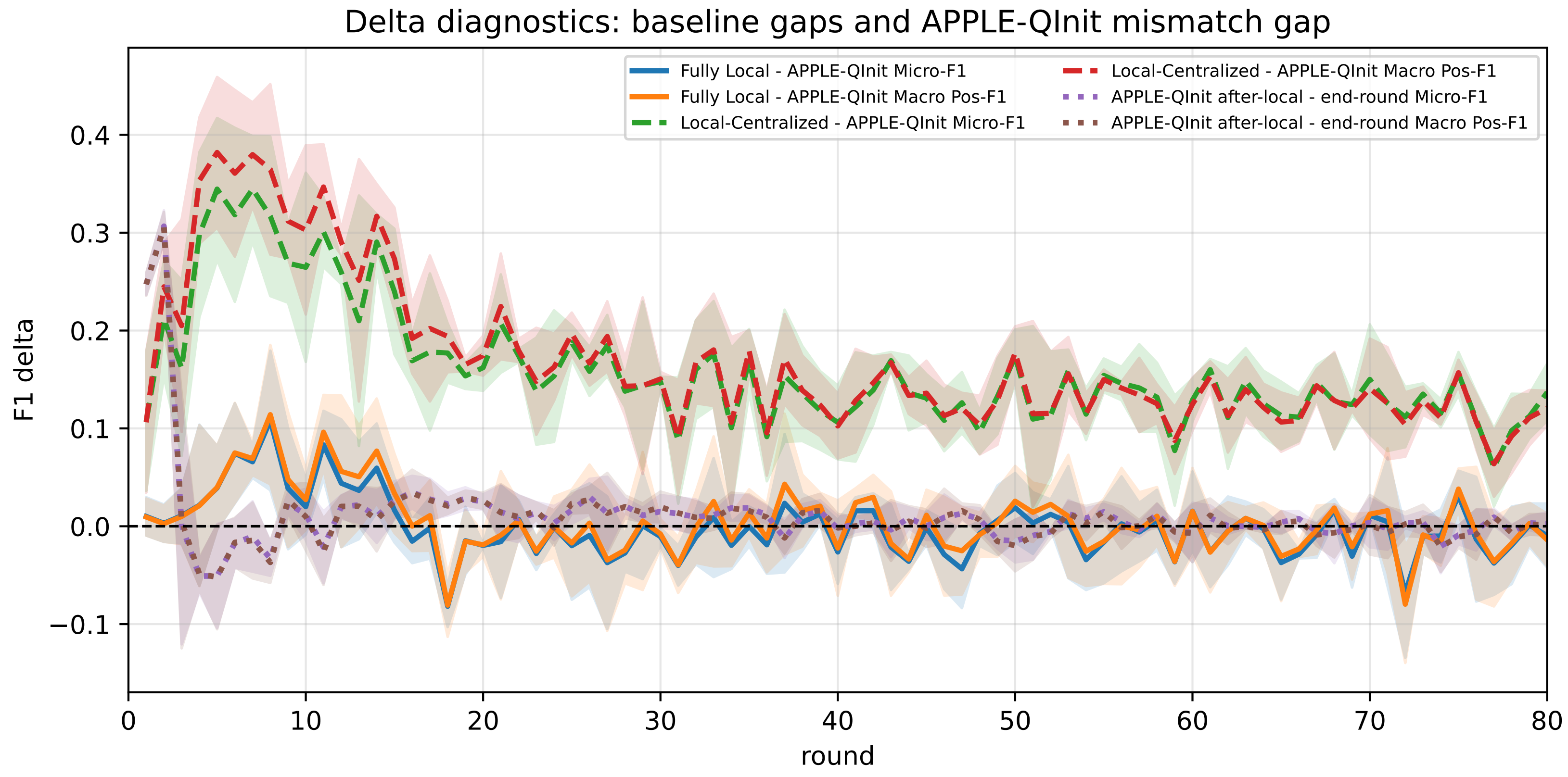
Pooled APPLE-QInit micro / macro F1 (after-local selected model)



Pooled APPLE-QInit micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QInit mismatch gap

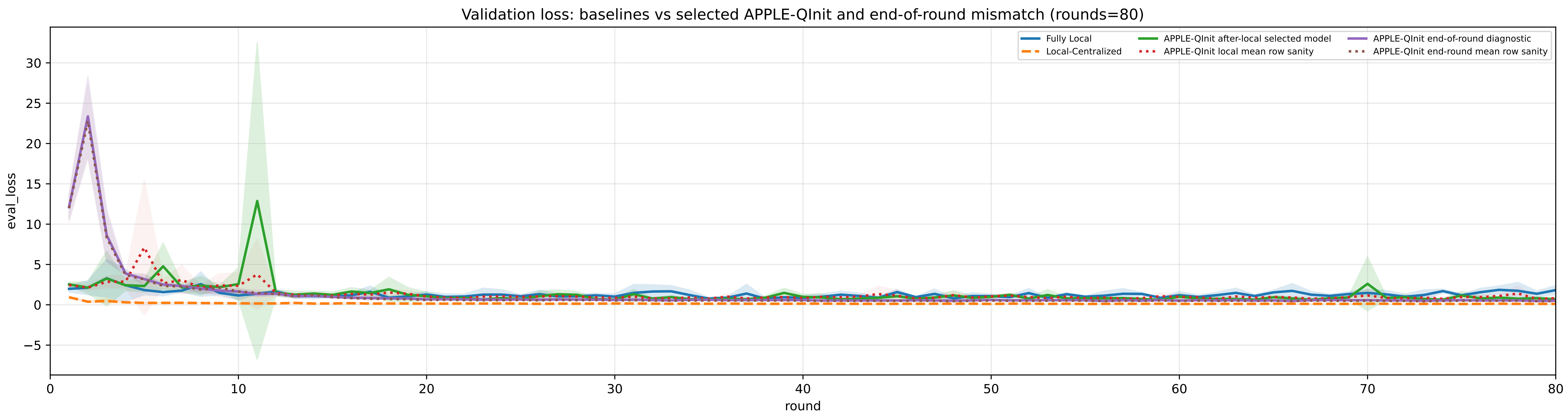


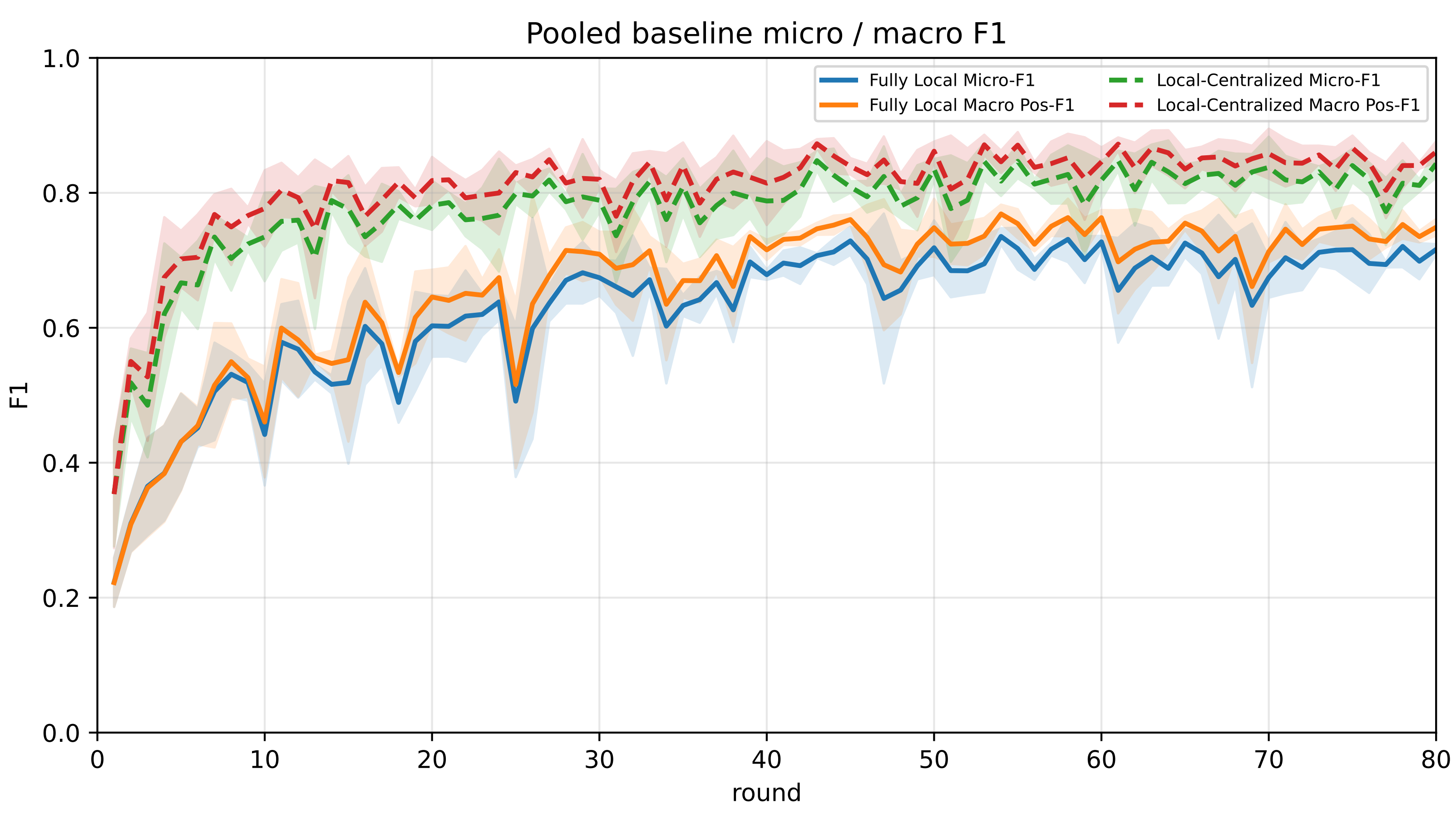
Controlled q-label heterogeneity: Mid | APPLE-QInit diagnostics | family=q_task_cycle2 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

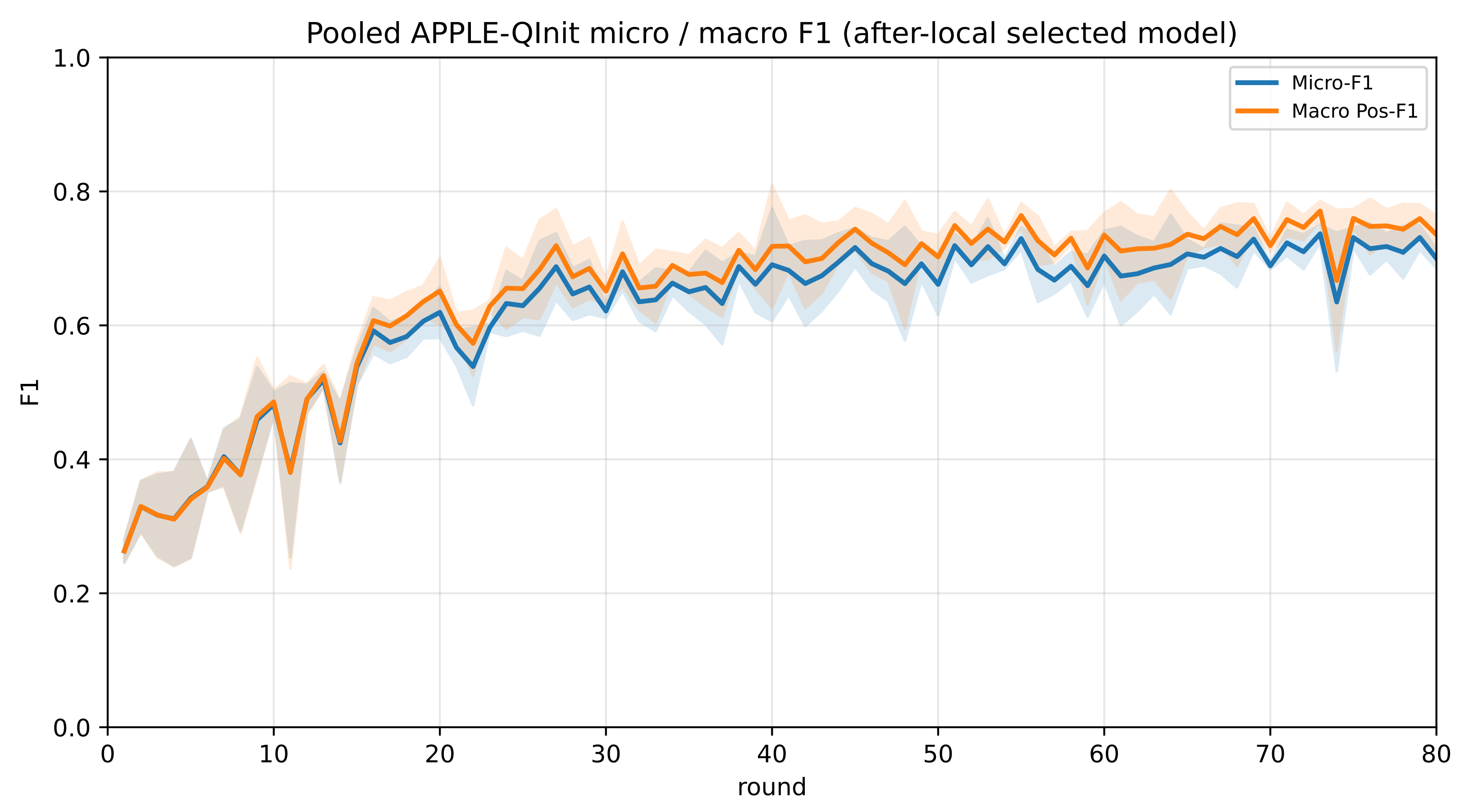
family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3% |
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

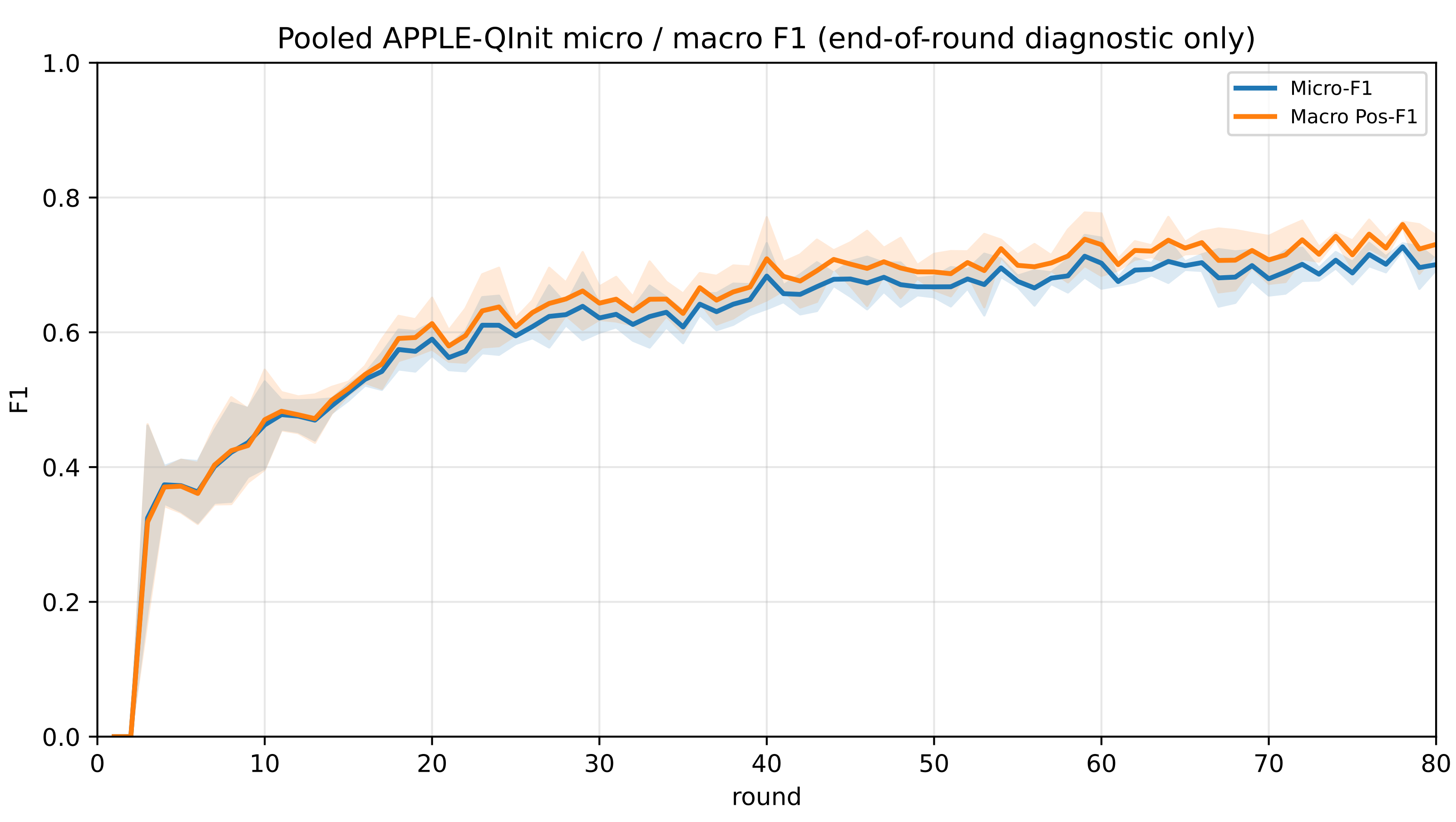
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

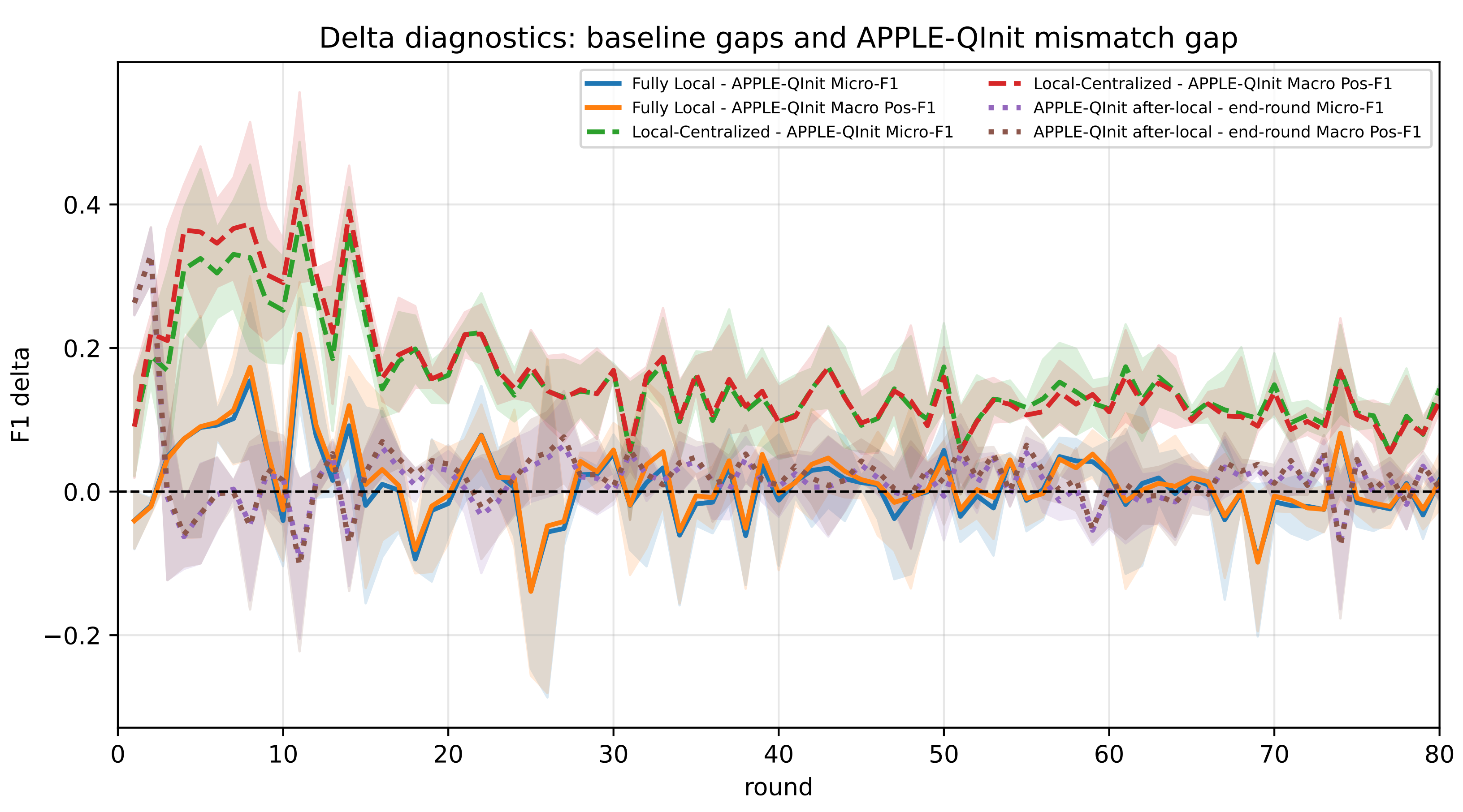
Algorithm	**Micro-F1**	**Macro Pos-F1**	**cycle2 pos-F1**	**cycle3 pos-F1**	**cycle4 pos-F1**	**cycle5 pos-F1**	**cycle6 pos-F1**
Fully Local	63.8% ± 5.4%	68.0% ± 5.8%	85.6% ± 8.0%	80.6% ± 4.3%	65.4% ± 6.2%	59.3% ± 9.8%	49.2% ± 2.7%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
APPLE-QInit	69.4% ± 5.3%	72.0% ± 6.1%	85.2% ± 8.8%	81.6% ± 11.2%	68.6% ± 7.6%	67.0% ± 1.0%	57.5% ± 2.5%









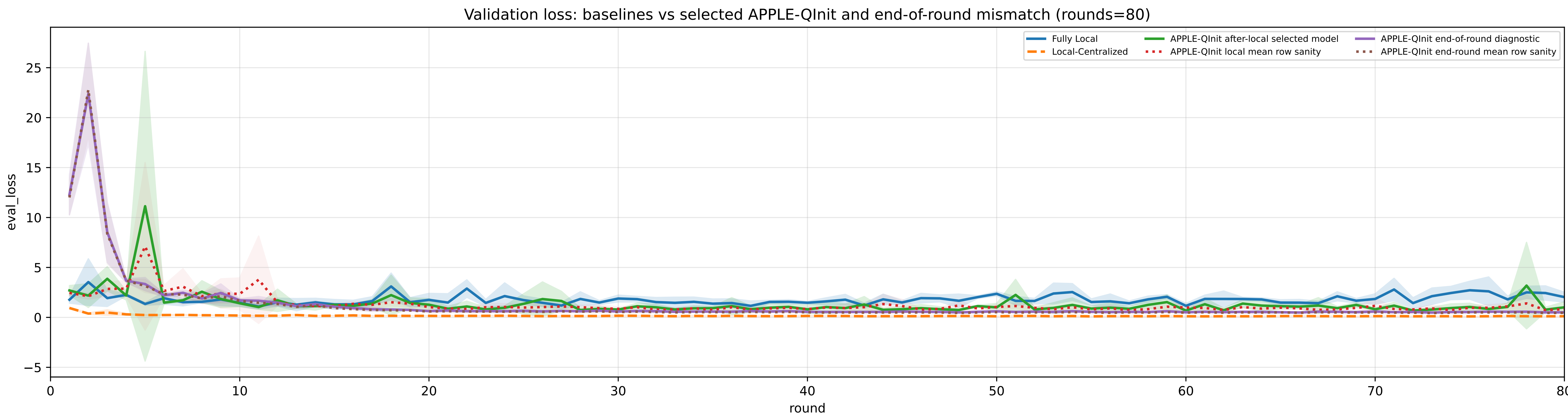


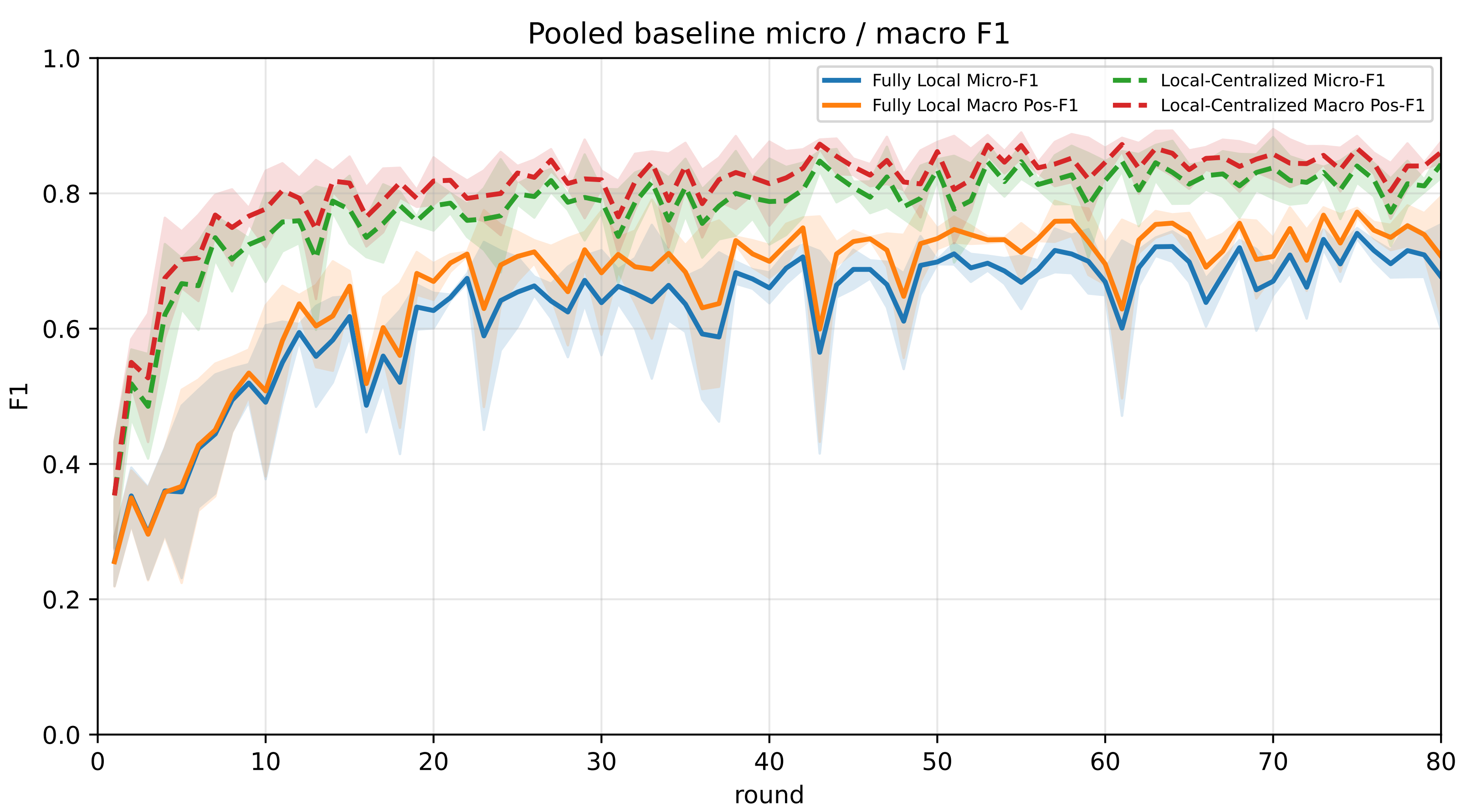
**Controlled q-label heterogeneity: Mid | APPLE-QInit diagnostics | family=q_task_cycle3 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti**

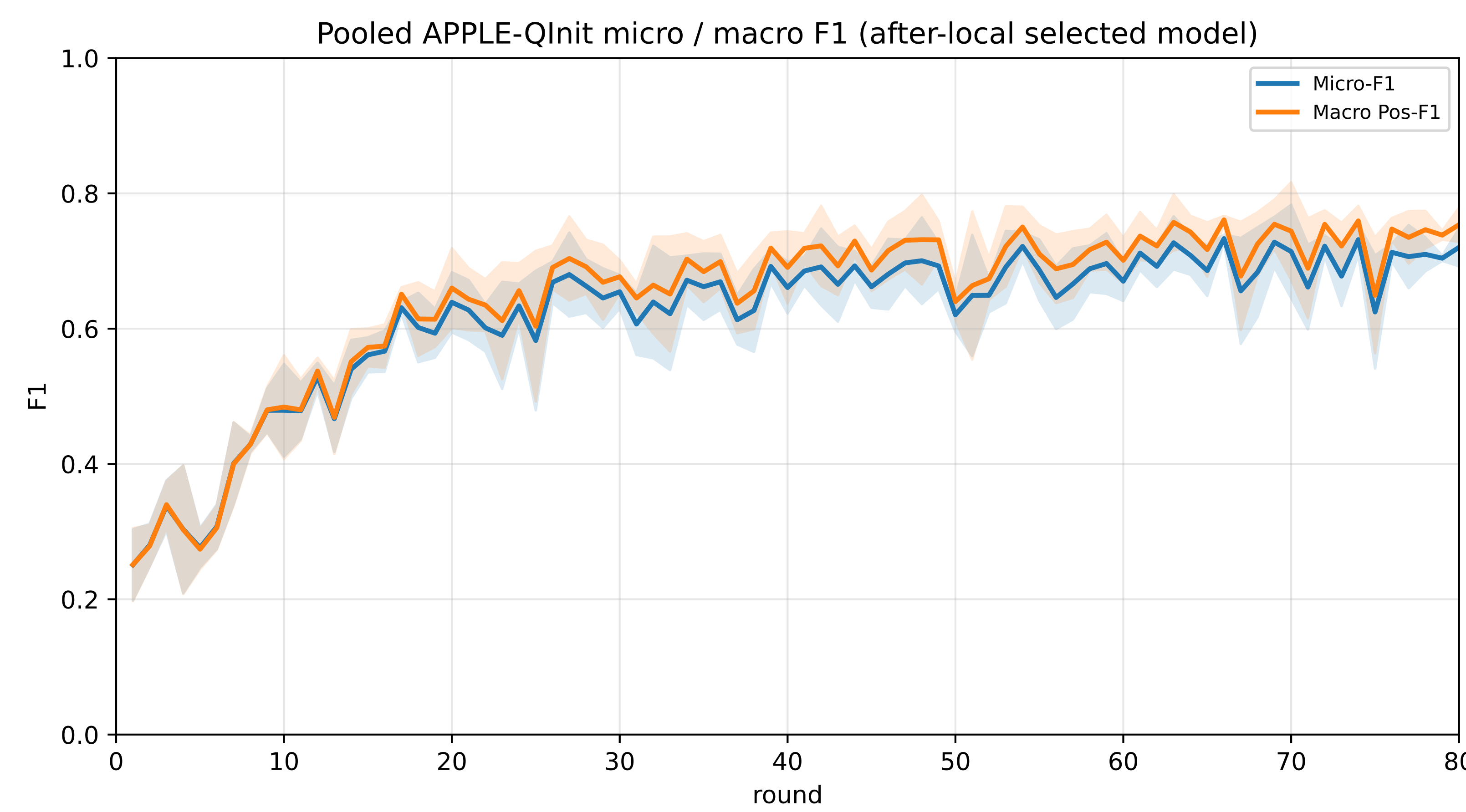
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

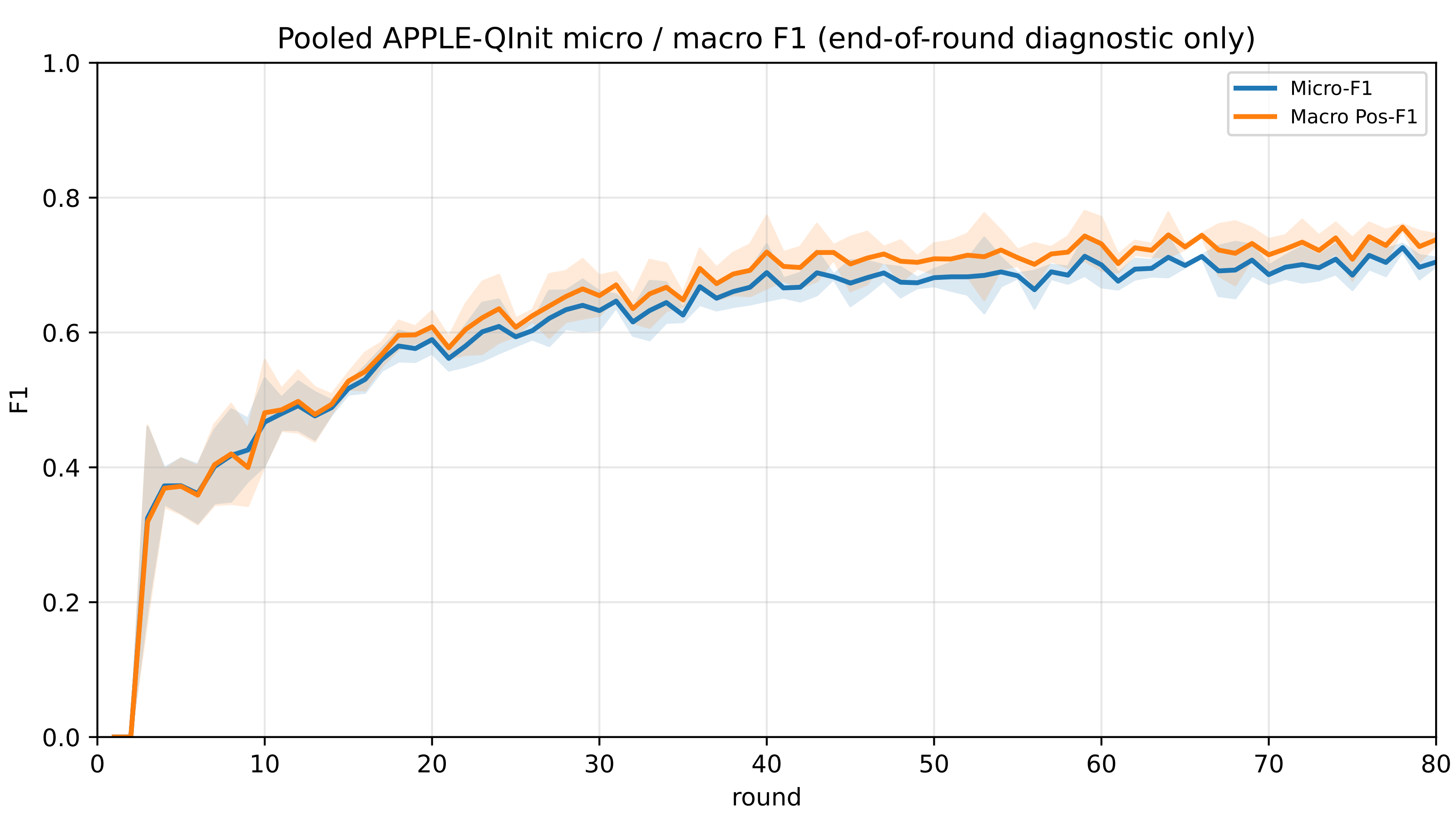
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

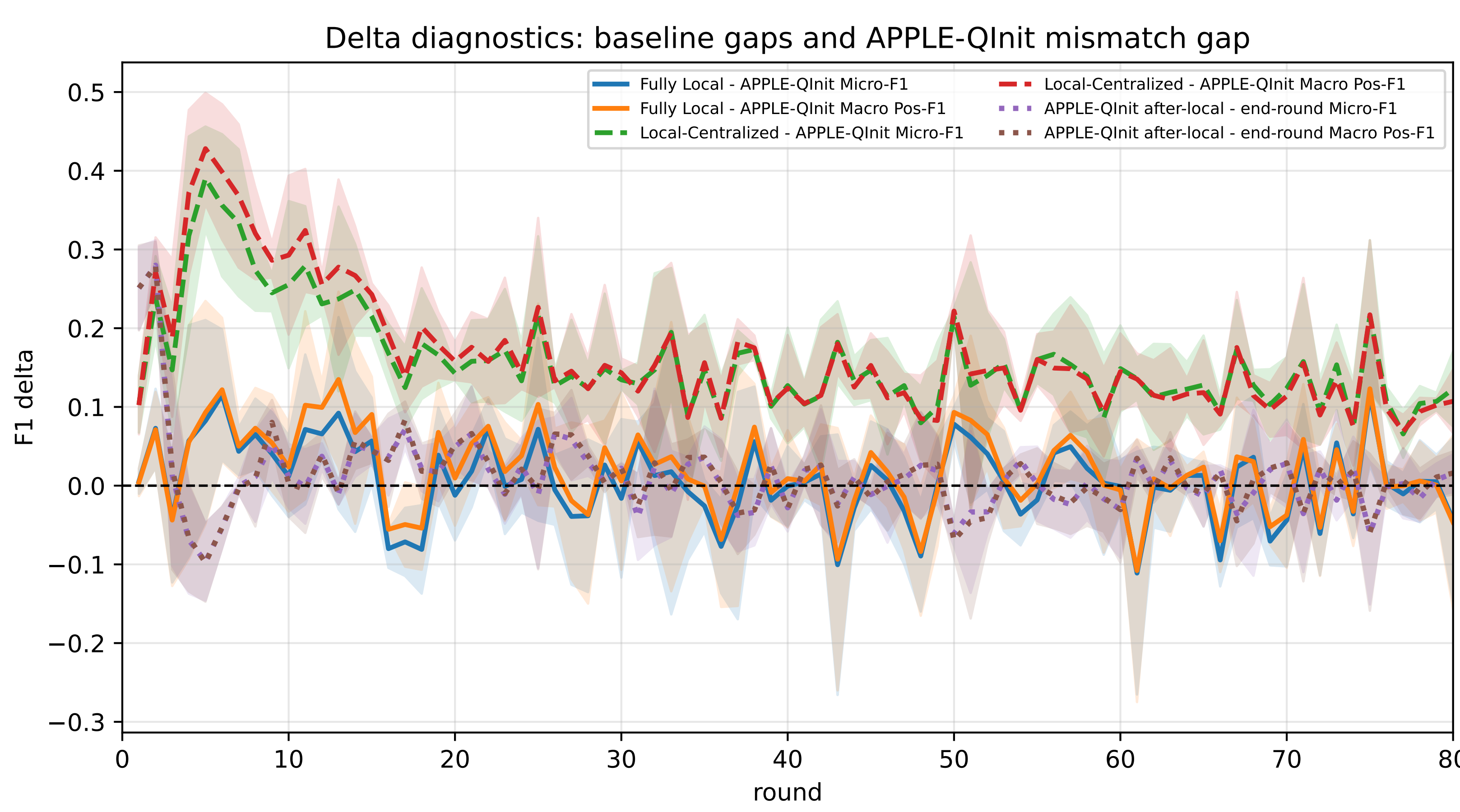
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	60.1% ± 11.6%	64.2% ± 10.9%	75.3% ± 14.5%	77.9% ± 12.0%	63.8% ± 6.1%	60.7% ± 11.5%	43.1% ± 11.1%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
APPLE-QInit	67.7% ± 4.4%	70.4% ± 3.9%	85.7% ± 0.8%	76.5% ± 7.3%	69.6% ± 6.9%	65.2% ± 3.0%	55.0% ± 4.5%









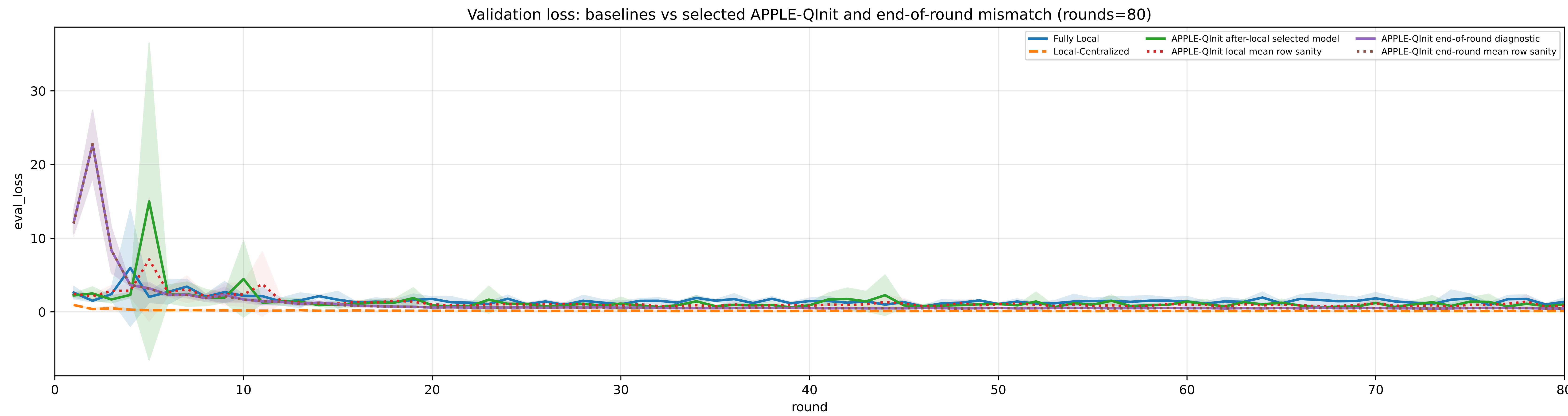


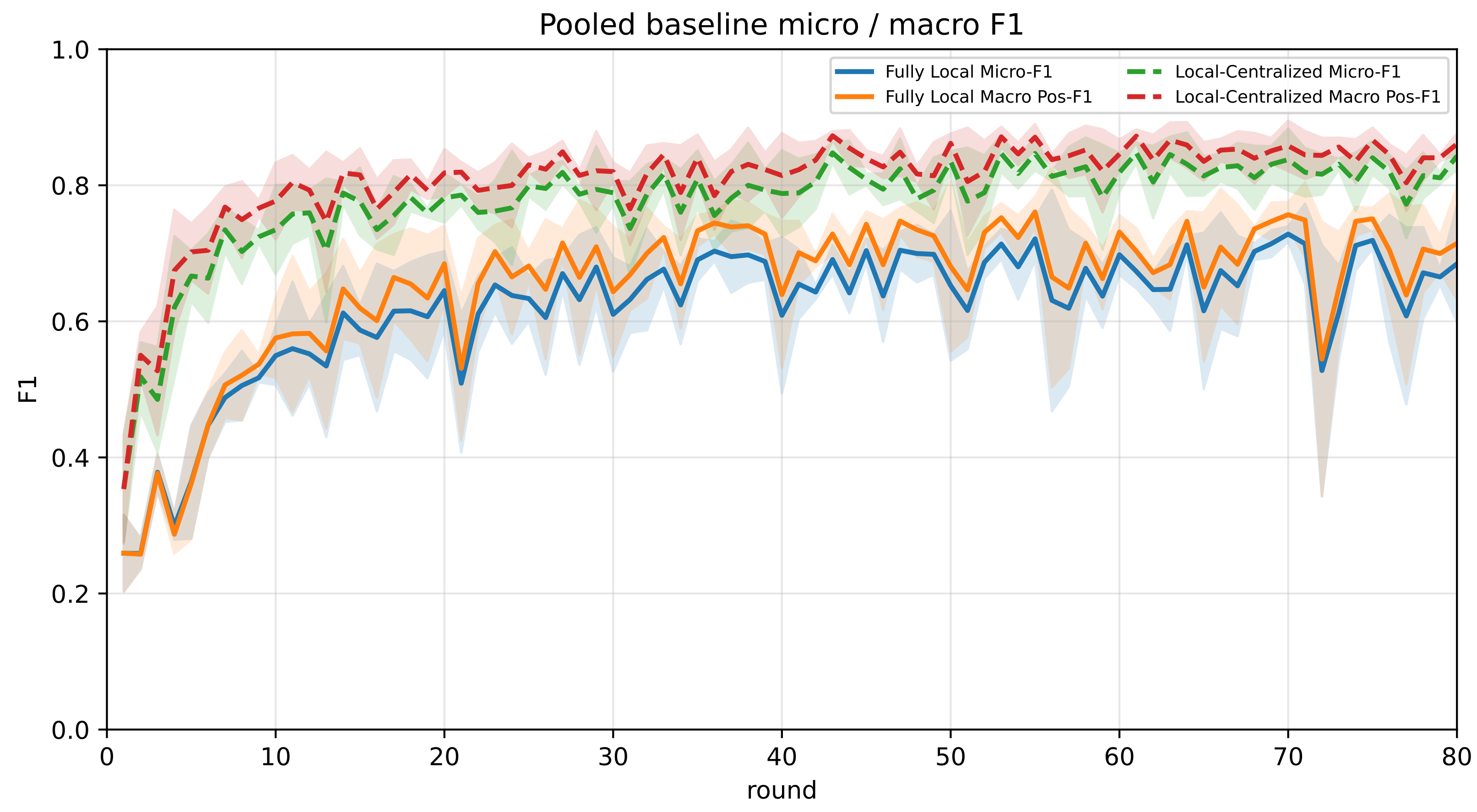
Controlled q-label heterogeneity: Mid | APPLE-QInit diagnostics | family=q_task_cycle4 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

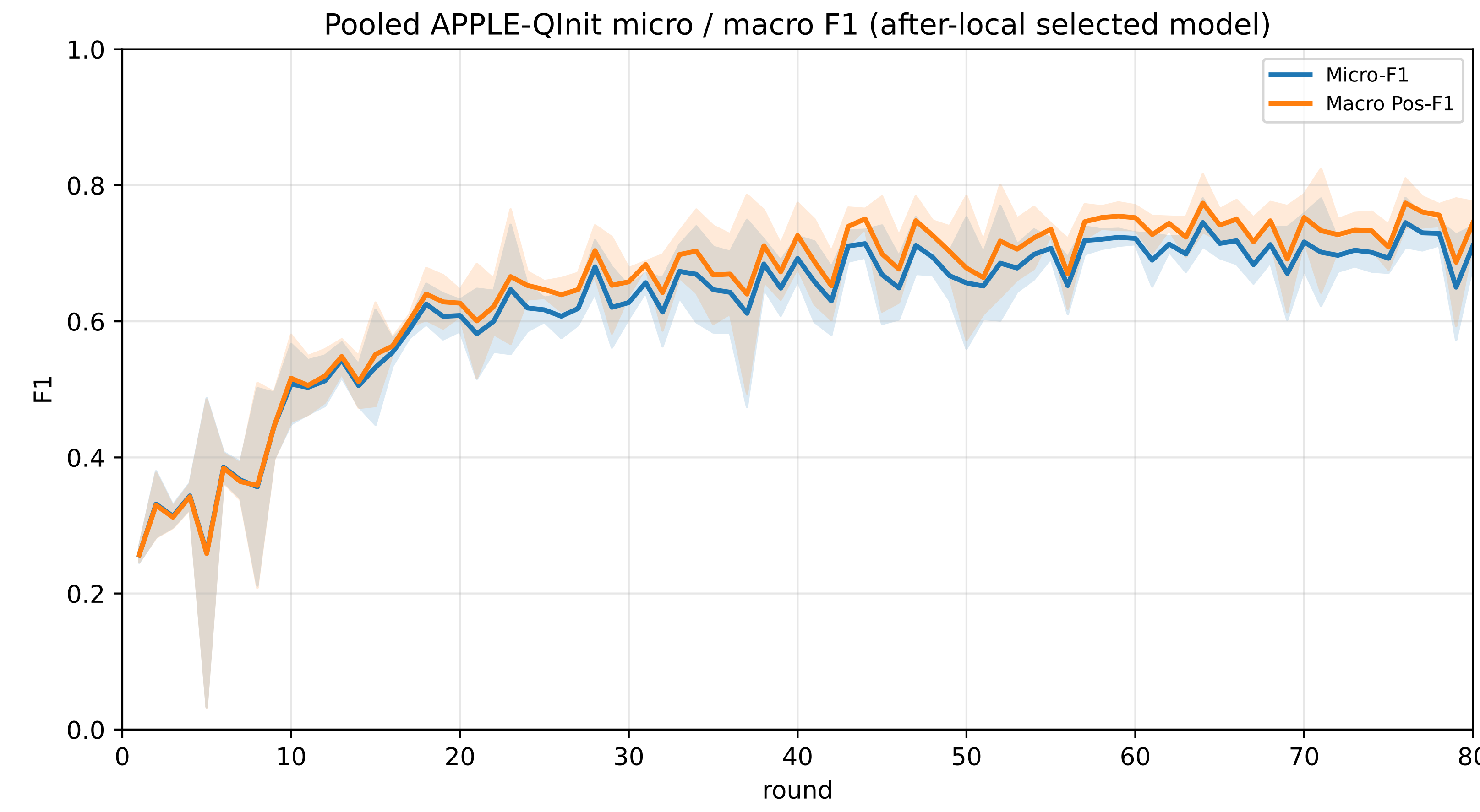
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

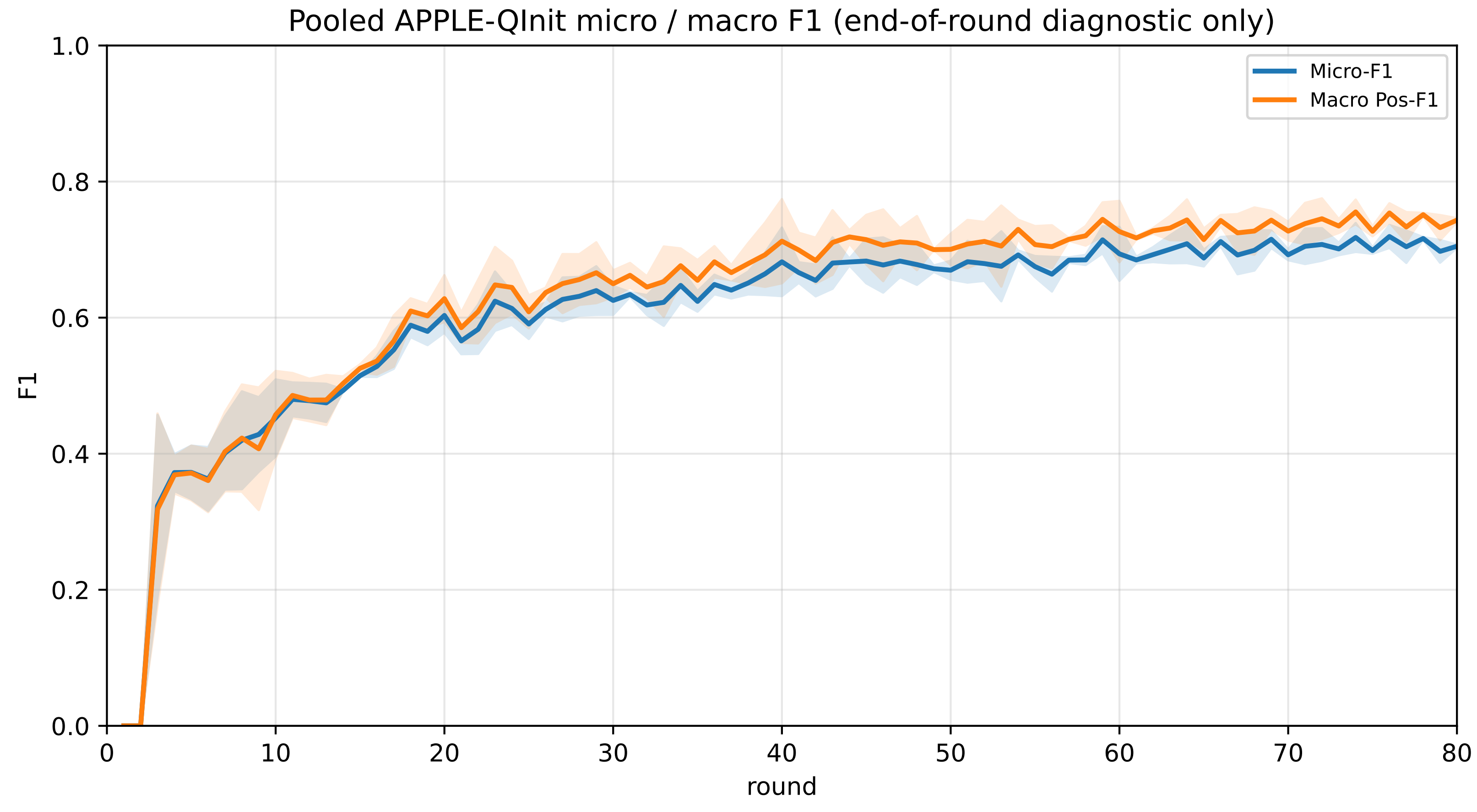
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

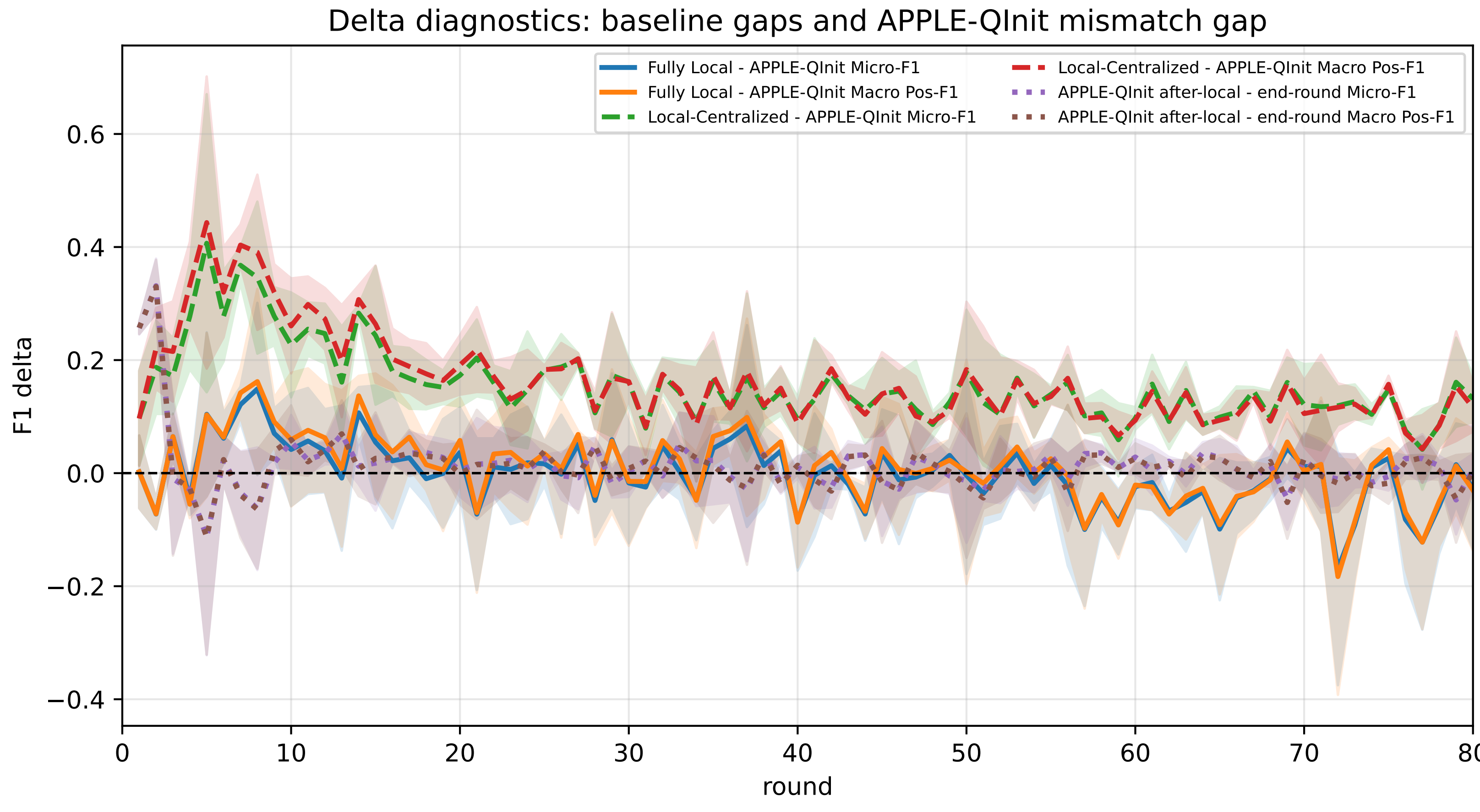
Algorithm	**Micro-F1**	**Macro Pos-F1**	**cycle2 pos-F1**	**cycle3 pos-F1**	**cycle4 pos-F1**	**cycle5 pos-F1**	**cycle6 pos-F1**
Fully Local	58.7% ± 5.5%	63.8% ± 5.1%	82.7% ± 2.7%	76.5% ± 5.8%	63.3% ± 5.5%	54.2% ± 7.8%	42.1% ± 3.8%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
APPLE-QInit	68.0% ± 4.0%	71.1% ± 4.0%	89.1% ± 4.0%	79.1% ± 7.4%	64.8% ± 2.9%	66.6% ± 2.0%	55.8% ± 4.3%









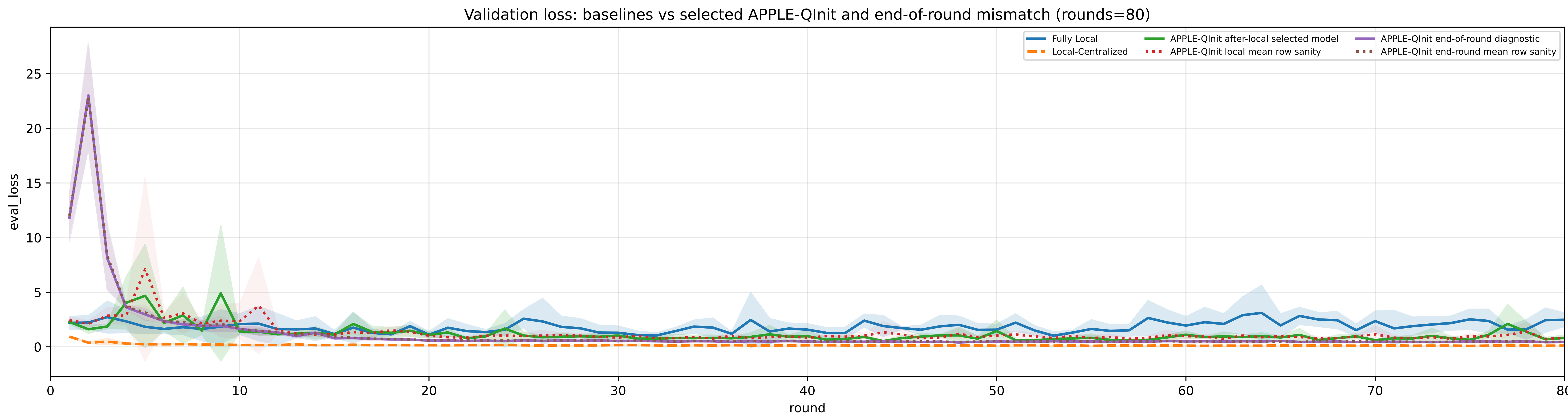


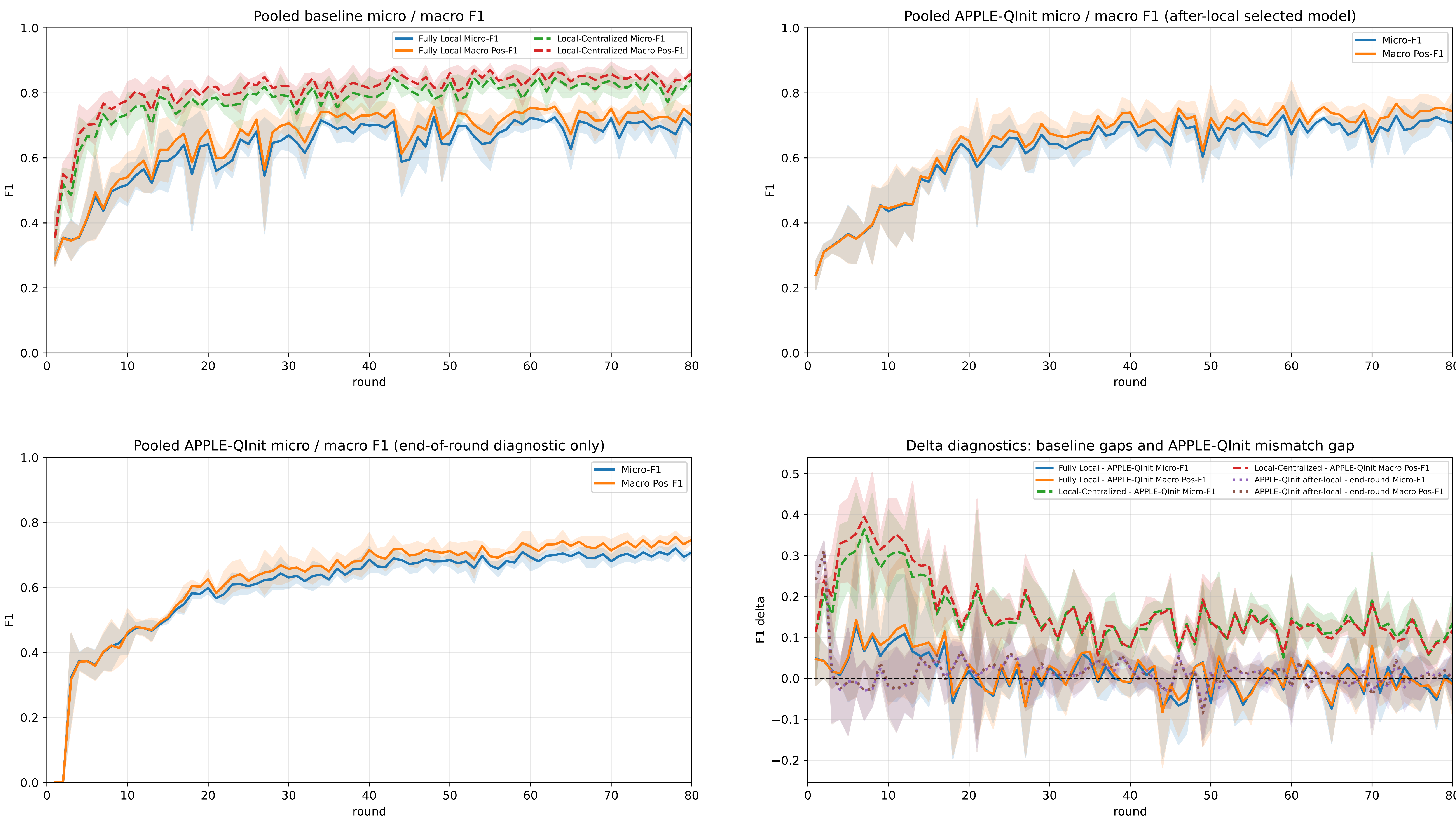
**Controlled q-label heterogeneity: Mid | APPLE-QInit diagnostics | family=q_task_cycle5 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti**

family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.2% ± 0.9%	66.3% ± 1.4%	86.3% ± 4.7%	79.8% ± 5.8%	63.6% ± 1.5%	58.1% ± 1.4%	43.8% ± 3.4%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
APPLE-QInit	66.0% ± 6.4%	68.7% ± 6.5%	80.9% ± 10.2%	75.0% ± 10.6%	70.7% ± 6.2%	64.8% ± 3.5%	52.2% ± 6.5%



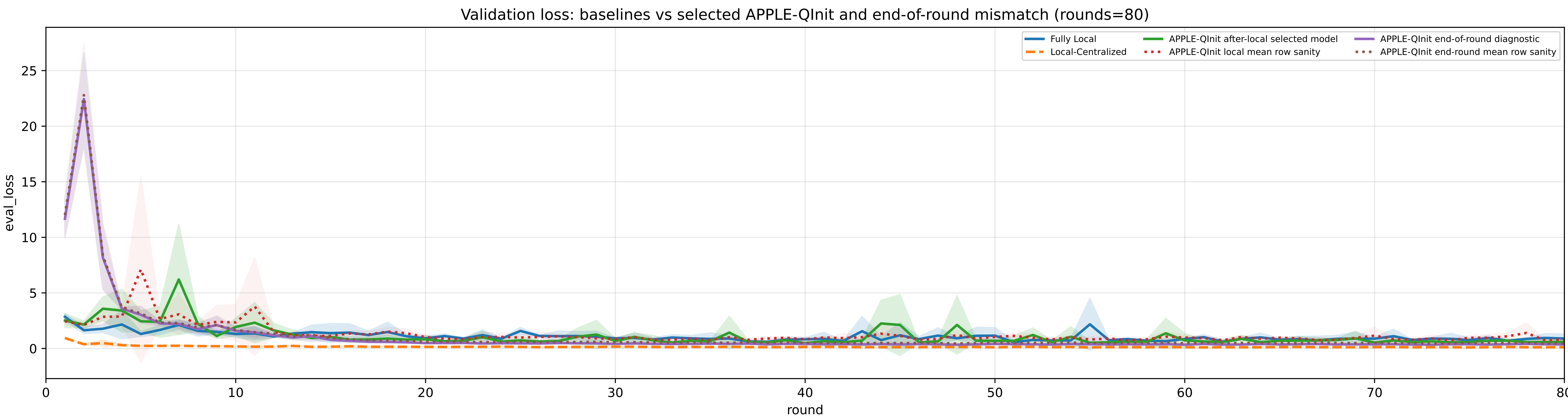


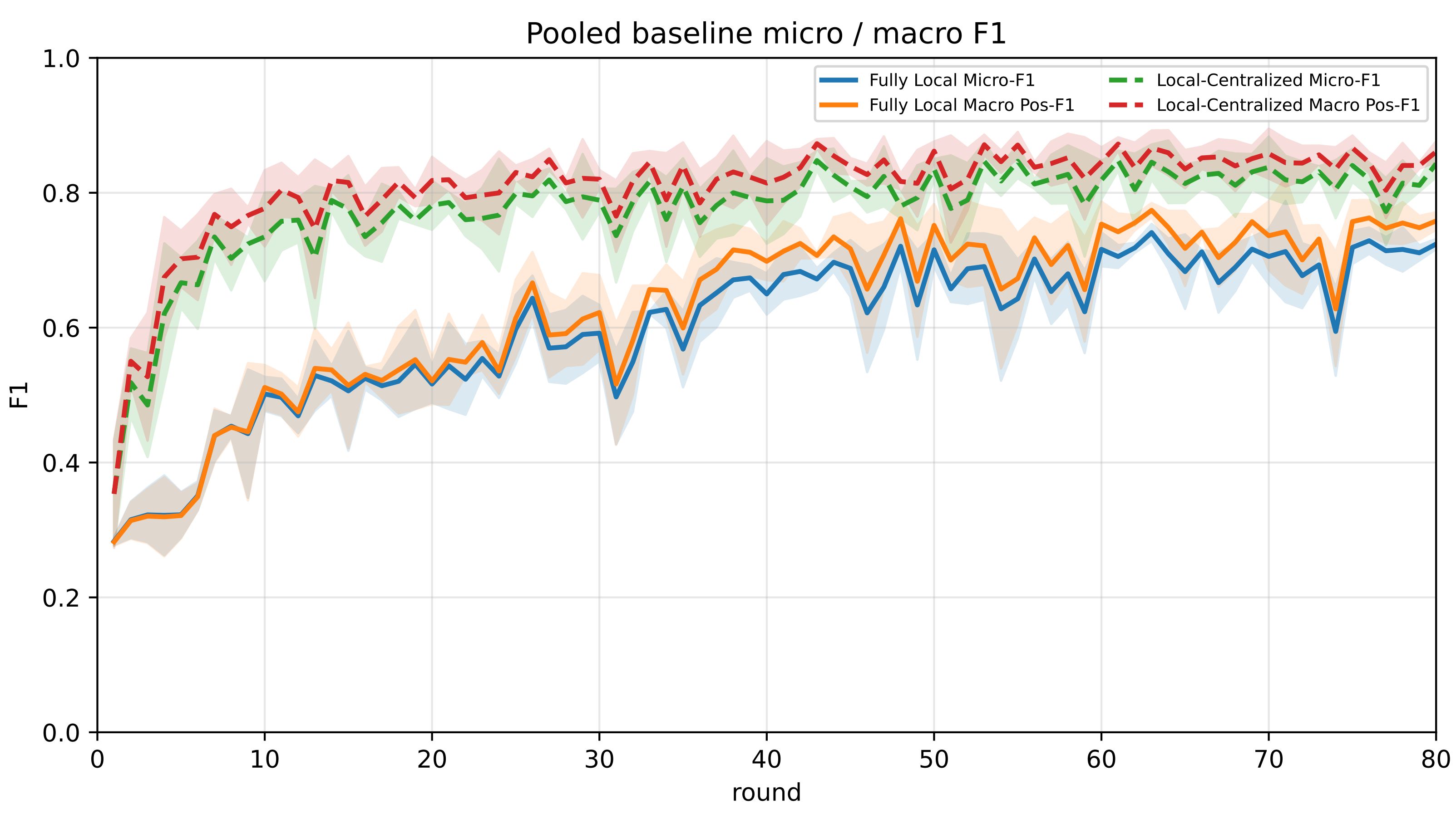
Controlled q-label heterogeneity: Mid | APPLE-QInit diagnostics | family=q_task_cycle6 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

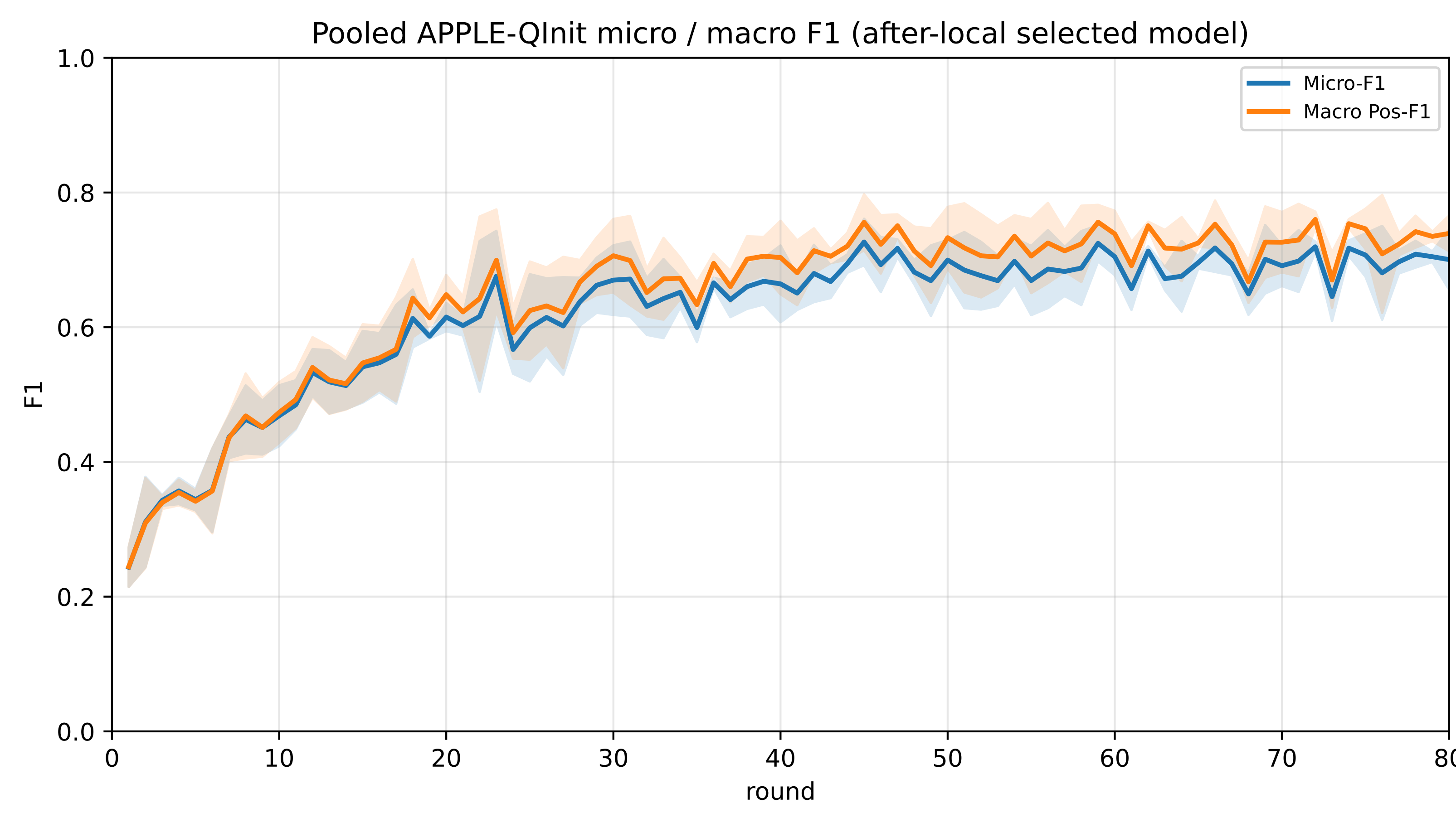
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

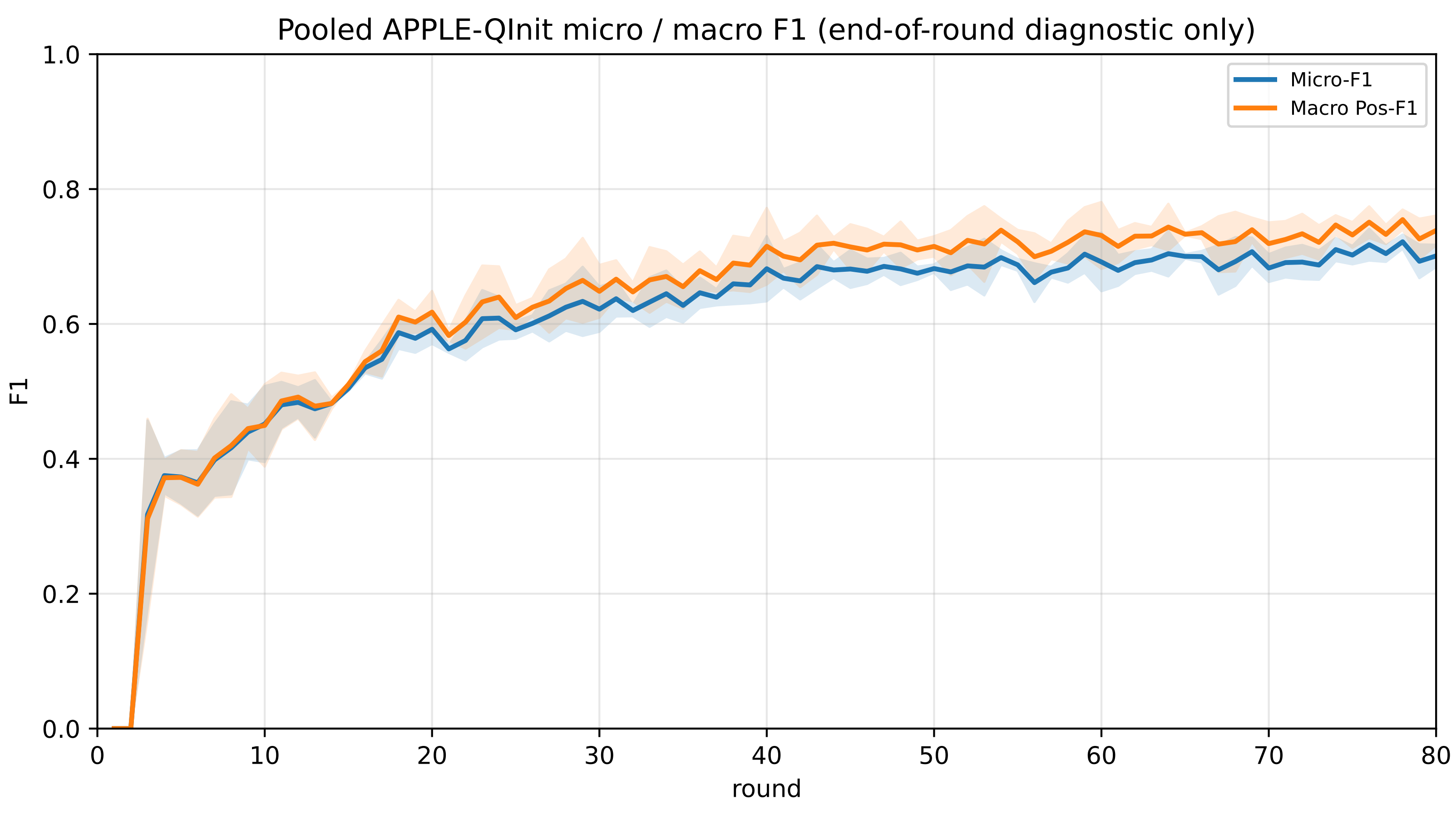
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

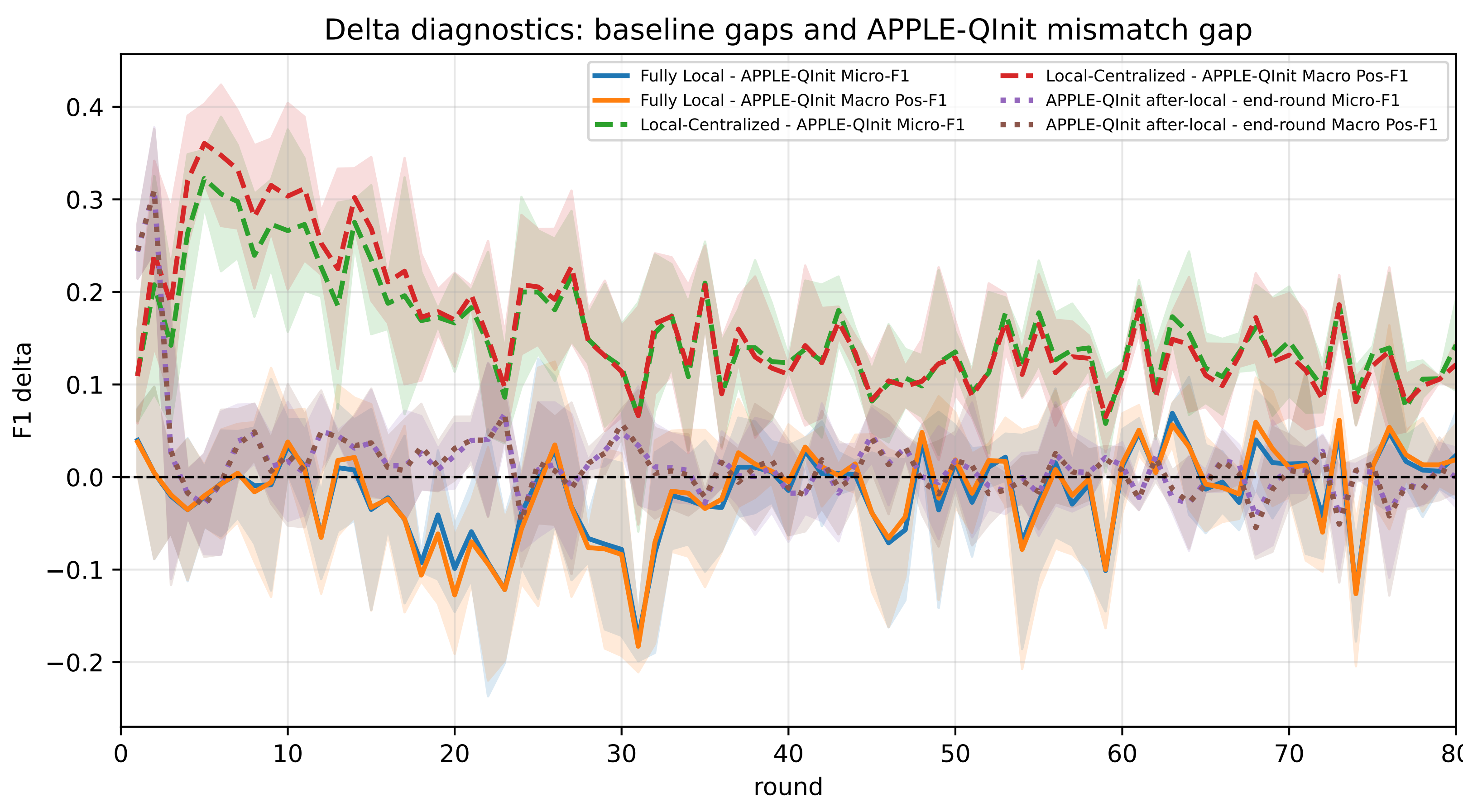
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.1% ± 1.8%	67.8% ± 2.4%	81.8% ± 8.2%	78.3% ± 6.8%	64.9% ± 1.4%	64.9% ± 4.4%	49.0% ± 3.2%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
APPLE-QInit	68.7% ± 2.7%	71.8% ± 2.8%	86.3% ± 5.1%	78.0% ± 7.6%	71.8% ± 4.2%	69.1% ± 0.7%	54.0% ± 3.2%





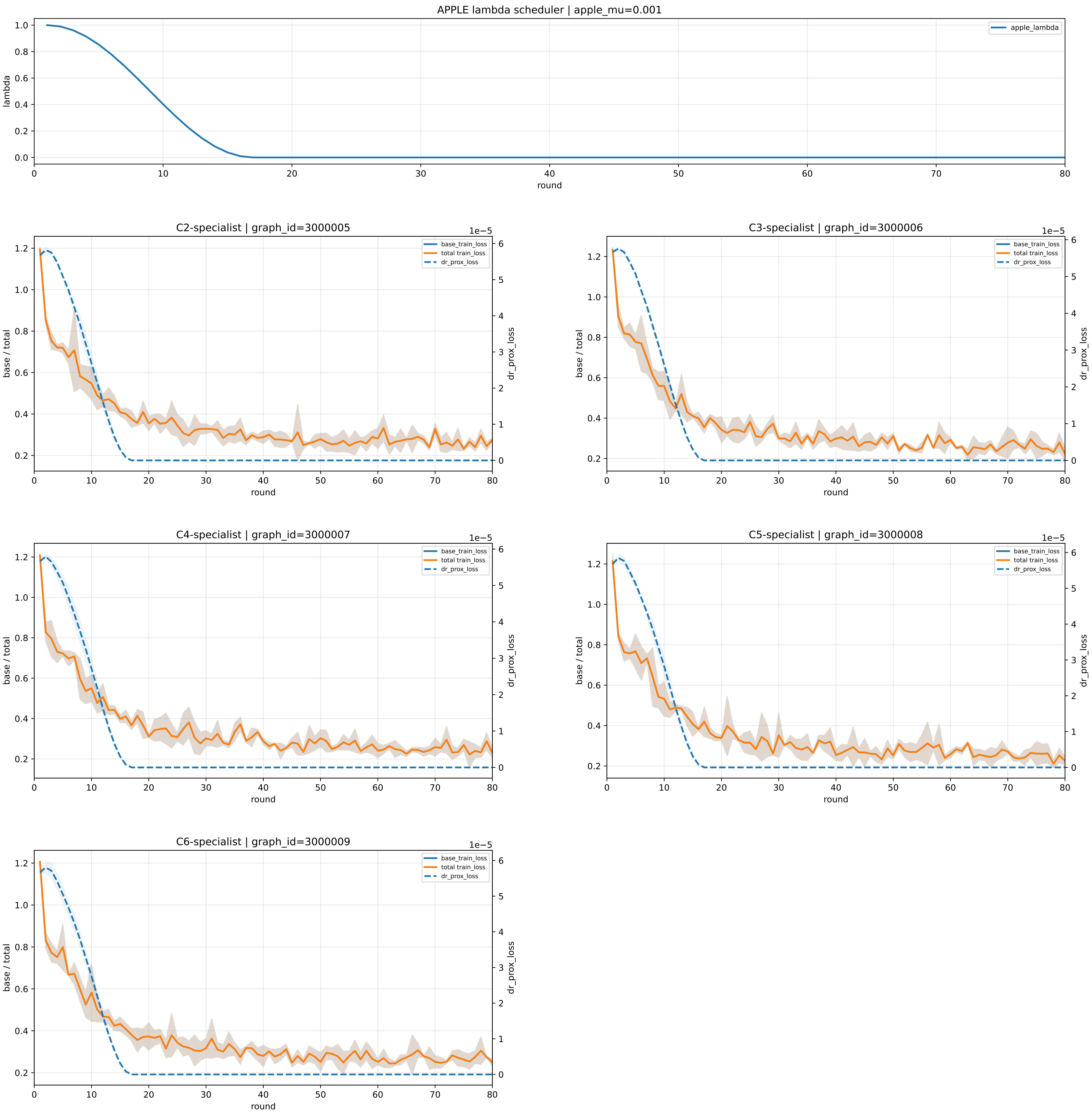


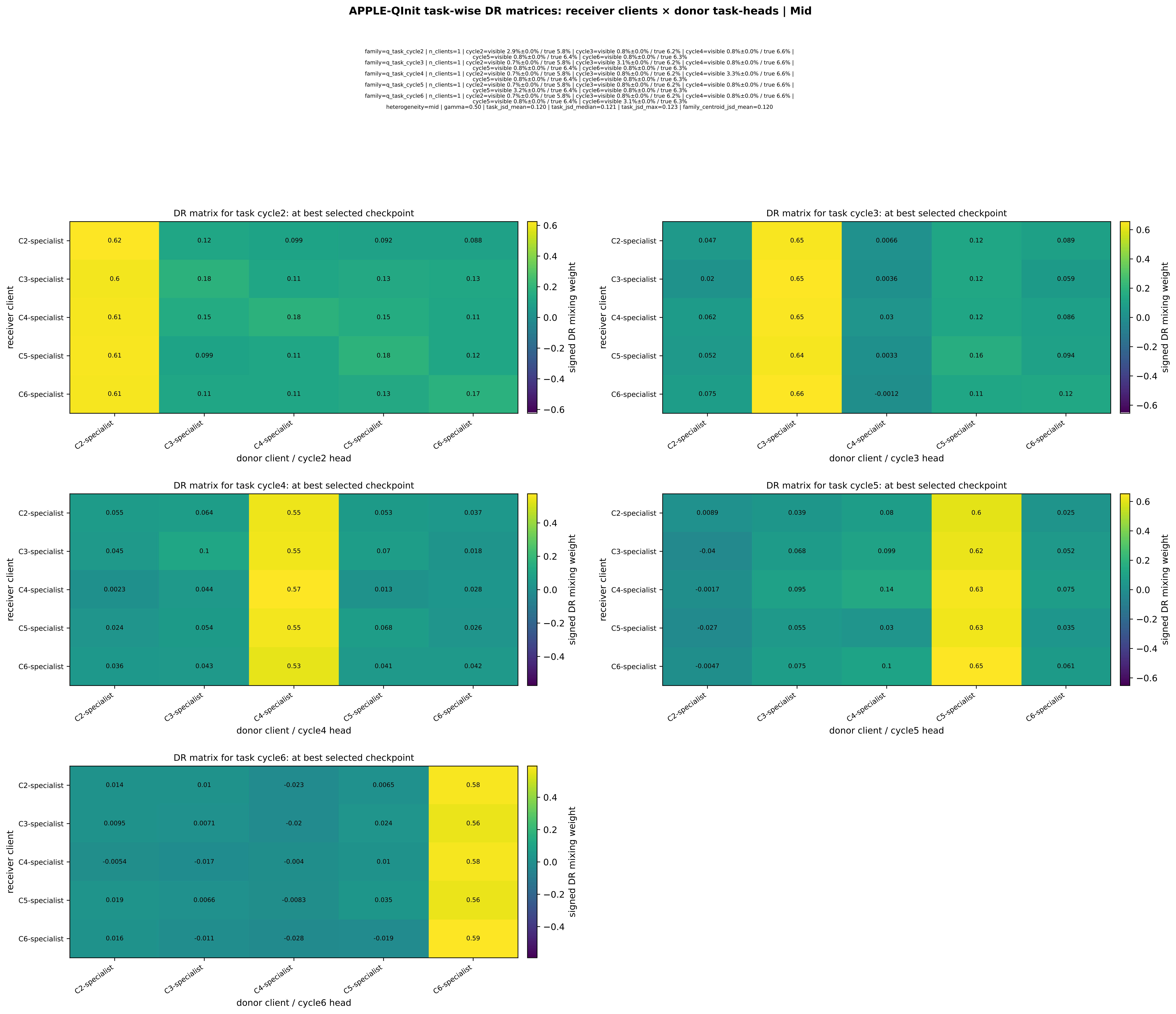




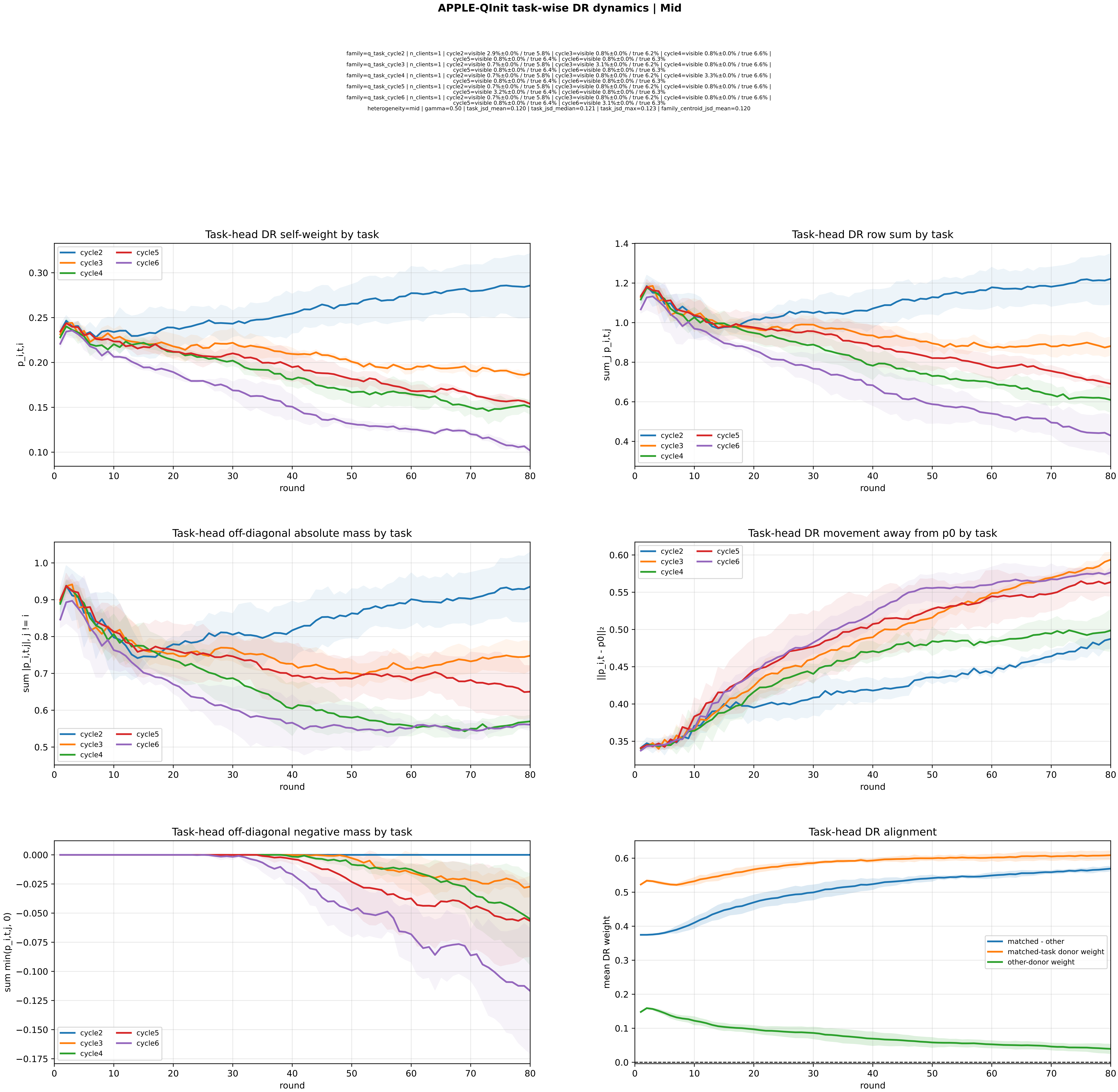
APPLE-QInit training-loss decomposition | Mid heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120





APPLE-QInit task-wise DR matrices: receiver clients × donor task-heads | Mid
family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120
DR matrix for task cycle2: at best selected checkpoint
C2-specialist 0.62 0.12 0.099 0.092 0.088
C3-specialist 0.6 0.18 0.11 0.13 0.13
C4-specialist 0.61 0.15 0.18 0.15 0.11
C5-specialist 0.61 0.099 0.11 0.18 0.12
C6-specialist 0.61 0.11 0.11 0.13 0.17
donor client / cycle2 head
DR matrix for task cycle3: at best selected checkpoint
C2-specialist 0.047 0.65 0.0066 0.12 0.089
C3-specialist 0.02 0.65 0.0036 0.12 0.059
C4-specialist 0.062 0.65 0.03 0.12 0.086
C5-specialist 0.052 0.64 0.0033 0.16 0.094
C6-specialist 0.075 0.66 -0.0012 0.11 0.12
donor client / cycle3 head
DR matrix for task cycle4: at best selected checkpoint
C2-specialist 0.055 0.064 0.55 0.053 0.037
C3-specialist 0.045 0.1 0.55 0.07 0.018
C4-specialist 0.0023 0.044 0.57 0.013 0.028
C5-specialist 0.024 0.054 0.55 0.068 0.026
C6-specialist 0.036 0.043 0.53 0.041 0.042
donor client / cycle4 head
DR matrix for task cycle5: at best selected checkpoint
C2-specialist 0.0089 0.039 0.08 0.6 0.025
C3-specialist -0.04 0.068 0.099 0.62 0.052
C4-specialist -0.0017 0.095 0.14 0.63 0.075
C5-specialist -0.027 0.055 0.03 0.63 0.035
C6-specialist -0.0047 0.075 0.1 0.65 0.061
donor client / cycle5 head
DR matrix for task cycle6: at best selected checkpoint
C2-specialist 0.014 0.01 -0.023 0.0065 0.58
C3-specialist 0.0095 0.0071 -0.02 0.024 0.56
C4-specialist -0.0054 -0.017 -0.004 0.01 0.58
C5-specialist 0.019 0.0066 -0.0083 0.035 0.56
C6-specialist 0.016 -0.011 -0.028 -0.019 0.59
donor client / cycle6 head
C2-specialist
C3-specialist
C4-specialist
C5-specialist
C6-specialist
receiver client
signed DR mixing weight

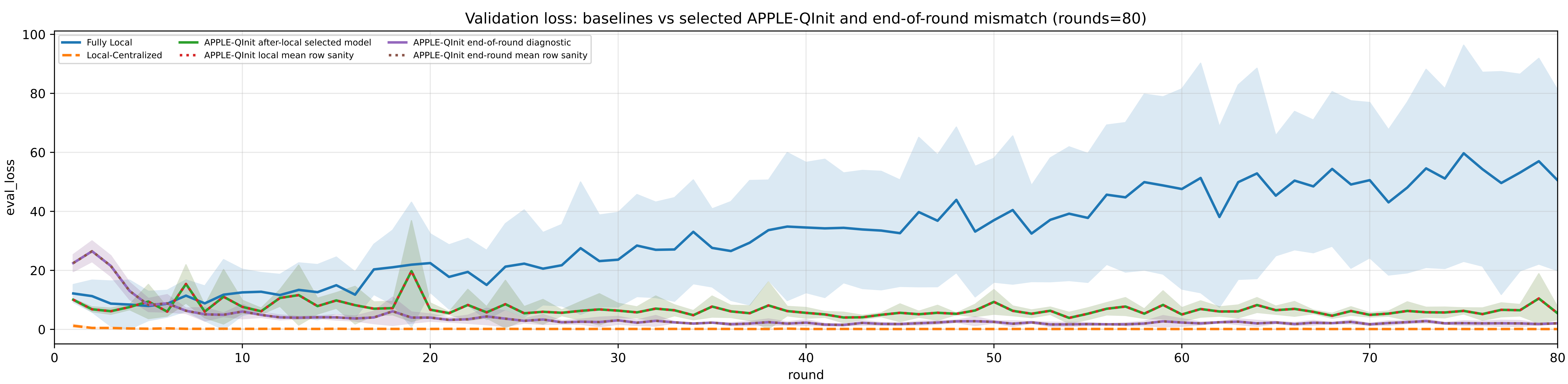

APPLE-QInit task-wise DR dynamics | Mid
family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120
Task-head DR self-weight by task
p_i,t,i
round
Task-head DR row sum by task
sum_j p_i,t,j
Task-head off-diagonal absolute mass by task
sum |p_i,t,j|, j != i
Task-head DR movement away from p0 by task
||p_i,t - p0||₂
Task-head off-diagonal negative mass by task
sum min(p_i,t,j, 0)
Task-head DR alignment
mean DR weight
cycle2
cycle3
cycle4
cycle5
cycle6
matched - other
matched-task donor weight
other-donor weight

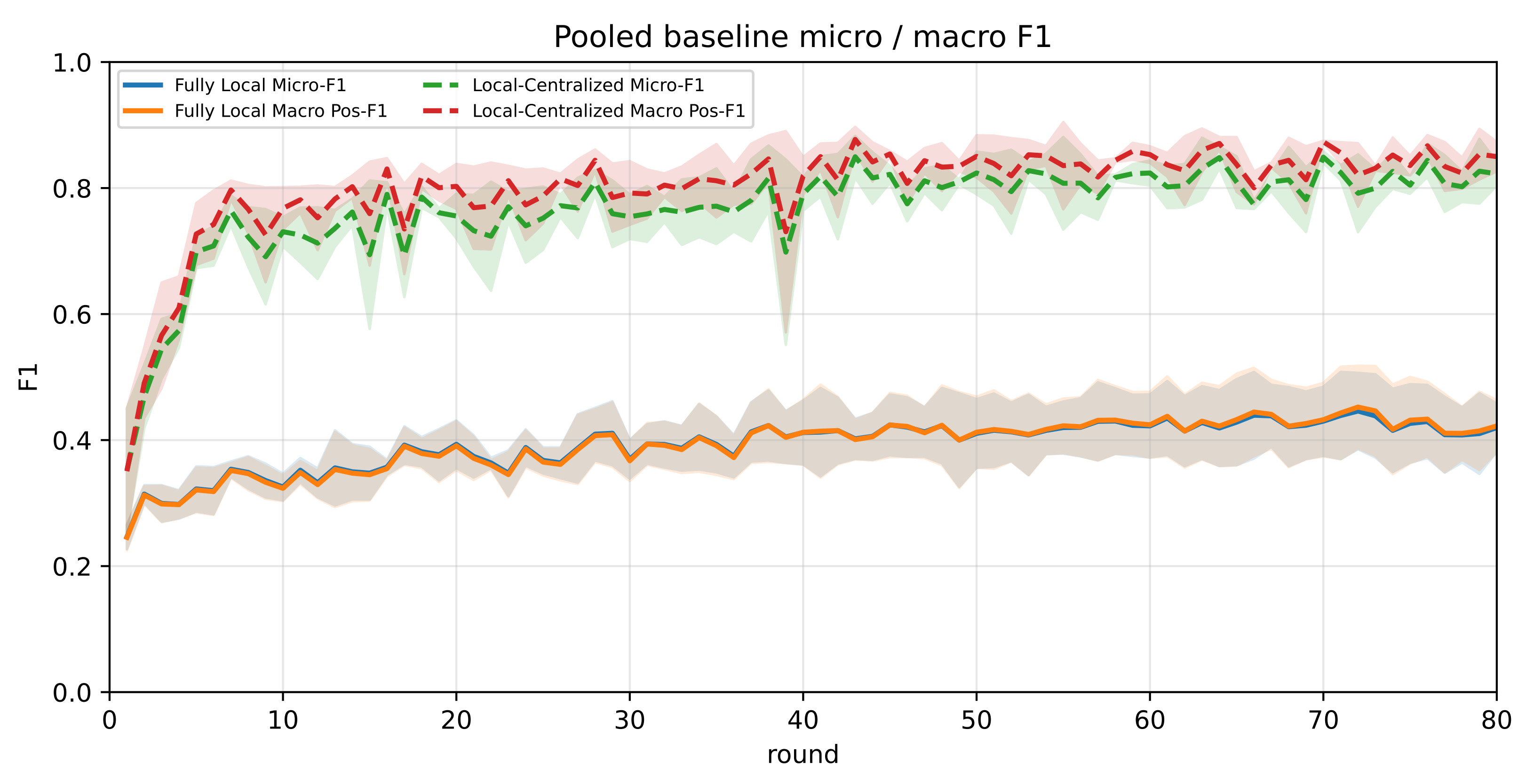
Controlled q-label heterogeneity: High | APPLE-QInit diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

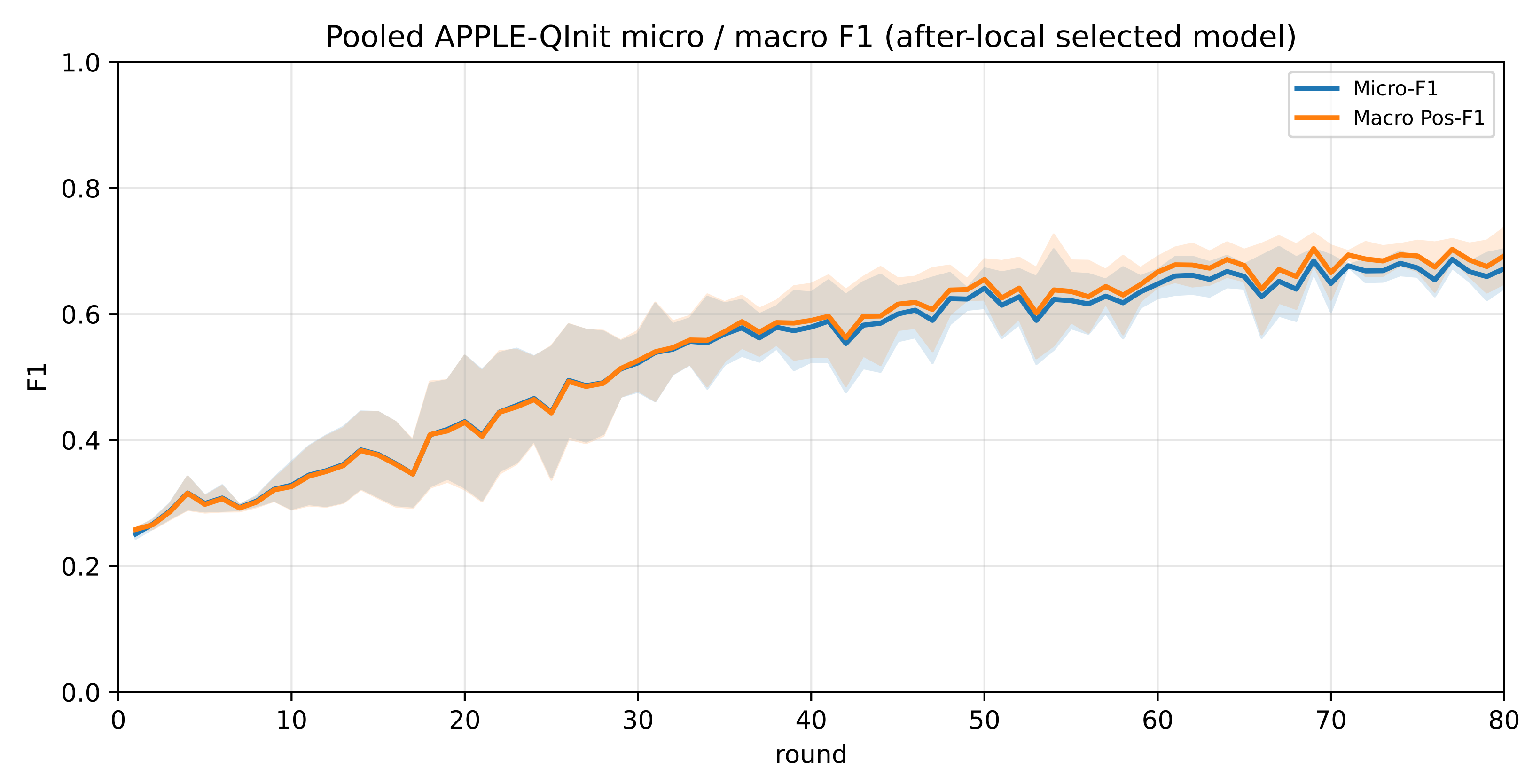
family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

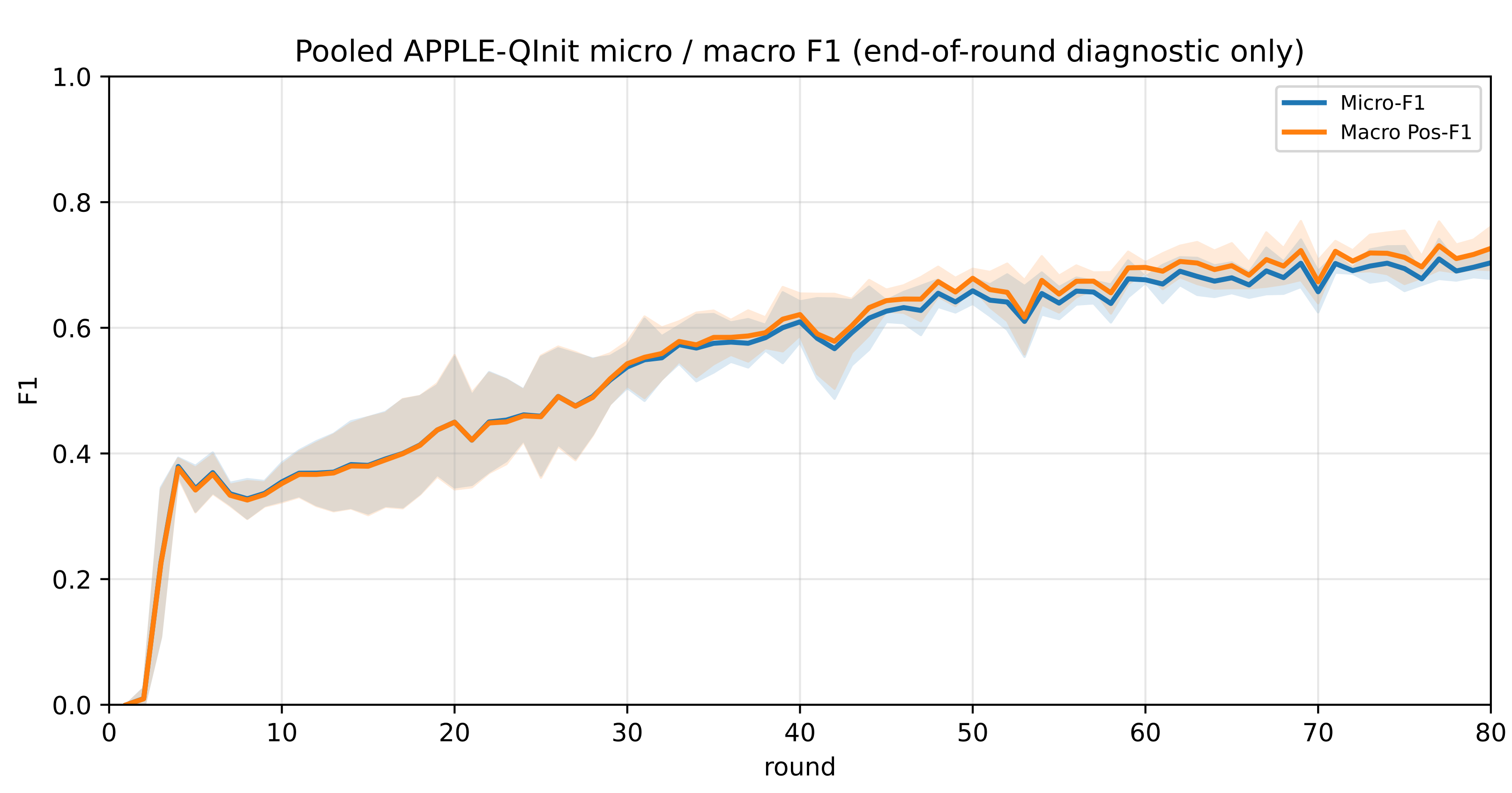
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

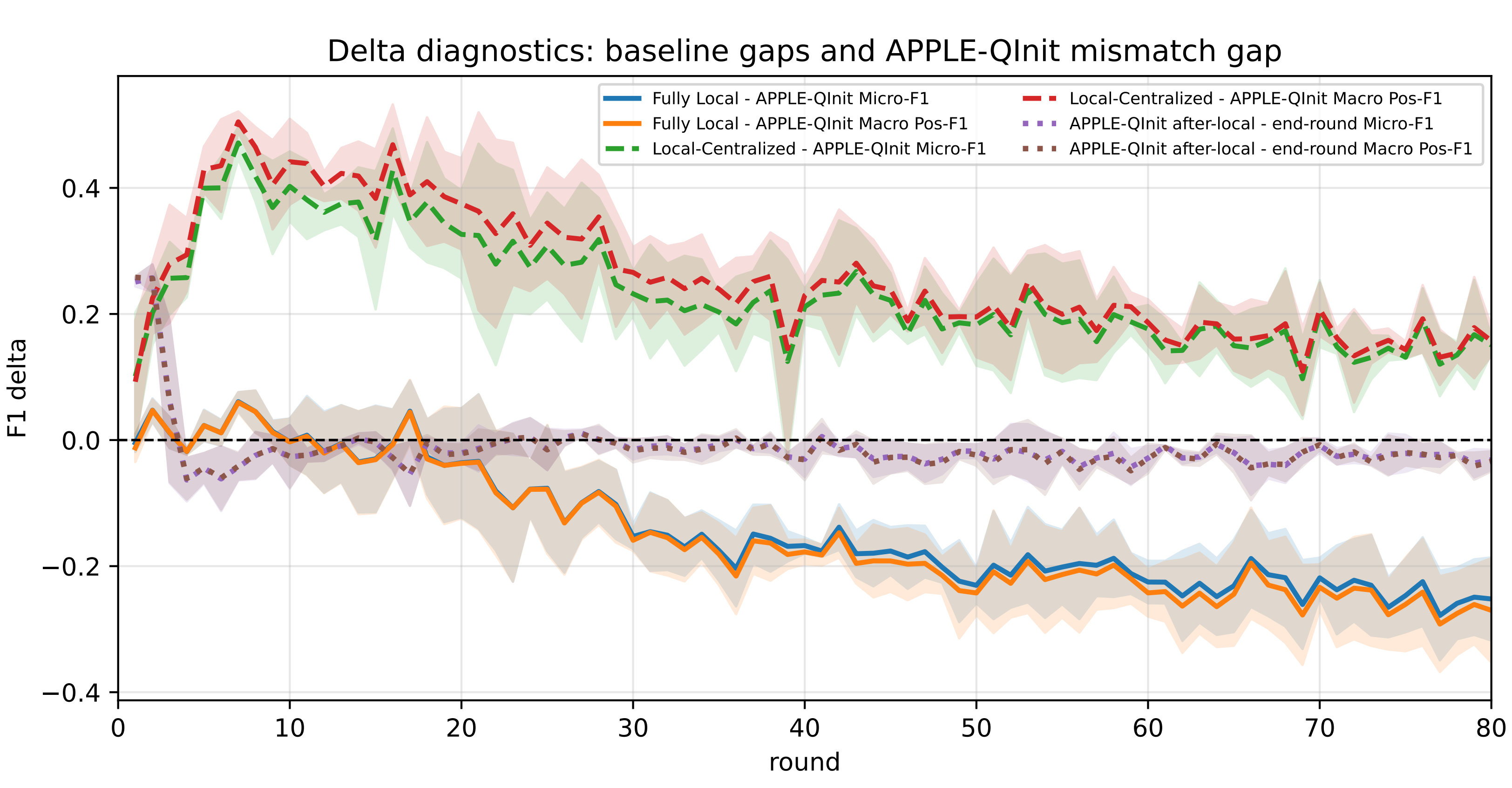
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.6% ± 1.8%	27.6% ± 1.8%	22.4% ± 1.5%	27.5% ± 2.1%	27.3% ± 1.2%	26.7% ± 1.2%	30.2% ± 1.5%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
APPLE-QInit	64.4% ± 6.6%	66.6% ± 7.5%	73.3% ± 14.1%	76.2% ± 9.9%	64.7% ± 4.4%	62.5% ± 8.2%	53.2% ± 4.4%









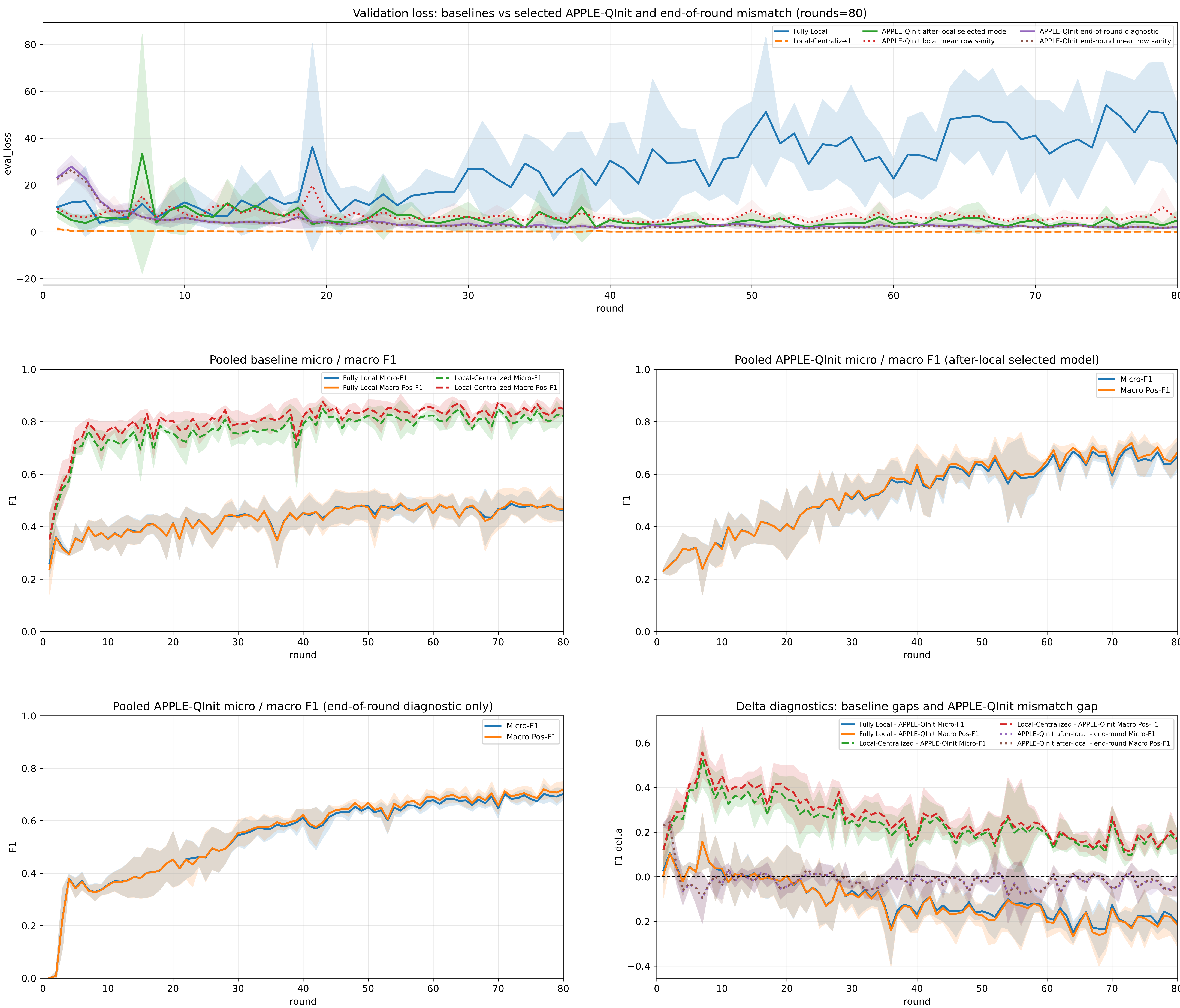


**Controlled q-label heterogeneity: High | APPLE-QInit diagnostics | family=q_task_cycle2 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti**

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

Algorithm	**Micro-F1**	**Macro Pos-F1**	**cycle2 pos-F1**	**cycle3 pos-F1**	**cycle4 pos-F1**	**cycle5 pos-F1**	**cycle6 pos-F1**
Fully Local	28.6% ± 0.7%	28.6% ± 0.7%	23.4% ± 1.7%	29.1% ± 1.2%	30.2% ± 0.8%	28.5% ± 1.4%	32.0% ± 1.3%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
APPLE-QInit	61.4% ± 6.7%	63.8% ± 7.0%	70.1% ± 11.0%	75.2% ± 13.6%	62.9% ± 1.5%	59.6% ± 10.5%	51.2% ± 5.9%

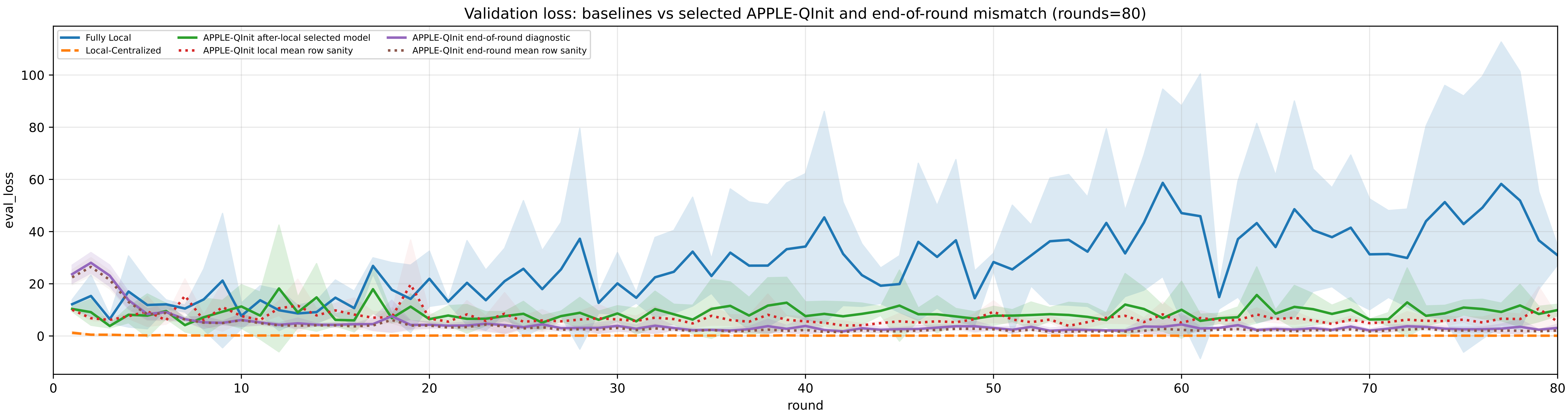


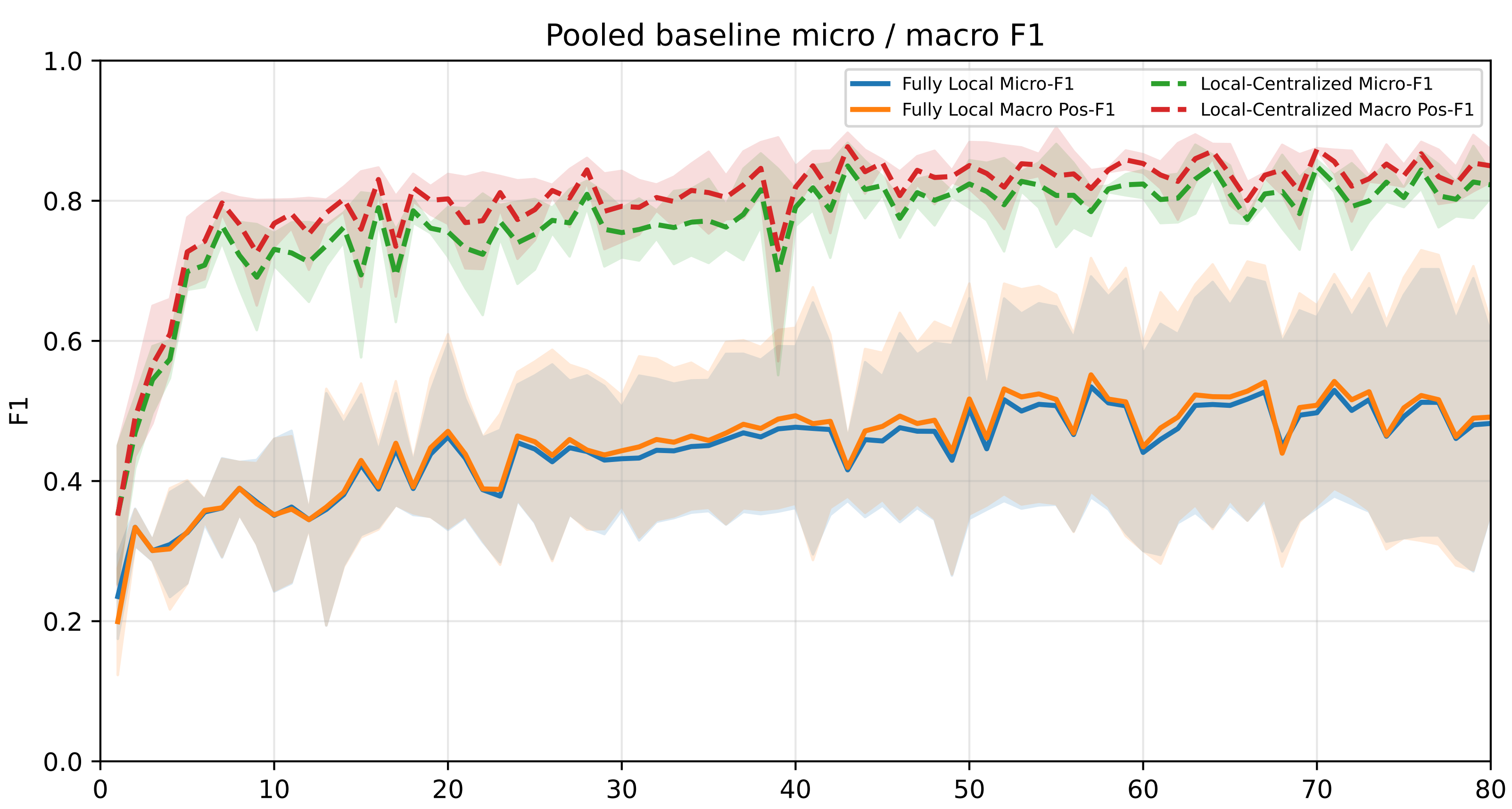
Controlled q-label heterogeneity: High | APPLE-QInit diagnostics | family=q_task_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

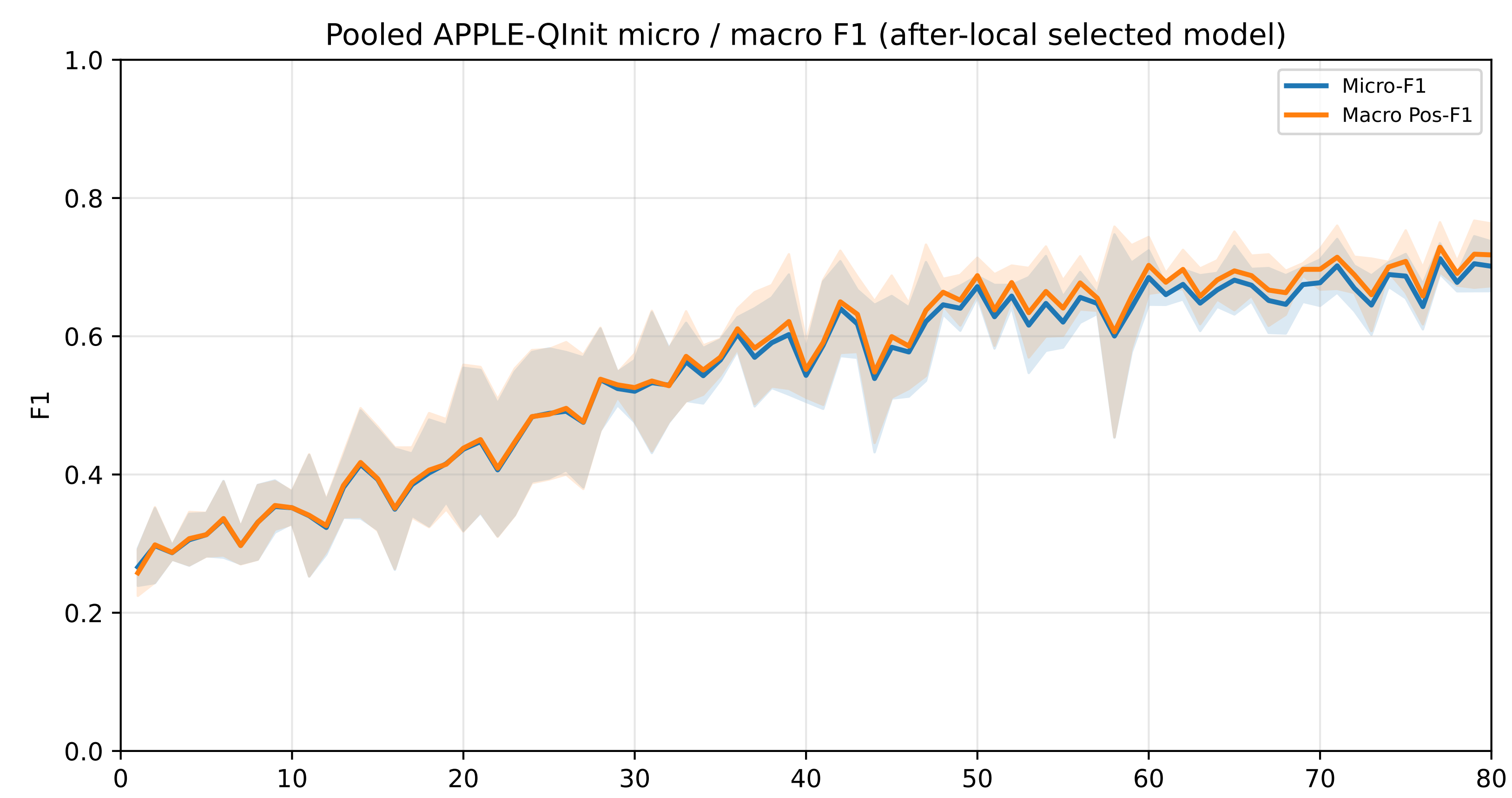
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

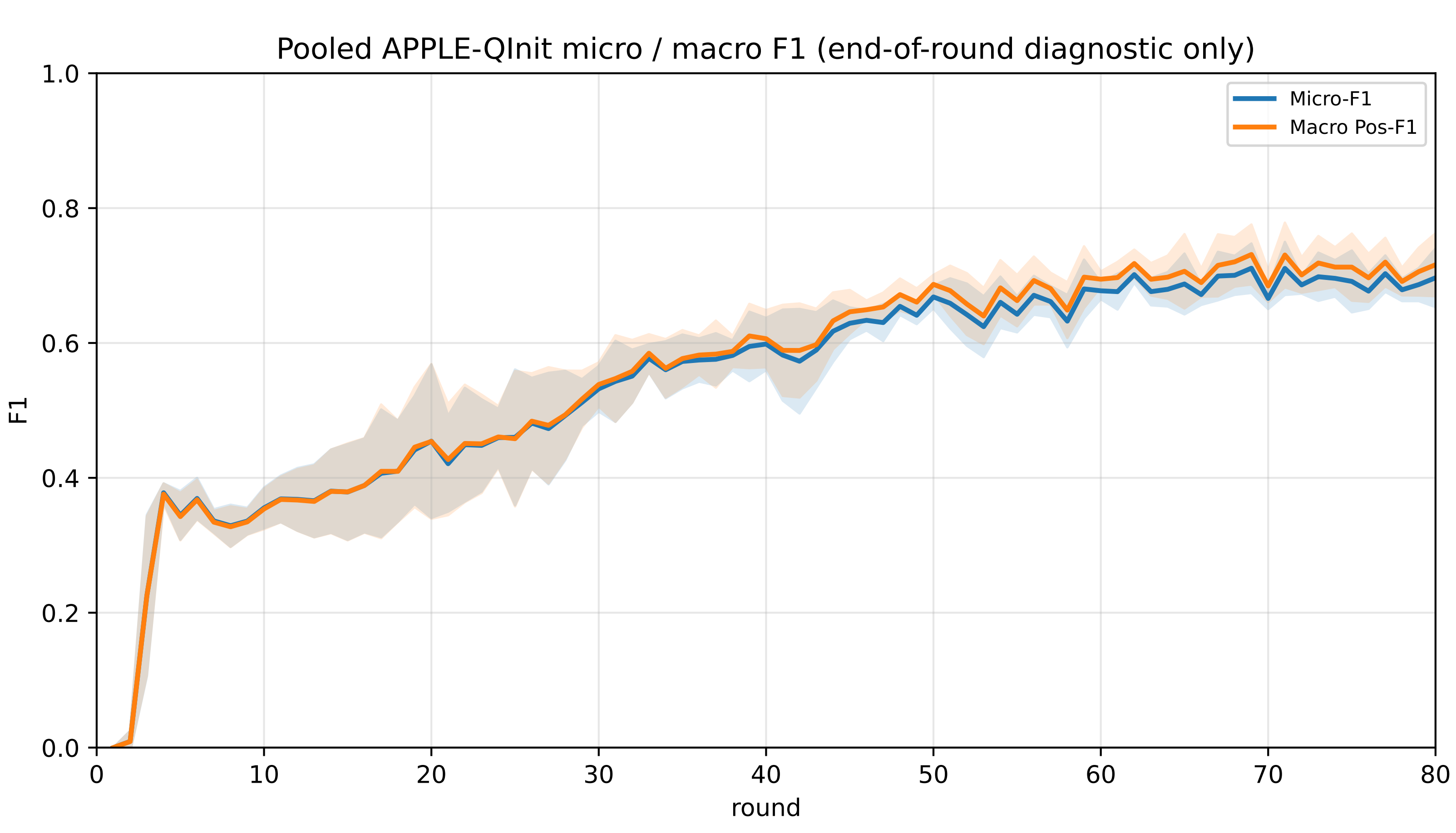
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

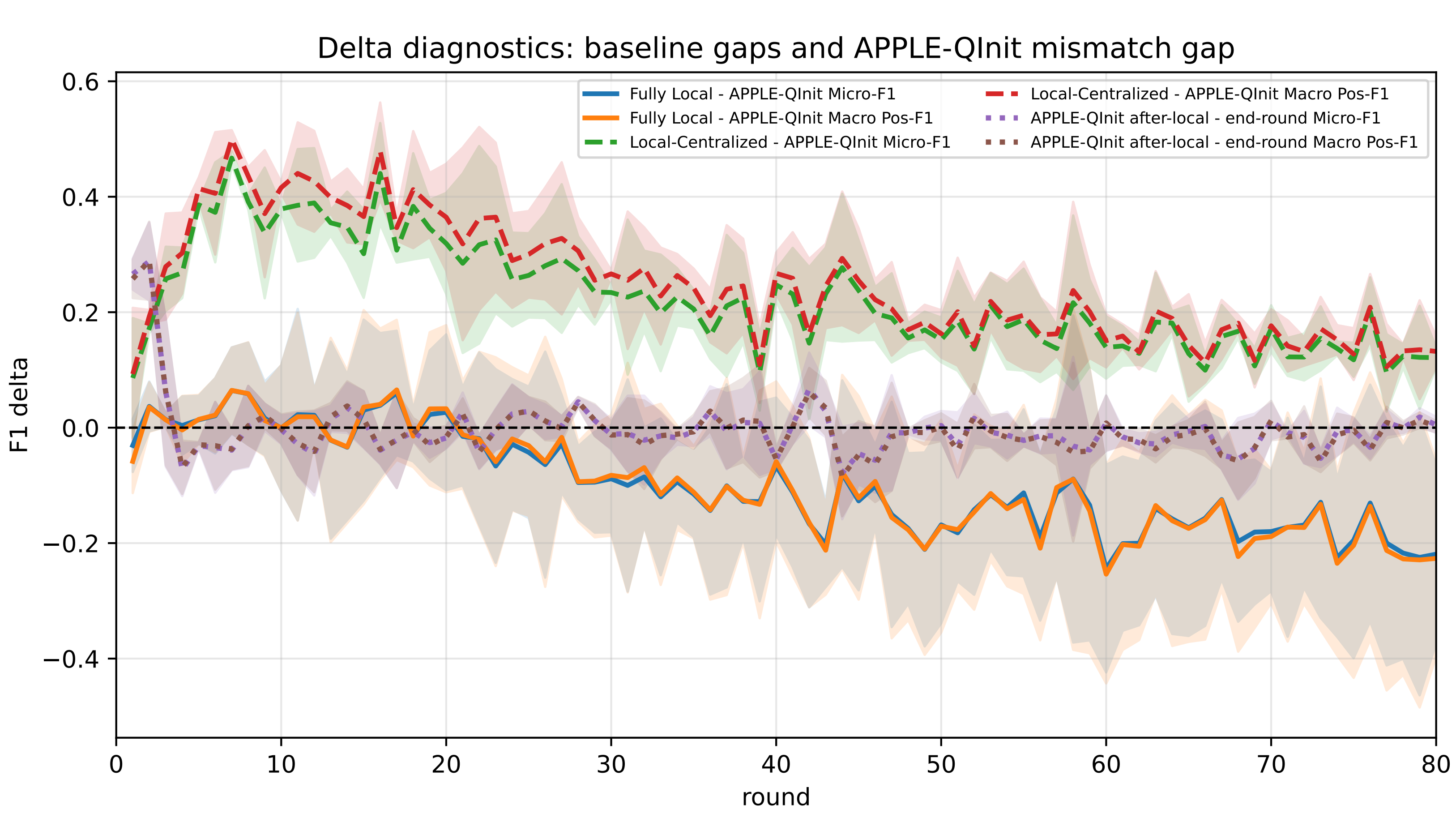
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	26.5% ± 4.7%	26.6% ± 4.7%	22.3% ± 4.2%	27.0% ± 7.0%	26.9% ± 5.3%	27.0% ± 4.1%	29.5% ± 3.2%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
APPLE-QInit	66.4% ± 5.6%	68.1% ± 5.9%	75.6% ± 8.9%	73.4% ± 4.4%	69.9% ± 5.2%	66.4% ± 9.2%	55.0% ± 5.0%











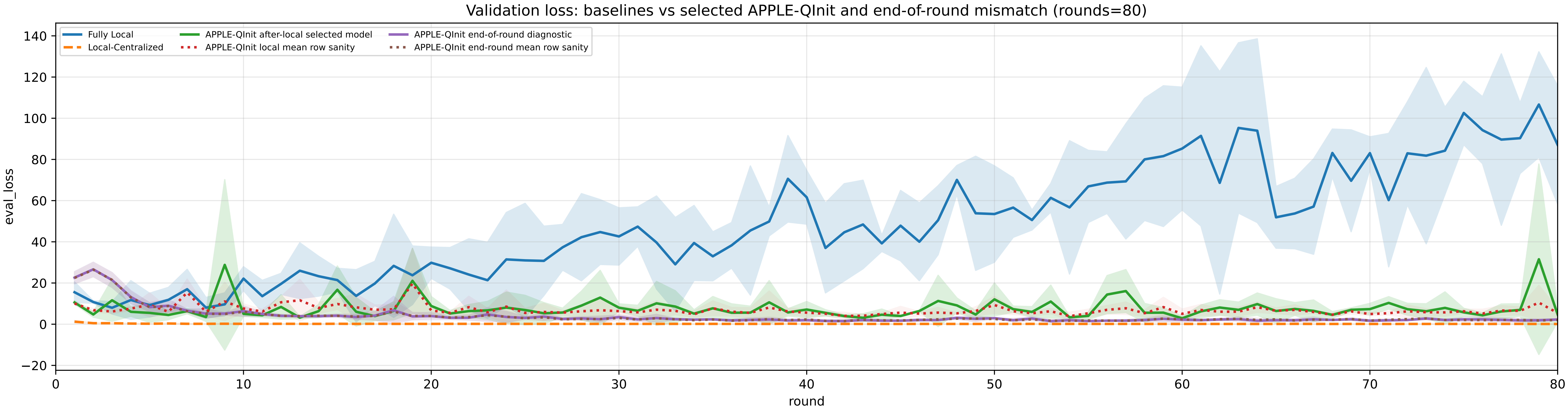
Controlled q-label heterogeneity: High | APPLE-QInit diagnostics | family=q_task_cycle4 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3% | heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

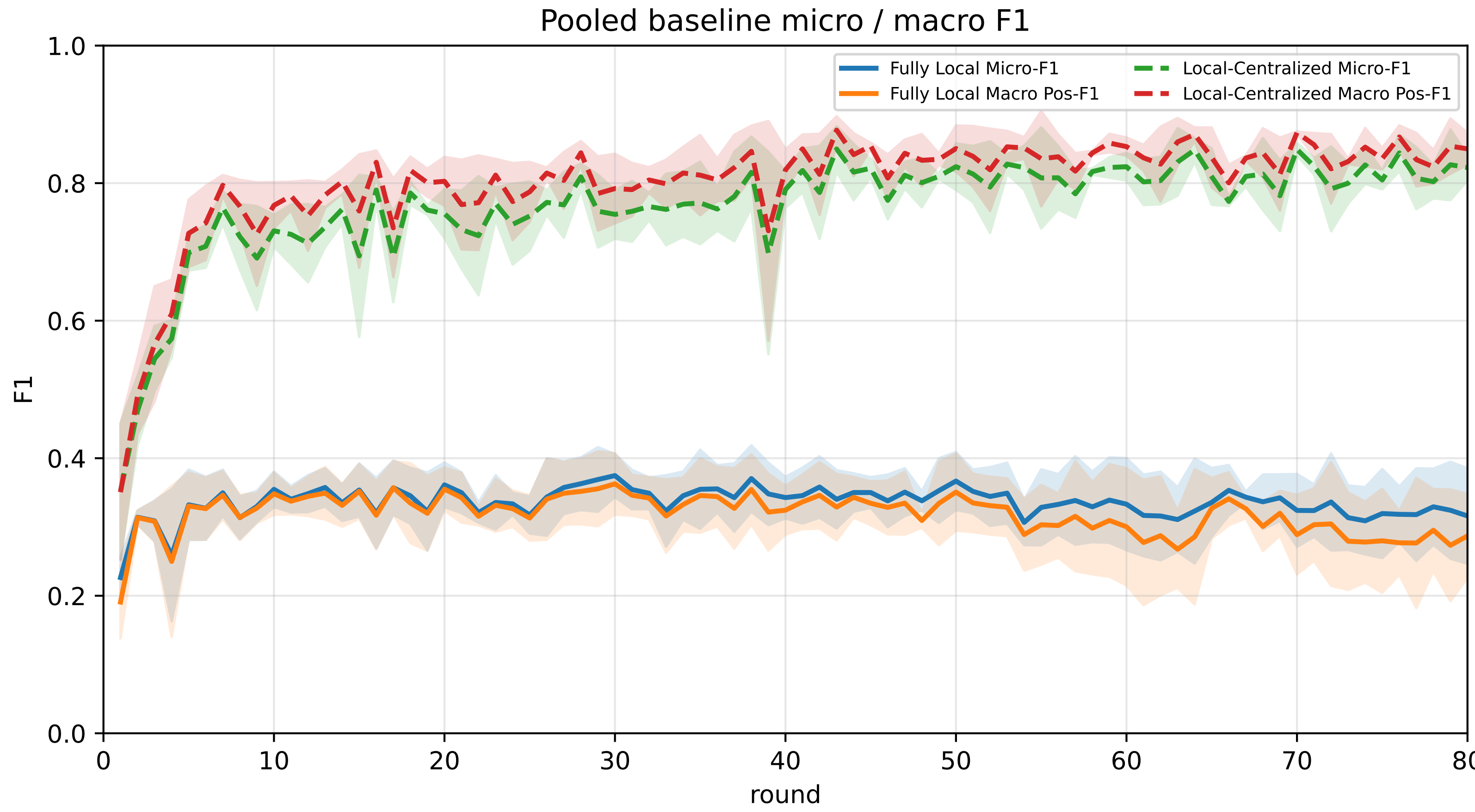
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.5% ± 1.5%	27.5% ± 1.4%	23.1% ± 1.8%	28.2% ± 0.7%	26.4% ± 1.8%	28.3% ± 1.7%	31.5% ± 2.1%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
APPLE-QInit	63.5% ± 11.6%	65.7% ± 13.0%	74.1% ± 21.6%	77.4% ± 15.5%	60.9% ± 10.9%	63.5% ± 12.7%	52.8% ± 7.6%

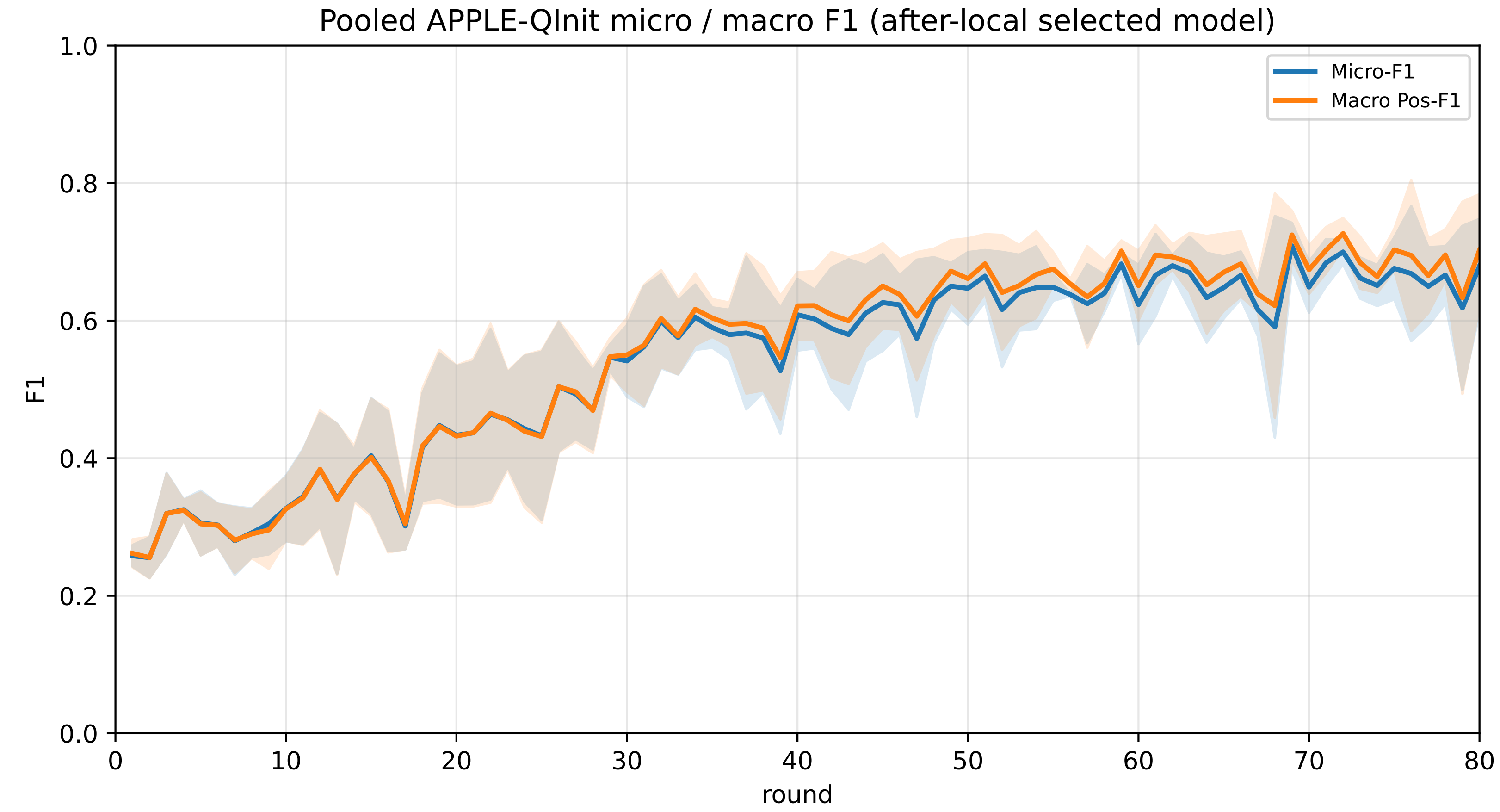
Validation loss: baselines vs selected APPLE-QInit and end-of-round mismatch (rounds=80)



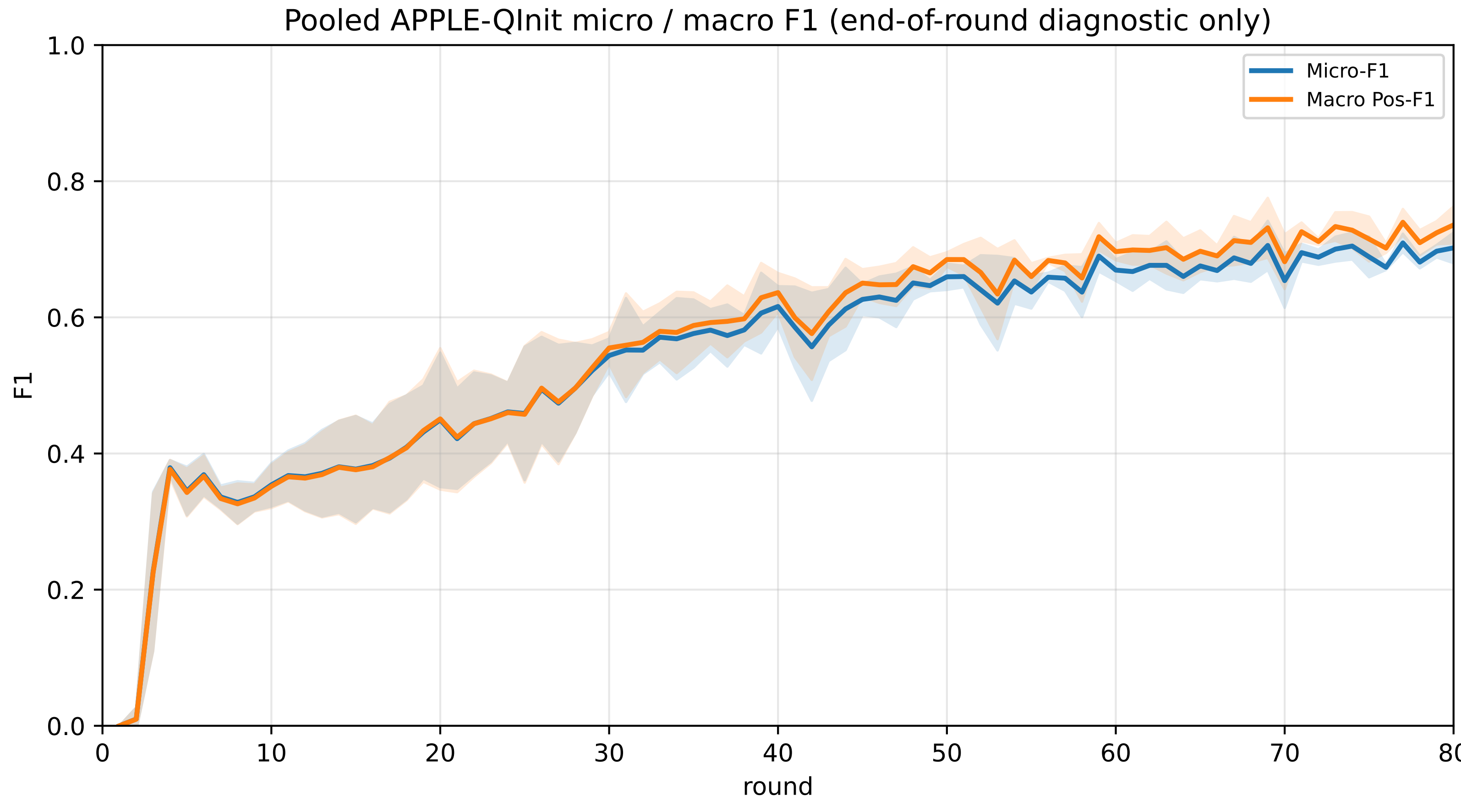
Pooled baseline micro / macro F1



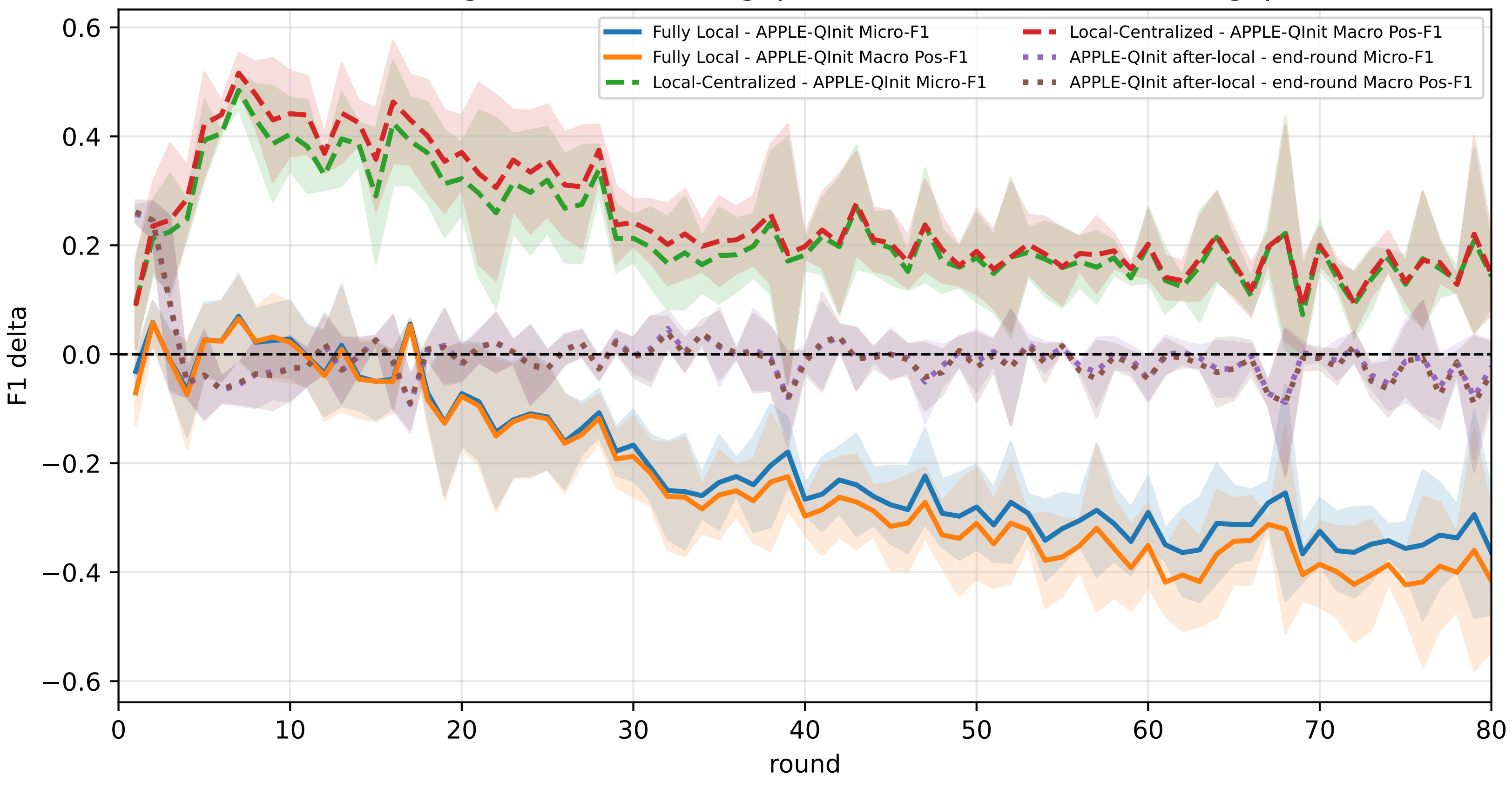
Pooled APPLE-QInit micro / macro F1 (after-local selected model)



Pooled APPLE-QInit micro / macro F1 (end-of-round diagnostic only)



Delta diagnostics: baseline gaps and APPLE-QInit mismatch gap

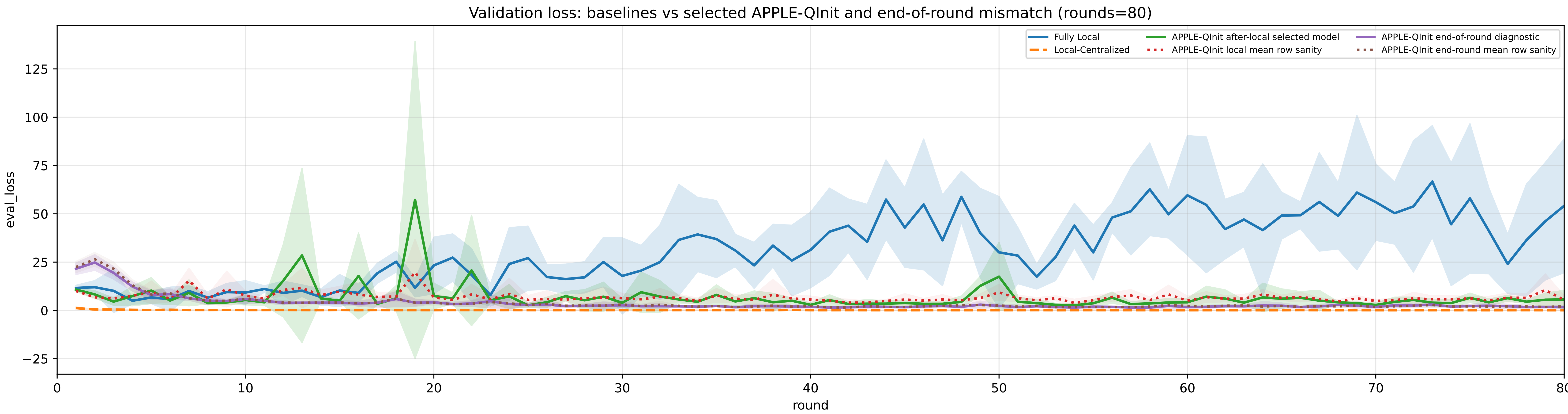


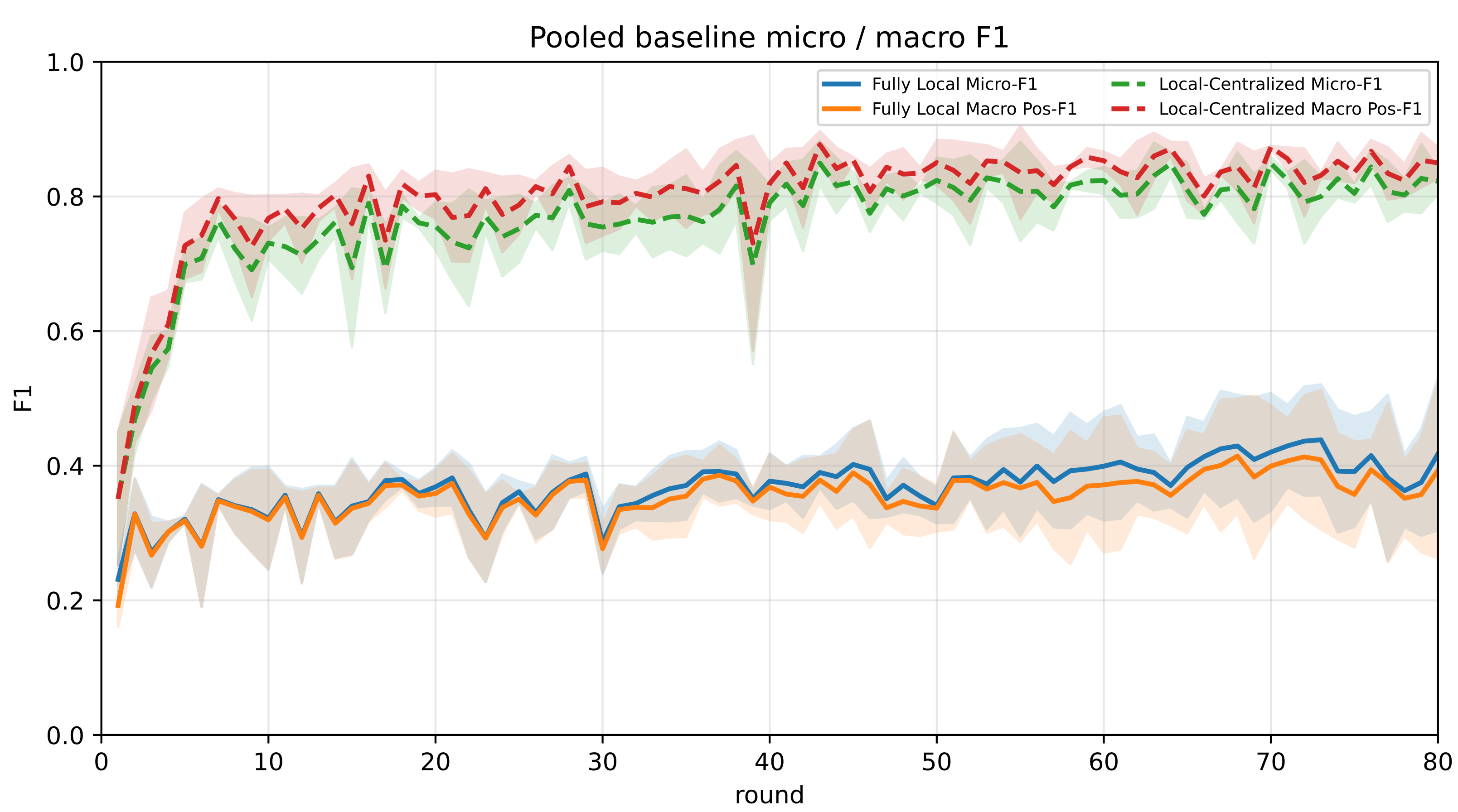
Controlled q-label heterogeneity: High | APPLE-QInit diagnostics | family=q_task_cycle5 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

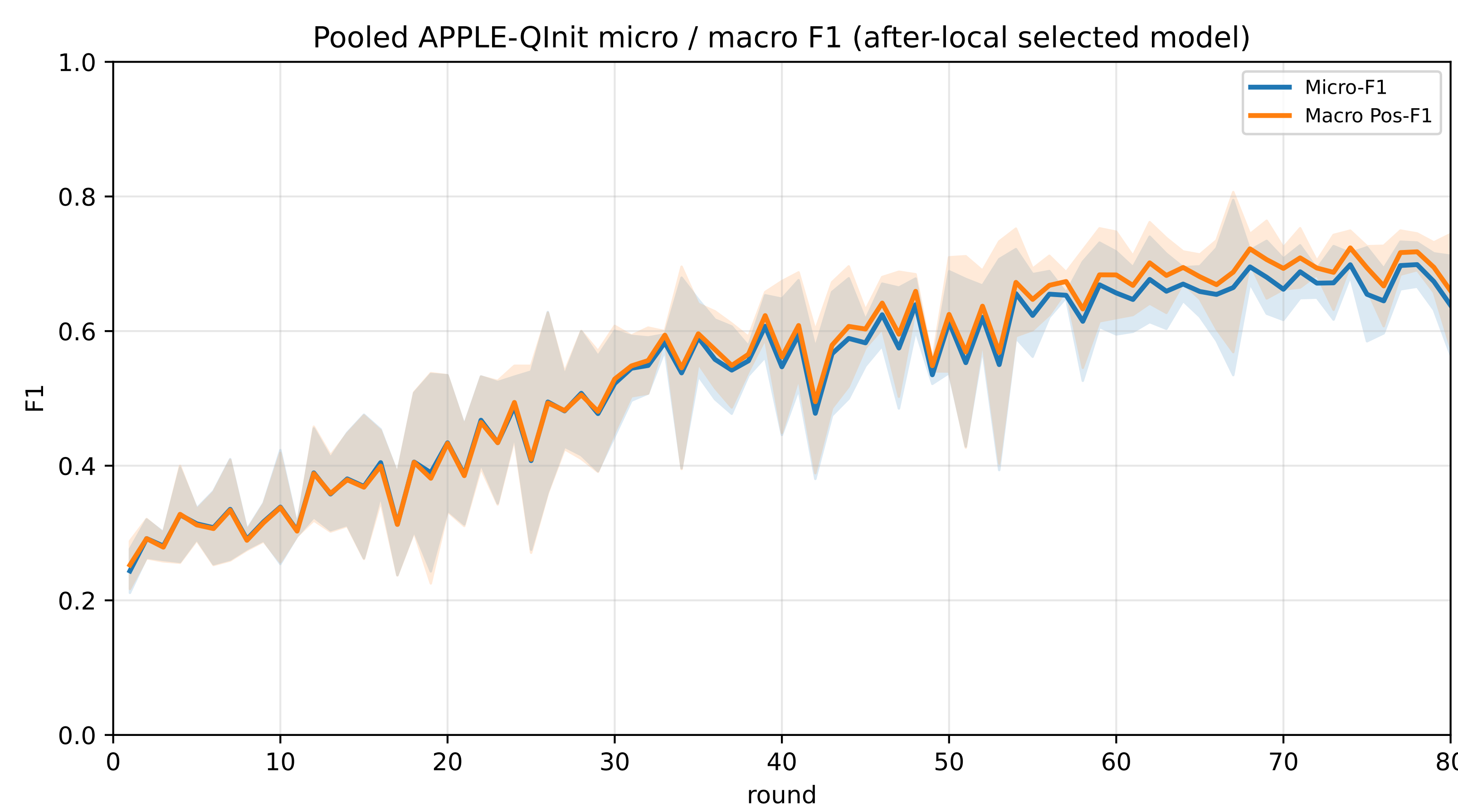
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

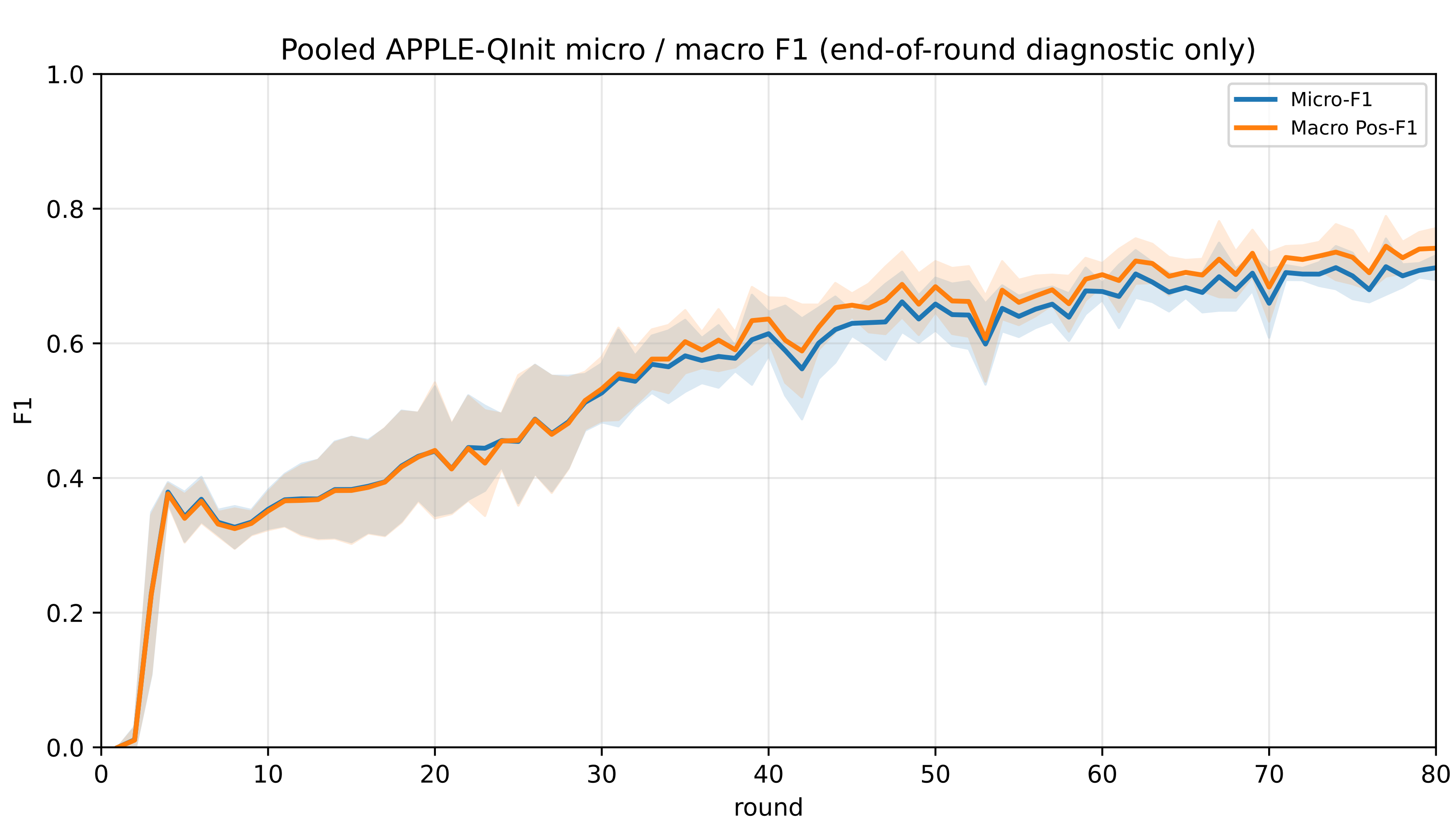
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

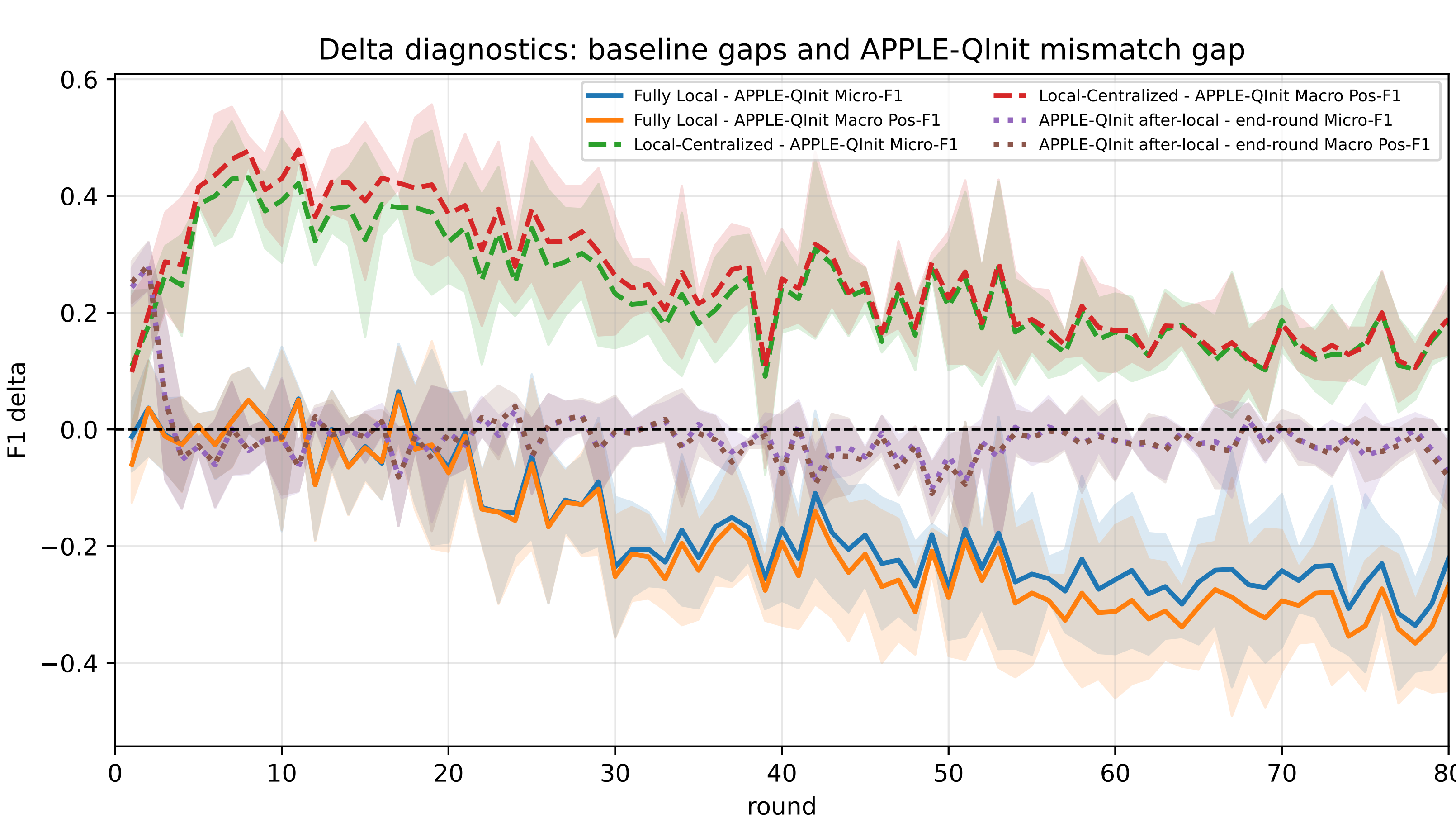
Algorithm	**Micro-F1**	**Macro Pos-F1**	**cycle2 pos-F1**	**cycle3 pos-F1**	**cycle4 pos-F1**	**cycle5 pos-F1**	**cycle6 pos-F1**
Fully Local	24.2% ± 1.1%	24.2% ± 1.1%	20.1% ± 0.7%	25.3% ± 1.1%	24.5% ± 1.6%	22.9% ± 1.3%	28.3% ± 1.1%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
APPLE-QInit	64.1% ± 8.0%	66.2% ± 8.9%	74.4% ± 17.2%	77.0% ± 10.2%	67.3% ± 5.0%	57.5% ± 9.8%	55.0% ± 4.3%









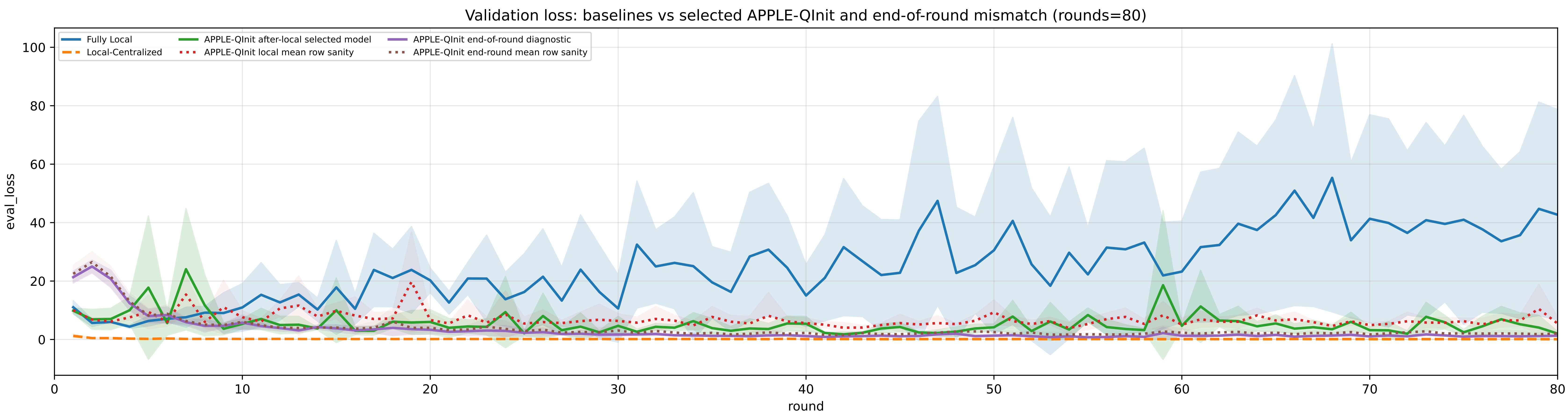


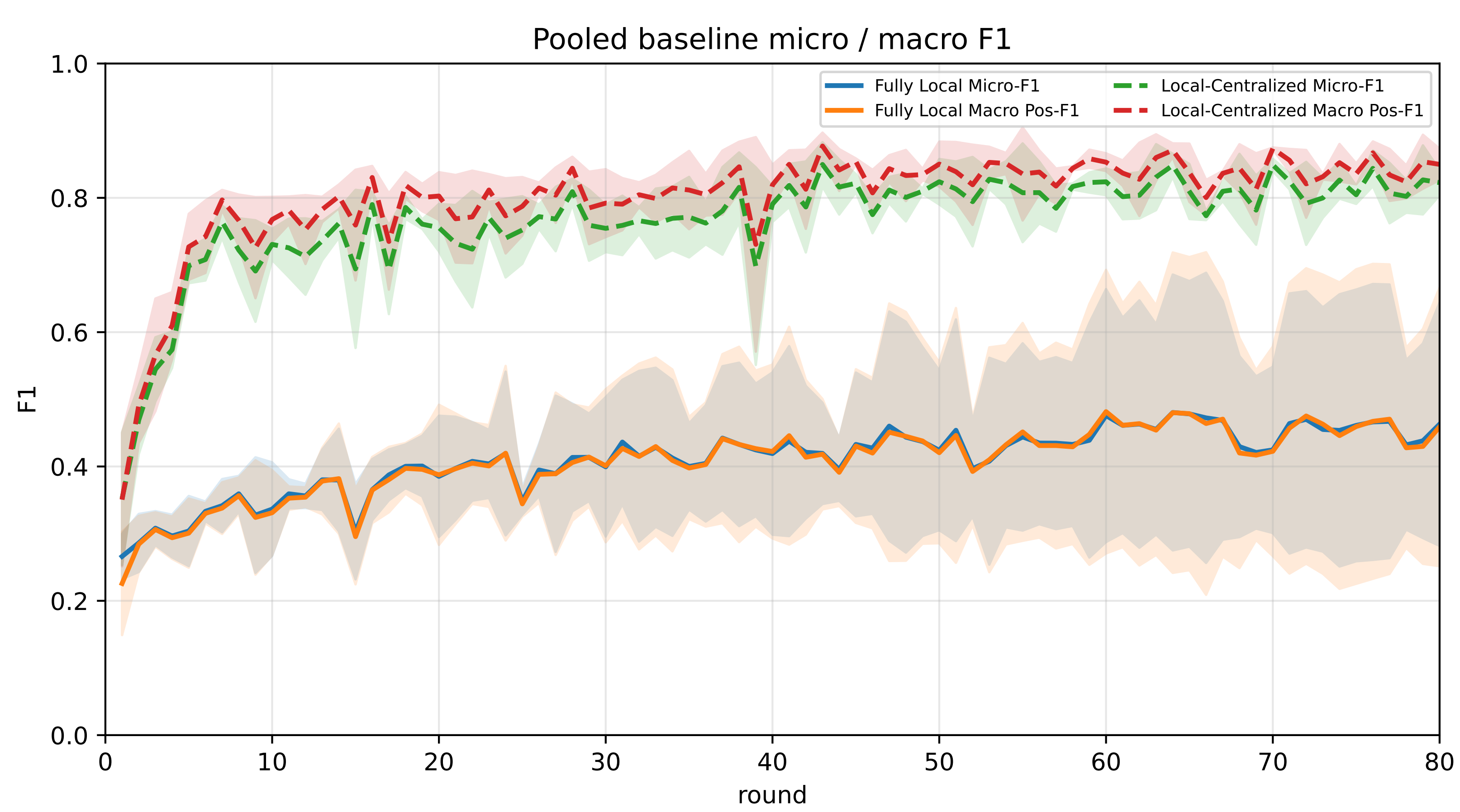
Controlled q-label heterogeneity: High | APPLE-QInit diagnostics | family=q_task_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

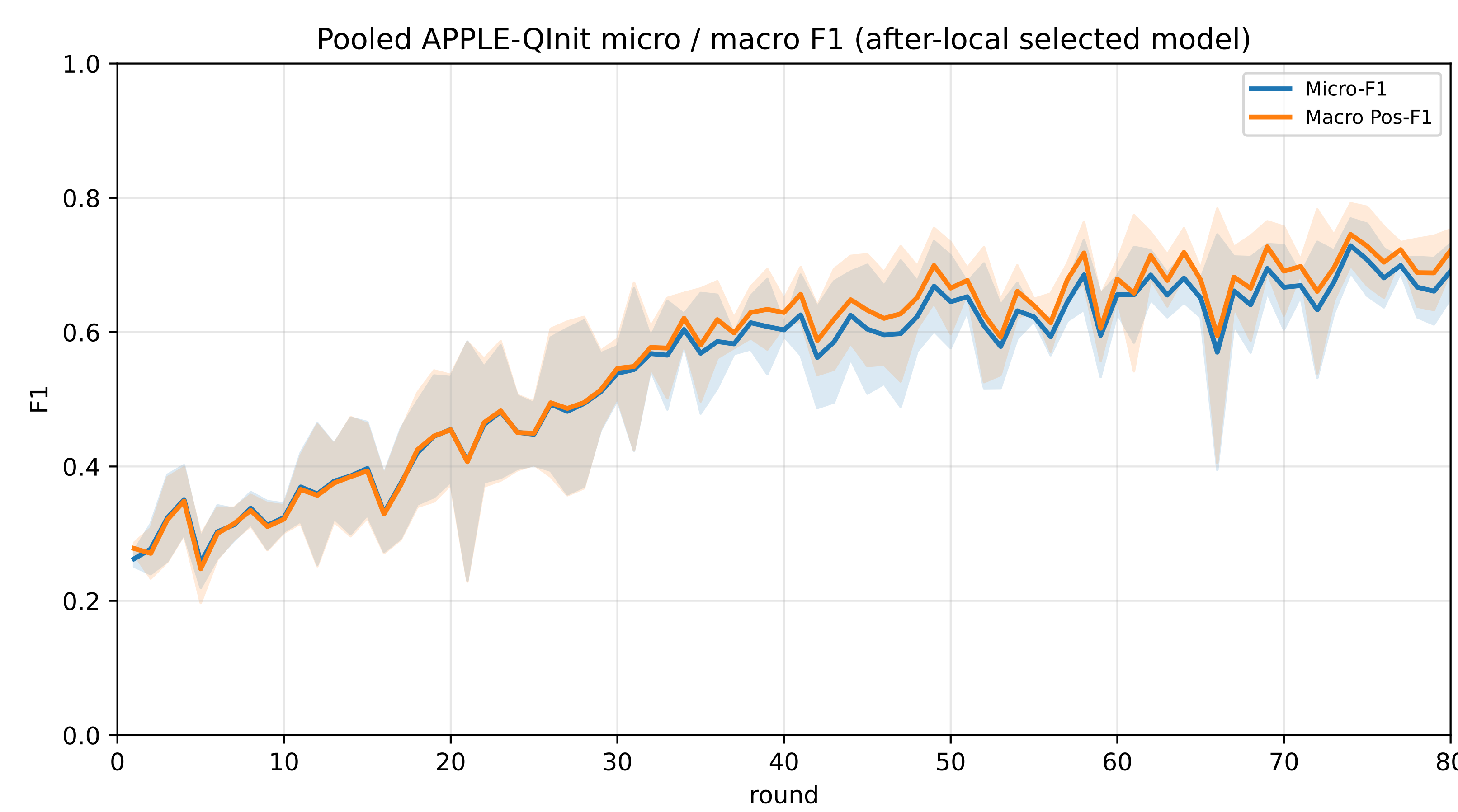
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

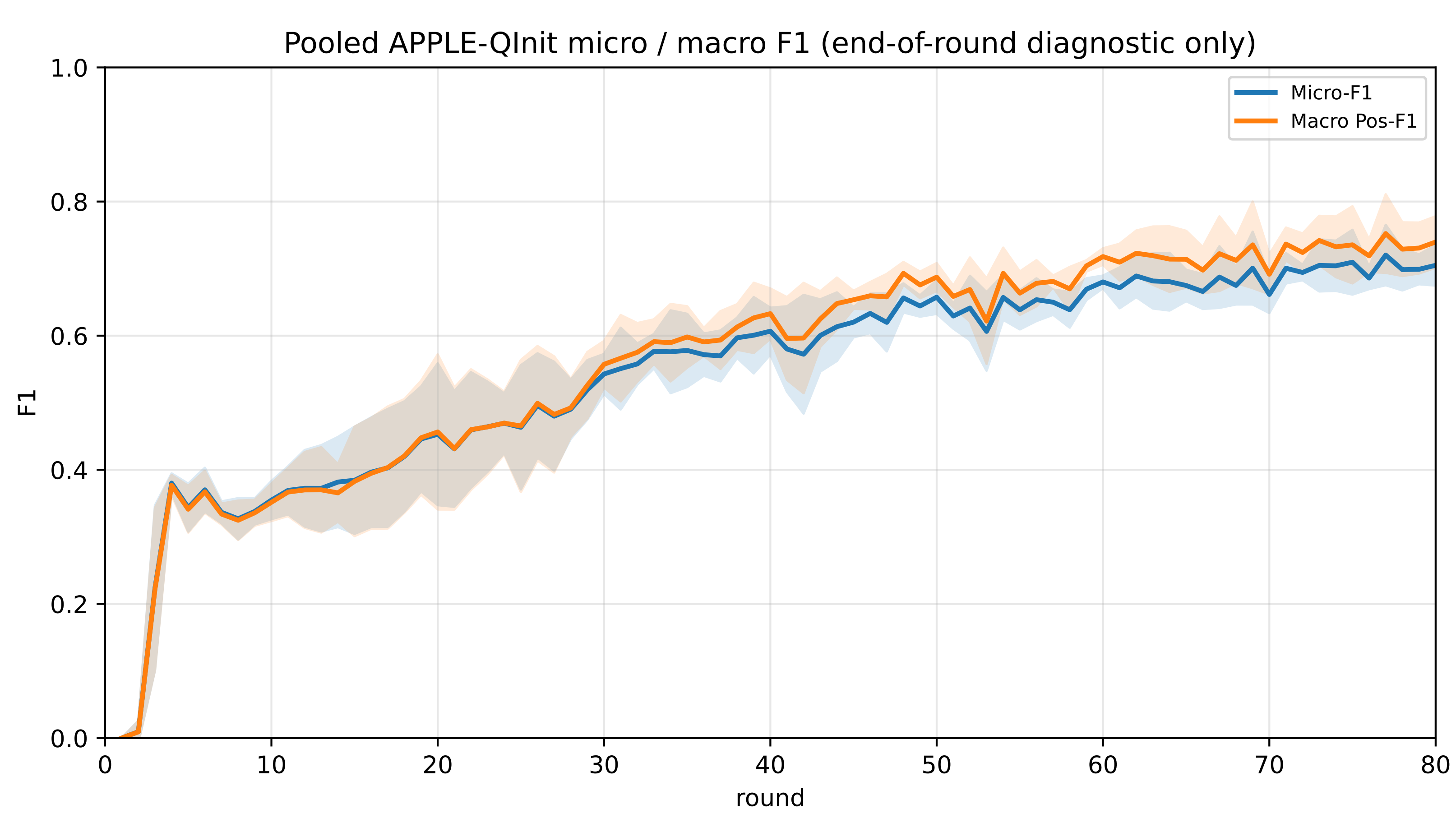
Best test performance (Fully Local vs Local-Centralized vs APPLE-QInit, rounds=80)

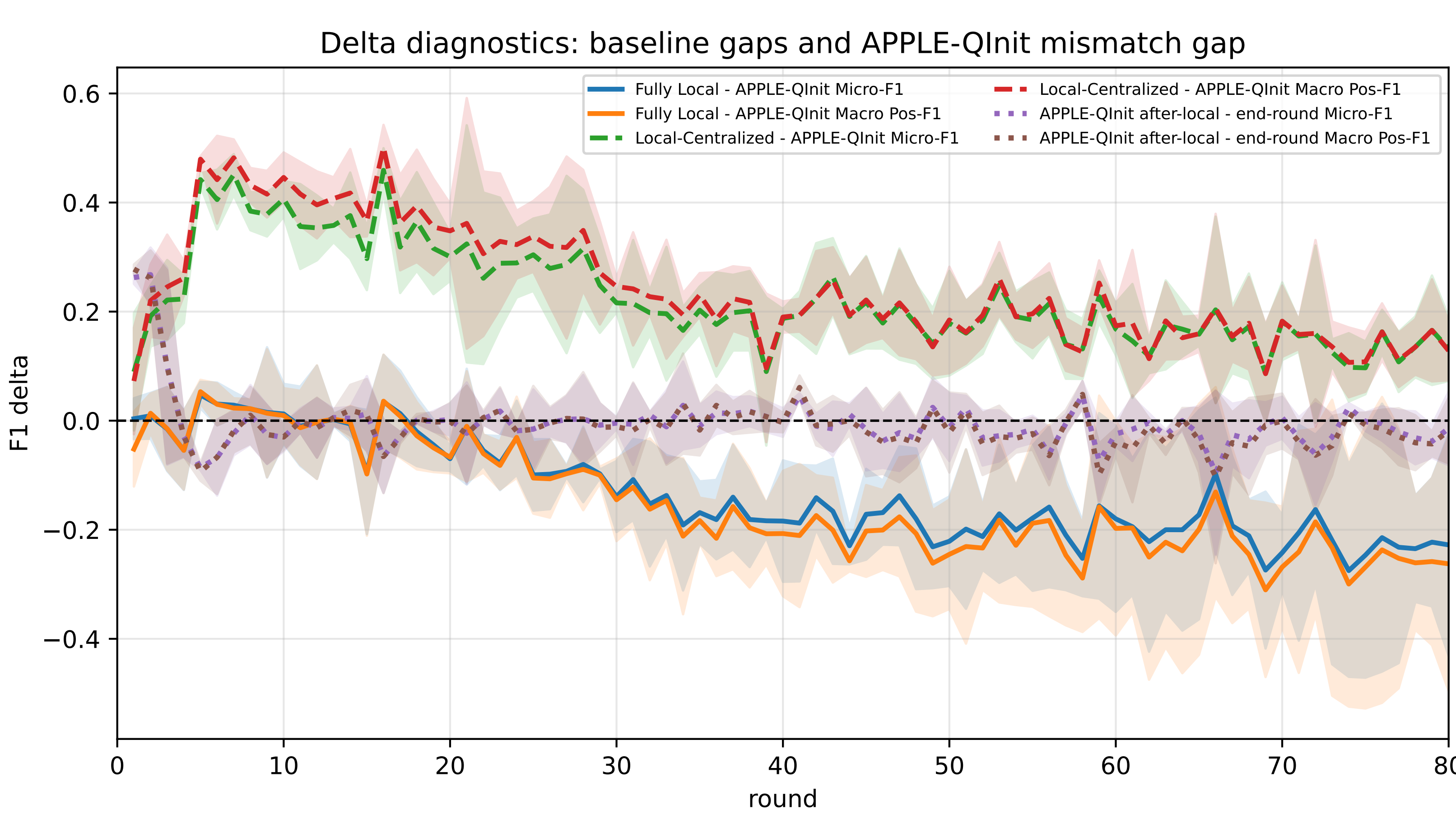
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	31.2% ± 8.8%	31.1% ± 8.9%	26.7% ± 8.1%	32.1% ± 9.5%	33.6% ± 11.2%	31.3% ± 9.2%	32.1% ± 6.8%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
APPLE-QInit	66.7% ± 4.1%	69.1% ± 4.6%	74.9% ± 10.9%	80.2% ± 6.5%	65.4% ± 3.3%	71.7% ± 2.3%	53.5% ± 2.6%





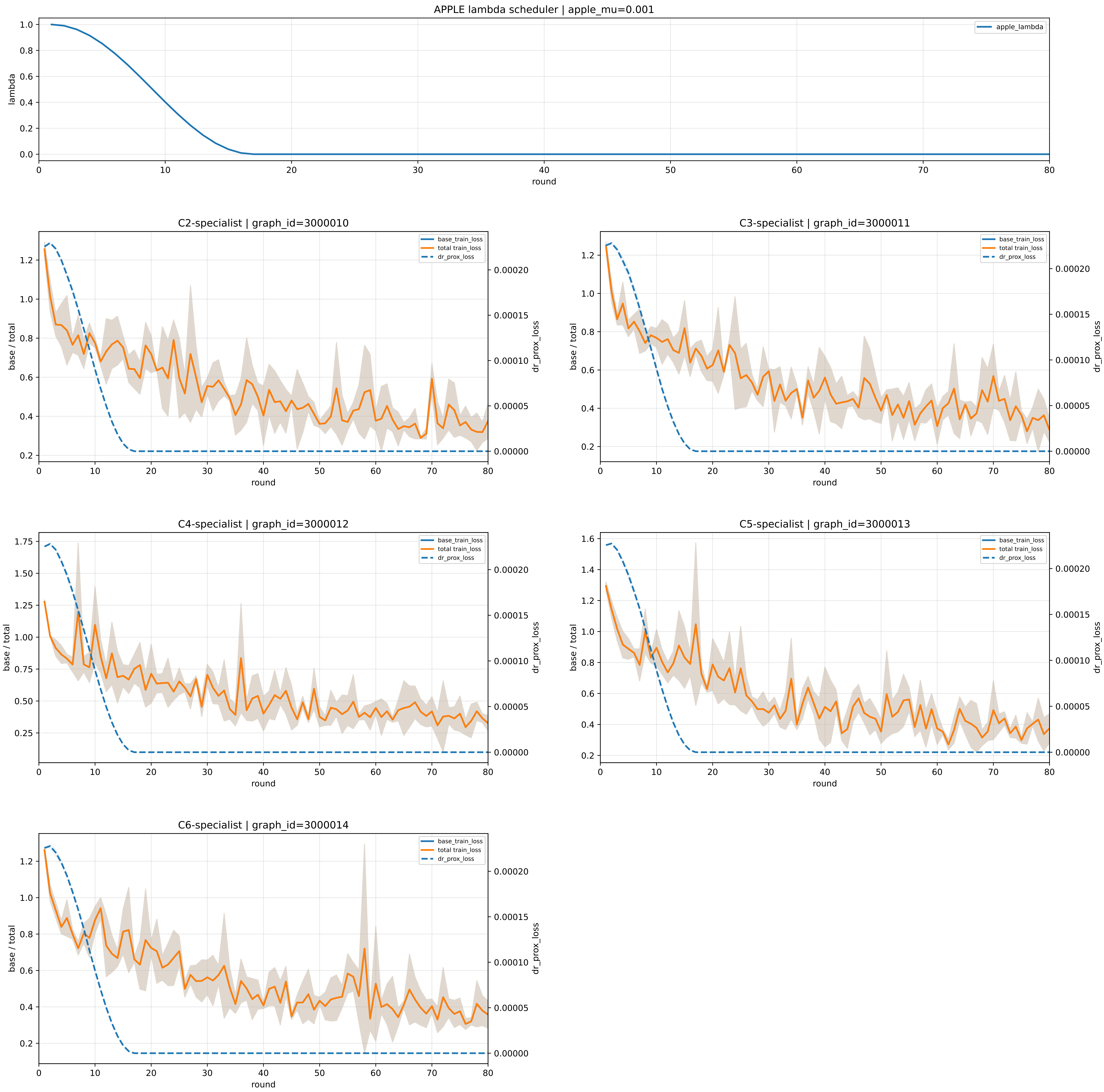


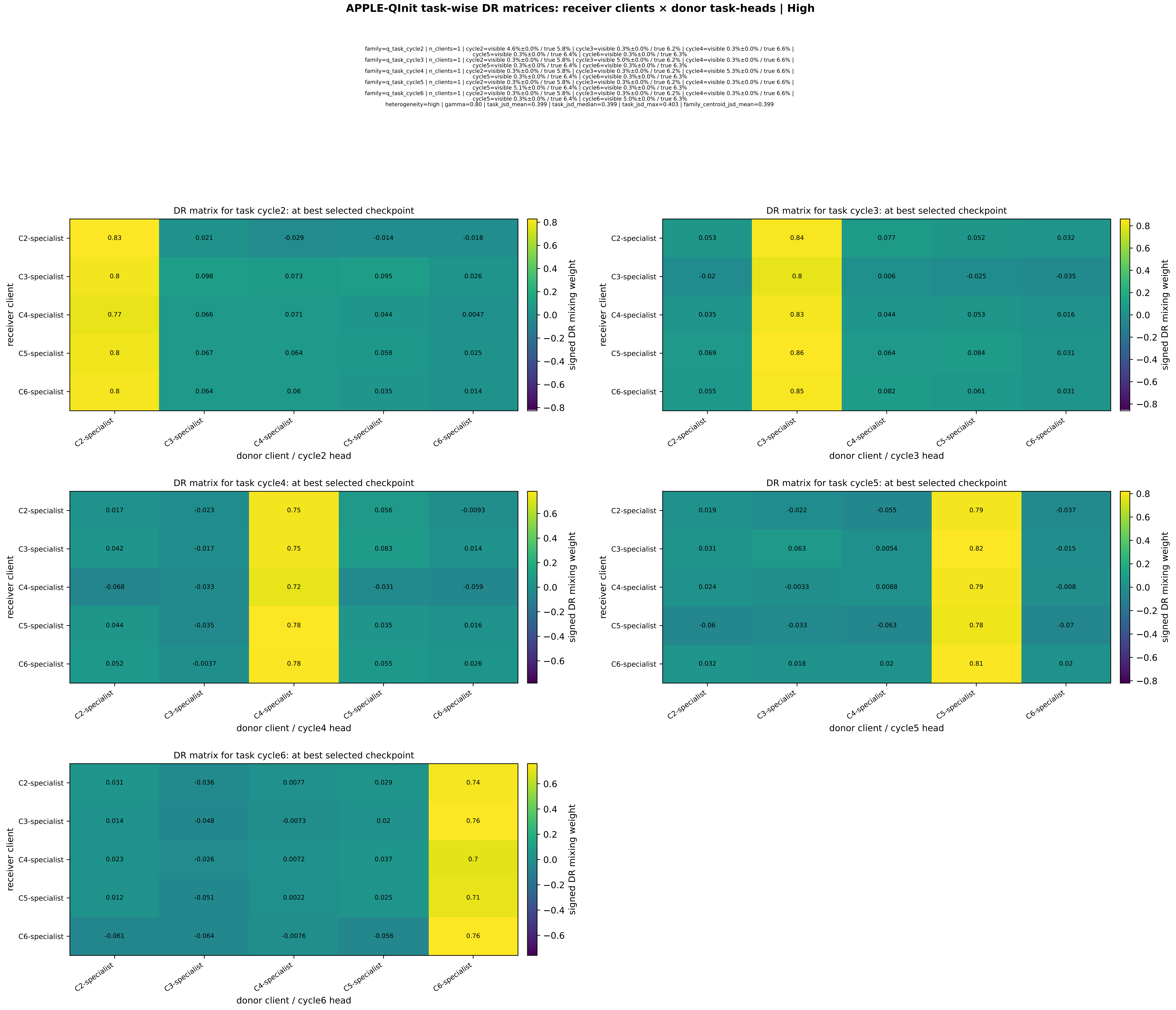


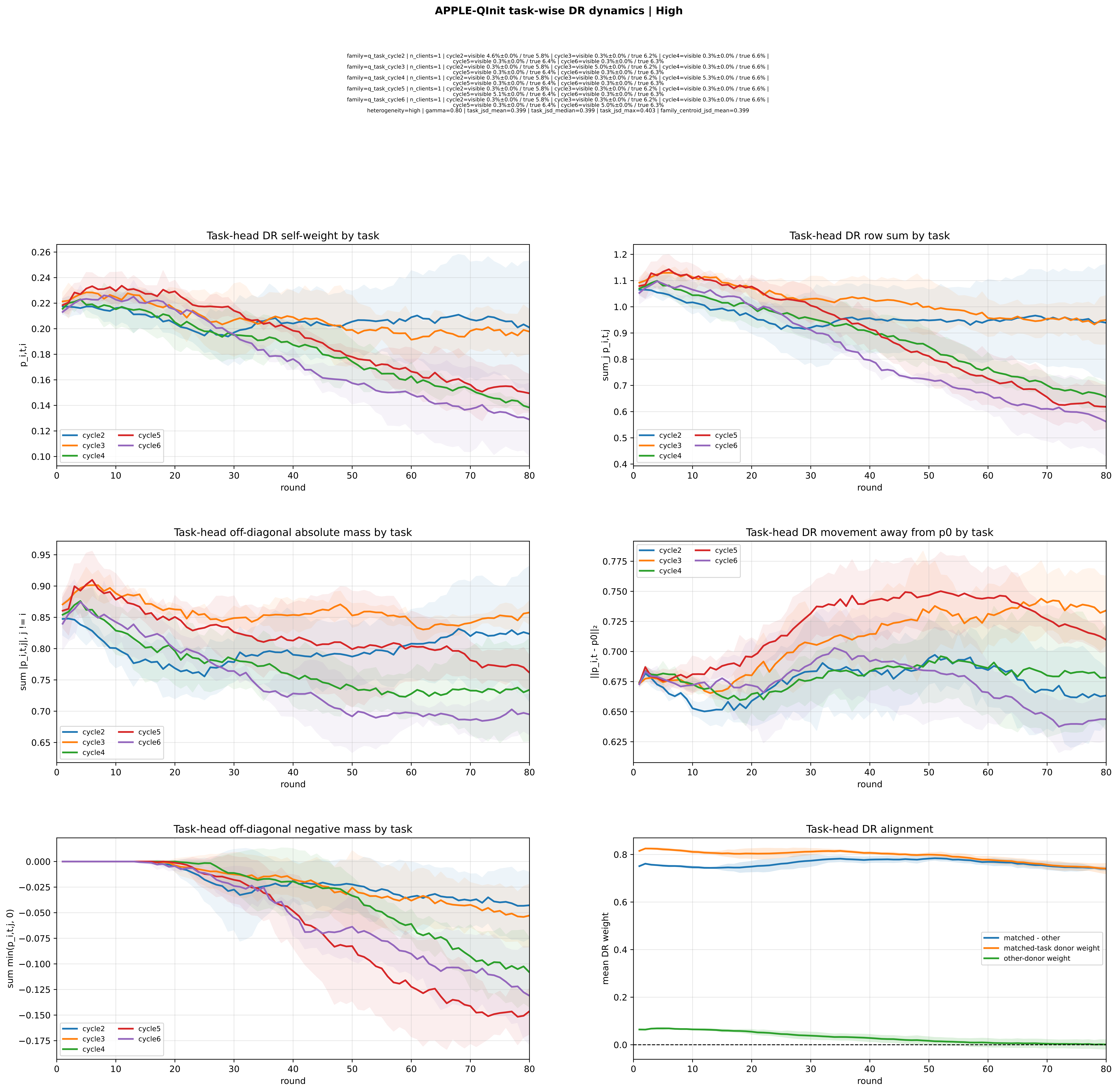


APPLE-QInit training-loss decomposition | High heterogeneity

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399




APPLE-QInit task-wise DR matrices: receiver clients × donor task-heads | High
family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399
DR matrix for task cycle2: at best selected checkpoint
receiver client
donor client / cycle2 head
signed DR mixing weight
C2-specialist 0.83 0.021 -0.029 -0.014 -0.018
C3-specialist 0.8 0.098 0.073 0.095 0.026
C4-specialist 0.77 0.066 0.071 0.044 0.0047
C5-specialist 0.8 0.067 0.064 0.058 0.025
C6-specialist 0.8 0.064 0.06 0.035 0.014
DR matrix for task cycle3: at best selected checkpoint
receiver client
donor client / cycle3 head
signed DR mixing weight
C2-specialist 0.053 0.84 0.077 0.052 0.032
C3-specialist -0.02 0.8 0.006 -0.025 -0.035
C4-specialist 0.035 0.83 0.044 0.053 0.016
C5-specialist 0.069 0.86 0.064 0.084 0.031
C6-specialist 0.055 0.85 0.082 0.061 0.031
DR matrix for task cycle4: at best selected checkpoint
receiver client
donor client / cycle4 head
signed DR mixing weight
C2-specialist 0.017 -0.023 0.75 0.056 -0.0093
C3-specialist 0.042 -0.017 0.75 0.083 0.014
C4-specialist -0.068 -0.033 0.72 -0.031 -0.059
C5-specialist 0.044 -0.035 0.78 0.035 0.016
C6-specialist 0.052 -0.0037 0.78 0.055 0.026
DR matrix for task cycle5: at best selected checkpoint
receiver client
donor client / cycle5 head
signed DR mixing weight
C2-specialist 0.019 -0.022 -0.055 0.79 -0.037
C3-specialist 0.031 0.063 0.0054 0.82 -0.015
C4-specialist 0.024 -0.0033 0.0088 0.79 -0.008
C5-specialist -0.06 -0.033 -0.063 0.78 -0.07
C6-specialist 0.032 0.018 0.02 0.81 0.02
DR matrix for task cycle6: at best selected checkpoint
receiver client
donor client / cycle6 head
signed DR mixing weight
C2-specialist 0.031 -0.036 0.0077 0.029 0.74
C3-specialist 0.014 -0.048 -0.0073 0.02 0.76
C4-specialist 0.023 -0.026 0.0072 0.037 0.7
C5-specialist 0.012 -0.051 0.0022 0.025 0.71
C6-specialist -0.061 -0.064 -0.0076 -0.056 0.76
C2-specialist C3-specialist C4-specialist C5-specialist C6-specialist


APPLE-QInit task-wise DR dynamics | High
family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399
Task-head DR self-weight by task
p_i,t,i
round
Task-head DR row sum by task
sum_j p_i,t,j
round
Task-head off-diagonal absolute mass by task
sum |p_i,t,j|, j != i
round
Task-head DR movement away from p0 by task
||p_i,t - p0||₂
round
Task-head off-diagonal negative mass by task
sum min(p_i,t,j, 0)
round
cycle2
cycle3
cycle4
cycle5
cycle6
Task-head DR alignment
mean DR weight
round
matched - other
matched-task donor weight
other-donor weight